Coefficient Extraction Operators in the Theory of Multidimensional Random Walk Enumeration

by

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Abstract

In this work we apply an operator approach to multidimensional random walk enumeration in order to automatize some of the process in solving these problems. In particular given a random walk model in any number of dimensions we can use operators to write down a functional equation $(1 - t\Gamma S)Q = q$ for the generating function Q where Γ is an operator and S is the Laurent step polynomial for the model. Further we use the operator perspective to further and unify some of the different kernel methods (orbit and obstinate methods in particular). This operator paradigm work for both classic random walk models and models with interactions with hyperplanes. Near the end we solve some two-dimensional problems with interactions.

Preface

This work intends to establish a new framework of notation and tools for the enumeration of random walks with or without interactions, and similar problems. The notation makes it much easier to explore the symmetries of the problem and write down summation formulas where it was not possible to do so before (with a simplified notation). The tools make the derivation of functional equations for various problems an almost trivial matter. The real power is in combining both approaches to gain insights into the generating function world of enumerating random walks on constrained regions in multidimensional spaces.

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Chapter 1

Preliminaries

In these pages we'll set up some of the basic tools and notation that we'll use in our study of random walk models. This chapter will not be comprehensive. For more information on each section please consult the cited literature.

1.1 Formal Series

We will work in the setting of formal power series and Laurent series. Issues with convergence will not be addressed in much detail. If we are given a ring R we may construct several formal objects with a commutative indeterminate x with coefficients in R

- $R[x]$: Ring of polynomials with monomials x^k for integers $k \geqslant 0$.
- $R[x, 1/x]$: Ring of Laurent polynomials with monomials x^k for integers k.
- $R[[x]]$: Ring of formal power series with monomials x^k for integers $k \geqslant 0$.
- $R((x))$: Field of formal Laurent series with monomials x^k for integers k.

We can also define their multivariate analogues like $R[x_1, \ldots, x_n]$, $R[x_1, \overline{x_1}, \ldots, x_n, \overline{x}_n]$ and their series counterparts $R[[x_1, \ldots, x_n]]$ and $R((x_1, \ldots, x_n))$. For a thorough treatment of the subject please see [1]. When we work with a combination of objects coming from different rings/fields over the indeterminate x we'll consider all objects as part of the most general ring/field at the time. If in doubt, we'll always work in the field of formal Laurent series. Given a sequence (w_n) we may consider its generating function $W(x)$ given by $\sum_n w_n x^n$ which lives appropriately in one of the rings over x. The multivariate case for a multi-index sequence $(w_{n_1,\ldots,n_m})$ is defined analogously

as

$$W(x_1, \ldots, x_d) = \sum_{n_1,\ldots,n_m} w_{n_1,\ldots,n_m} x_1^{n_1} \ldots x_m^{n_m} \tag{1.1}$$

whenever this series is a formal Laurent series. When we solve enumeration problems via generating functions we often need to extract particular coefficients from the formal polynomial/series. In the one-variable case we have the coefficient extraction operators on formal series

$$[x^n]W(x) = w_n \tag{1.2}$$

The multivariate case is handled analogously with the caveat that whenever the number of variables extracted is less than the number of variables present we extract by treating these remaining variables as parameters. For a more detailed discussion please consult [2]. When we deal random walk generating functions we often work with functional equations. In the broadest sense in this thesis we'll discuss functional equations which may involve a function F, the function evaluated at various arguments and various coefficient extractions of F. This will be done via operators so in the end we'll have a functional equation involving F and operators. We'll write the identity operator as 1.

1.2 Random Walk Models

In this work we'll investigate random walk models on the n-dimensional lattice $\mathbb{Z}^n$. A specific model $\mathcal{M}$ will consist of several ingredients

- $\mathcal{S}$: a finite set of steps that we can take from a point in $\mathbb{Z}^n$ to another point in $\mathbb{Z}^n$. In general this step set may differ from point to point (e.g., a honeycomb lattice embedded in $\mathbb{Z}^2$). The steps themselves will be written as vectors $s = (s_1, \ldots, s_n)$. For example $\mathcal{S}$ could be a set of small steps defined by coordinates among $\{-1, 0, 1\}$. Another example is when we have more than one step of a given kind.

- $\mathcal{D}$: a domain for the random walks. This is the region in $\mathbb{Z}^n$ where the random walks will live. The easiest cases involve the whole space or a half-space. Starting with quarter-plane regions the analysis becomes rather complicated.

- $\mathcal{D}_0$: an interaction domain for the random walks. This is a sub-region in $\mathcal{D}$ and we'll want to count how many times random walks interact with it by stepping on coordinates that are part of it. We may count different points from $\mathcal{D}_0$ in different ways. This less-studied paradigm may involve marking visits to the origin, axes in two dimensions, various higher-dimensional hyperplanes, a cone, infinite strip, the sides of a box and so on.

- $\mathcal{Q}$: a set of random walks acting as an initial condition from which all other random walks evolve. This could be as trivial as starting with the empty walk ending at the origin to a set of infinitely many walks with some pattern.

It is natural to consider the generating function that will keep track of all the information characterizing the random walks in the model. For each random walk w its length $|w|$ will be the total number of steps that it was taken and we'll keep track of it via a parameter t. We'll keep track of its endpoint coordinates in $\mathbb{Z}^n$ by the indeterminates $x_1, \ldots, x_n$ and its interactions with $\mathcal{D}_0$ via marking parameters a_m where m is some specific subscript index. The marking is highly dependent on the geometry of $\mathcal{D}_0$ and the way we want to count interactions. If the end coordinates of the random walk are $(m_1, \ldots, m_n)$ and if the marking monomial (a product of a_m's) is $a(w)$ we can write the generating function for the random walk model as

$$Q(x_1, \ldots, x_n; t) = \sum_{w \in \mathcal{M}} a(w) x_1^{m_1} \ldots x_n^{m_n} t^{|w|} \tag{1.3}$$

Other important objects are the step polynomial (or characteristic polynomial) $S(x_1, \ldots, x_n)$ defined by

$$S(x_1, \ldots, x_n) = \sum_{s \in \mathcal{S}} x_1^{s_1} \ldots x_n^{s_n} \tag{1.4}$$

and the starting polynomial

$$q(x_1, \ldots, x_n; t) = \sum_{w \in \mathcal{Q}} a(w) x_1^{m_1} \ldots x_n^{m_n} t^{|w|} \tag{1.5}$$

Usually we see S a Laurent polynomial of degree (or valuation) 1 in each variable and we have $q = 1$. Note that in all of these quantities the variables a_m have been supressed from the notation. We can include those too whenever necessary especially if there are only finitely many of them. Furthermore it'll be useful to use the quantities $b = a - 1$ and $c = b/a = 1 - \overline{a}$.

Example 1. To illustrate the notation we'll investigate a random walk model on the plane. We'll look at random walks starting at the origin, restricted to the quarter plane, and which have only small diagonal steps. We'll count their interactions with the axes. In our terminology we have a random walk model $\mathcal{M}$ with $\mathcal{S}$ given by

$$\mathcal{S} = \{(-1, -1), (1, -1), (-1, 1), (1, 1)\}. \tag{1.6}$$

The region $\mathcal{D}$ consists of all points in $\mathbb{Z}^2$ with non-negative coordinates. Next $\mathcal{D}_0$ is composed of

the two non-negative axes with overlap at the origin - we'll count visits to the origin as visits to each axis. Finally $\mathcal{Q}$ consists of the empty walk. In the language of generating functions we have $q = 1$ and

$$S = S(x_1, x_2) = x_1 + \overline{x}_1 + x_2 + \overline{x}_2 = (x_1 + \overline{x}_1)(x_2 + \overline{x}_2) \tag{1.7}$$

We have a choice in naming the variables counting the interactions so we'll go with their defining quality (as this proves notationally simpler later on) - we mark the locations where one coordinate is zero. That is mark walks ending on $x_k = 0$ by incrementing the power of a_k in the generating function. In this model the only first step we can take is $(1, 1)$ which contributes a monomial $x_1 x_2 t$. From here on we can build on the single random walk of size one by taking a step in any of the four directions. This action lead to the following monomials

- $(1, 1)$: monomial $x_1^2 x_2^2 t^2$ since there are no interactions.
- $(1, -1)$: monomial $a_2 x_1^2 t^2$ since the walk lands on the x_1-axis with coordinate $x_2 = 0$.
- $(-1, 1)$: monomial $a_1 x_2^2 t^2$ since the walk lands on the x_2-axis with coordinate $x_1 = 0$.
- $(-1, -1)$: monomial $a_1 a_2 t^2$ since the walk lands on the origin which is on both axes.

Since we've seen all the random walks of size at most 2 we see that the generating function $Q = Q(x_1, x_2; t)$ starts as

$$Q = 1 + x_1 x_2 t + (a_1 a_2 + a_1 x_2 + a_2 x_1 + x_1 x_2) t^2 + \cdots \tag{1.8}$$

$$= 1 + x_1 x_2 t + (a_1 + x_1)(a_2 + x_2) t^2 + \cdots \tag{1.9}$$

The fact that the first few coefficients factor in this case may be unexpected but it is due to the factoring of $S(x_1, x_2)$ itself. We'll see a more general result later.

1.3 Thesis Notation

In the interest of generalizing the problem-solving strategies involved in random walk enumeration we will need powerful notation to handle all the intricacies of the problem. In fact it is the notation itself that paved the way towards generalization because it is suggestive and points towards the deeper connections between the generating functions and quantities involved. With this thesis we aim to make this kind of notation more common in higher-dimensional random walk analysis. Typically in multivariate setting we see multi-index notation where a bolded, vectorial or even supressed

notation denotes the collection of variables involved. We would use this notation as well but it is insufficient when working with restricted geometries for $\mathcal{D}$ and $\mathcal{D}_0$.

Definition 2. *Let $\mathbb{O} := (0, \ldots, 0)$ and $\pm\mathbb{I} := (\pm 1, \ldots, \pm 1)$ where we have as many terms in these sequences as needed in the specific context and let $[n] = \{1, \ldots, n\}$*

Definition 3. *For sequences $I = (i_1, \ldots, i_m)$ and $J = (j_1, \ldots, j_m)$ and a symbol x we denote the monomial*

$$x_I^J = x_{i_1}^{j_1} \cdots x_{i_m}^{j_m}. \tag{1.10}$$

with the special cases

$$x_I^{\mathbb{I}} = x_I, \quad x_I^{-\mathbb{I}} = \frac{1}{x_I} = \overline{x}_I \tag{1.11}$$

Definition 4. *For sequences I, J as before and indeterminate x we denote the coefficient extraction*

$$[x_I^J]F = [x_{i_1}^{j_1} \cdots x_{i_m}^{j_m}]F = F_I^J \tag{1.12}$$

with special cases

- *$F_I^J = F$ if I is empty.*
- *$F_I^J = F_J$ if I is the full set of variables ordered unambiguously.*

We'll primarily be using this notation for extractions w.r.t. the x variables in the generating functions but sometimes we'll be using such a coefficient extraction w.r.t. to other variables, most commonly t. We'll never mix such extractions with this notation.

Example 5. Here we'll illustrate the notation. If $Q(x_1, x_2; t)$ is given as in the previous example we have

$$Q_1^0 = [x_1^0]Q(x_1, x_2) = Q(0, x_2) = 1 + a_1(x_2 + a_2)t^2 + \cdots \tag{1.13}$$

$$Q_{1,2}^{0,0} = Q_{0,0} = Q(0, 0) = 1 + a_1 a_2 t^2 + \cdots \tag{1.14}$$

Furthermore we may write

$$S_1^{-1} = [\overline{x}_1](x_1 + \overline{x}_1)(x_2 + \overline{x}_2) = x_2 + \overline{x}_2 \tag{1.15}$$

and $S_{1,2}^{\pm 1, \pm 1} = 1$

Chapter 2

Introduction

Why do we study random walks? Apart from being an interesting mathematical object that leads to a lot of interesting mathematics, they have some real world applications. A lot of processes can be modeled via random walks - from stock market objects to proteins and polymers. There is a rich theory of continuous and discrete random walks. Specifically in random walk enumeration we attempt to find out things about the numbers of walks of some length and of certain type with certain geometrical constraints. The topic often leads to the use of generating functions as a powerful tool in the analysis. In the end we gain insight into what makes certain random walk models "easy" and others more difficult. In this thesis we look at the enumeration for constrained random walks with interactions in order to gain insight into the deeper workings of random walk enumeration problems.

2.1 Recent History

With the terminology from autorefch:Preliminaries in place we are ready to discuss some of the recent history of random walk enumeration. In the most basic case random walks starting at the origin on d dimensions have no restrictions $\mathcal{D}$ is the whole space and there are no interactions. We can enumerate these walks easily by writing the full generating function

$$Q(x_1, \ldots, x_d; t) = \frac{1}{1 - tS(x_1, \ldots, x_d)} \tag{2.1}$$

$$= 1 + tS(x_1, \ldots, x_d) + t^2 S(x_1, \ldots, x_d)^2 + \cdots \tag{2.2}$$

This works since we can use each individual steps and every walk is uniquely represented by a sequence of steps from $\mathcal{S}$. This works for arbitrary initial conditions too since we can start the walks at any point and evolve them. The coefficients of each power of t are Laurent polynomials

in the variables $x_1, \ldots, x_d$. If we have the generating function for a random walk model we have a lot of information at our hands. Replacing x_k with 0 for each k gives us the generating function for walks returning to the origing. Similarly taking higher derivatives and evaluating at zero allows us to find the number of walks ending at specific points. If we set all the $x_k = 1$ inf Q we obtain the length generating function for the random walks in the plane. There are numerous questions we can ask about each of those characteristics of the model.

The nest step up is to deal with half-spaces - here $\mathcal{D}$ is the region where one of the $x_k \geqslant 0$. These cases are not as simple to solve but they are doable. Check [3], [4], [5] and [6] for work on the one-dimensional case that can be extended to general d. The generating functions we can obtain are algebraic. Naturally we are led to consider cases where $\mathcal{D}$ is an intersection of half-spaces. Even for $d = 2$ the problem has generated a rich theory of quadrant (or quarter-plane) walks. We obtain functional equations for the function Q that are not easy to solve in general - often the generating functions here are not even D-finite although there are D-finite examples. The main object of study in these cases has been a group G of bi-rational transformations preserving the polynomial $K = 1 - tS$ called the kernel of the model.

When G is finite (in the 23 non-equivalent cases) we obtain a D-finite generating function (full?) as can be seen the classifications [7],[8] and [9], [10] for specific cases. The main tool in the study of the finite group cases has been the algebraic kernel method in which we sum different functional equations produced by group actions by elements of G in order to cancel the majority of the terms in the functional equation. See [11] for treatment of models with repeated steps with finite groups.

There are 56 non-equivalent cases of quarter-plane walks with an infinite group. For 51 of them both the full generating function [12] and the length generating function for walks returning to the origin are not D-finite [13]. The remaining 5 cases have been dealt with in [14] and [15]. The main tool used to tackle these 5 cases is the iterated kernel method in which we use the roots of the kernel K to show the generating function is not D-finite.

Naturally we are led to invesitigating higher-dimensional variants of this problem. See [16] and [17] for when $\mathcal{D}$ is a Weyl chamber. In three dimensions we have the recent partial classification [18] of random walk models with up to six steps where the authors extend the algebraic kernel method to handle the majority of finite group cases encountered.

Another generalization is to consider walks with interactions and/or self-avoiding walks (or SAWs). These two types of generalizations have important applications in physics because they can be used to model polymers, proteins and other chains. For some research on SAWs with various interactions see [19],[20] [21], [22], [23] and [24]. The topic of SAWs doesn't lend itself as nicely to using generating functions like the ones considered so far so as a simpler first step we can consider regular random walks with interactions. See [25] and [26] for walks with interactions in an octact.

See [27], [28], and [29] for walks with interactions in the plane.

A long-standing dream goal in random walk enumeration research is the automatization of the process of determining characteristics of the models and similar problems. There has been some progress in this direction. See for example [30] and [31] for details on symmetric functions, [32] for a holonomic approach to combinatorics, [33] for automatic classification of lattice walks, and finally [34] for some of the computational challenges involved in automation.

A related subject of study is the asymptotics of random walks where some automation can be performed. See [35], [36] and [37] for examples of the theory. A lot of the works mentioned so far have also discussed asymptotics.

2.2 Thesis Objectives

This thesis grew out of work on regular quadrant random walks with interactions with the axes. The derivation of the functional equation in the non-interacting case is a simple inclusion-exclusion process but doing the same for the interacting case is somewhat more involved. It is possible to further simplify the equation via coefficient extraction and back-substitution. It works out beautifully to give us a much more compact functional equation. However, the process is quite painstaking even for three dimensions, let alone for higher-dimensions, although it is possible.

The mathematicians who study random walk enumeration have as a goal to classify random walks according to various characteristics like step set, dimension, group size associated to the kernel, and perhaps most importantly - D-finitenes. They also dream of being able to make the process automatic whenever it is possible. In this thesis we aim to further the study of random walk models towards this goal of automatization. We might as well be as optimistic as possible and suppose that a general algorithm exists that takes an arbitrary random walk model - in arbitrarily many dimensions, with arbitrary finite step sets and with arbitrary constraints, and outputs various data about the model - functional equation for the generating function, a solution and information about the model's D-finiteness. It is only logical that if such an algorithm were to exist it would not have difficulty with the particulars of the random walk model. While this kind of algorithm may be too strong to envision (it may be the case that there are atrocious random walk models which we cannot solve), the idea of the existence of such an algorithm has been a strong motivator in our pursuit of generalization. This thesis succeeds in demonstrating that the first task is achievable. For a vast assortment of random walk models we can almost-automatically write down a functional equation for its generating function counting length, position and interactions of the random walks.

The equation looks like

$$(1 - t\Gamma S)Q = q \tag{2.3}$$

where Γ is an operator on the variables of Q, S is the Laurent step polynomial (keeping inventory of the steps in polynomial form) and q is some Laurent polynomial initial condition. The operator idea is powerful and we'll use it as much as we can in this thesis. A secondary achievement is the furthering of the theory of the orbit summation method which along with the obstinate kernel method combine in a nicer theory of operators consisting of group actions. However time and space have not been sufficient to go too far in these lines. Near the end we present some solutions and partial solutions to random walk problems in two dimensions with interactions.

2.3 Layout of Thesis

In the first few chapters we go through motivating examples which will serve as a springboard into further generalizations. First we'll start off with one-dimensional random walks with and without interactions in Chapter 3. Then we'll quickly move on to two-dimensional random walks in Chapter 4. On the way we'll develop some of the basic operator theory involved in the random walk models. In Chapter 4 we'll make a detour to random walks on a honeycomb lattice - a problem that will show some of the group-theoretic and operator tools. In Chapter 5 we'll tackle the d-dimensional random walk model functional equations using operators. In chapter 6 we'll finally go through operators in a much generality as we can, deriving all sorts of properties for them along the way. In chapter 7 we'll discuss generalizations of the random walk models and we'll push the operator notation as far as we can to see what kind of problem we can hope to tackle with the theory.

In Chapter 8 we'll become more practical and deal with group-theoretic operators relevant to both two-dimensional and higher-dimensional random walk models. In Chapter 9 we extend the persistent kernel method as much as we can for the two-dimensional interacting walks. Then in Chapter 10 we attempt to solve as many highly-symmetrical two-dimensional interacting models as we can via the vanishing kernel method. In Chapter 11 we work on some models with higher group orders.

Chapter 3

One-dimensional random walks

3.1 Derivation of the functional equation

In the interest of working out higher dimensional walk problems it is very useful to first visit the 1-dimensional case which will illustrate a lot of the methoDS that will later come into play. The 1-dimensional case will also give us a clearer picture as to how things work in higher dimensions (although the picture becomes more complicated). We'll treat the problem using generating functions. We'll use lower case letters and not quite the full notation because it is not necessary. In one dimension, on one axis, we'll use steps of length at most 1. If we have a random walk ending at coordinate i we can either step back adding -1 to the coordinate, step forward adding $+1$ to the coordinate or staying in one place, adding 0 to the coordinate. This can be somewhat formalized by using polynomials. Ending a walk at coordinate i will be encoded with a monomial x^i. Then stepping backwards, forwards or staying in one place will be denoted with x^{i-1}, x^i and x^{i+1} respectively. This is the same as multiplying by $1/x = \overline{x}, 1, x$ in each case. We can encode this information is a Laurent step polynomial s defined as

$$S = S_{-1}\overline{x} + S_0 + S_1 x\,. \tag{3.1}$$

Here S_i will be either 0 or 1 for any specific i, tracking whether we have allowed those kinds of steps or not. Keeping this in mind, there are 8 possible step sets because we have two options for each S_i. We keep track of the length of a random walk with powers of t and we keep track of itS endpoint with powers of x. We'll also mark every interaction with the origin by multiplying by a every time a walk steps on the origin. We'll keep the walks on the non-negative axis.

Long story short we can find the generating function in a very simple, brute force manner, even

for higher dimensions. We start with the empty walk denoted by 1 since it starts and ends on the origin without taking a step, so no t, x or mark a. Then try out each step from s and eliminate those that take us past the origin in the negative direction. If we step on the origin we multiply by a. Those walks that remain are then multiplied by t since they are not of length 1. We can proceed iteratively k tiems to obtain random walks on the non-negative axis of length k. Adding up the resultS allows us to write the generating function up to $O(t^k)$. It will prove very useful to think about operators so we can formalize this notion in terms of operators. For example, here we are taking walks of length $k-1$, trying all options of s (this is like multiplying by s), removing all the walks that end up on the negative axis (this is like removing the $\overline{x}$ term) and marking those that end on the origin (this is like multiplying the x^0 coefficient by a). Let D stand for the operator that cleans up the walks and marks the valid ones. Then if Q_k stands for the marked walks of length k, then the k-th term of Q is $Q_k t^k$ and we have

$$Q_{k+1}t^{k+1} = tD(SQ_k t^k)\,. \tag{3.2}$$

Thus we can write the generating function Q iteratively as

$$Q = \sum_{k=0}^{\infty} t^k (DS)^k (1) \tag{3.3}$$

where we're iteratively applying DS to 1 to obtain all marked walks. We can treat this operator DS as a formal variable and write

$$Q = \sum_{k=0}^{\infty} (tDS)^k (1) = \frac{1}{1-tDS}(1)\,, \tag{3.4}$$

and then this gives us a functional equation for Q

$$(1-tDS)Q = 1\,. \tag{3.5}$$

There is another way of deriving this equation while not treating DS as a formal variable. Instead of applying the operator iteratively we can apply it on the generating function as a whole. The product tSQ is the generating for all random walks that end with x-coordinate greater or equal to -1 where some steps end up on x-coordinate 0 without being marked. We can apply D on this product to remove the terms ending at x-coordinate -1 and mark those that end up on x-coordinate 0. This leaves us with all the valid walks encoded in the generating function Q without the empty walk 1.

Therefore we can write a functional equation for Q as

$$Q = 1 + D(tSQ) = 1 + tDS(Q) \tag{3.6}$$

and we ca re-write this as $(1 - tDS)Q = 1$ as before. Now, we have to say a few words about what the operator D actually is. First we are removing all the walks that end on x-coordinate equal to -1 which can be handled by the operator $E = 1 - \overline{x}[\overline{x}]$ which removes those walks. Then the markling can be handled by the operator $L = 1 + (a-1)[x^0] = 1 + b[x^0]$ where $b = a - 1$. Therefore we can write $D = LE$

$$\begin{aligned} D = LE &= (1 - \overline{x}[\overline{x}])(1 + b[x^0]) && (3.7)\\ &= 1 + b[x^0] - \overline{x}[\overline{x}] - \overline{x}[\overline{x}]b[x^0] && (3.8)\\ &= 1 + b[x^0] - \overline{x}[\overline{x}] && (3.9) \end{aligned}$$

since the last term vanishes on all generating functions. We could have also applied the operators in reverse to get the same result.

Example 6. Suppose that $S = x + \overline{x}$. Then starting with $Q_0 = 1$,

$$\begin{aligned} Q_1 = LEs(1) &= (1 + b[x^0] - \overline{x}[\overline{x}])(x + \overline{x}) = x + \overline{x} - \overline{x} = x && (3.10)\\ Q_2 = LEs(Q_1) &= (1 + b[x^0] - \overline{x}[\overline{x}])(x^2 + 1) = x^2 + 1 + b = a + x^2 && (3.11)\\ Q_3 = LEs(Q_2) &= (1 + b[x^0] - \overline{x}[\overline{x}])(a\overline{x} + ax + x + x^3) && (3.12)\\ &= a\overline{x} + ax + x + x^3 - a\overline{x} = (1 + a)x + x^3\,. && (3.13) \end{aligned}$$

Therefore

$$\begin{aligned} Q &= Q_0 + Q_1 t + Q_2 t^2 + Q_3 t^3 + \cdots && (3.14)\\ &= 1 + xt + (a + x^2)t^2 + ((1 + a)x + x^3)t^3 + \cdots && (3.15) \end{aligned}$$

This commutativity makes sense because on the level of the function they are operator interacting with different monomials. We can now expand the functional equation to the following

$$\begin{aligned} 1 = Q - tLE(SQ) &= Q - t(SQ + b(SQ)_0 - \overline{x}(SQ)_{-1}) && (3.16)\\ &= q - t(SQ + b(S_{-1}Q_1 + S_0Q_0 + S_1Q_{-1}) - \overline{x}(S_{-1}Q_0)) && (3.17)\\ &= Q - t(SQ + b(S_{-1}Q_1 + S_0Q_0) - \overline{x}(S_{-1}Q_0)) && (3.18) \end{aligned}$$

since $Q_{-1} = 0$. We can rewrite this equation in the following form

$$(1 - tS)Q - t(bS_0 - \overline{x}S_{-1})Q_0 - btS_{-1}Q_1 = 1\,. \tag{3.19}$$

Here $Q_0 = Q(0)$ and $Q_1 = Q'(0)$ where $Q = Q(x)$. In the interest of simplifying the equation we can eliminate the Q_1 term by extracting the x^0 coefficient from the whole equation

$$1 = Q_0 - t(S_{-1}Q_1 + S_0Q_0 + S_1Q_{-1}) - tbS_0Q_0 - btS_{-1}Q_1 \tag{3.20}$$

$$= (1 - atS_0)Q_0 - atS_{-1}Q_1\,. \tag{3.21}$$

We can solve this equation for Q_1

$$tS_{-1}Q_1 = \overline{a}((1 - atS_0)Q_0 - 1) \tag{3.22}$$

and substitute the result into the functional equation (for now we don't worry about the possibility of $S_{-1} = 0$ because it leads to the same thing anyway). Then

$$1 = (1 - tS)Q - t(bS_0 - \overline{x}S_{-1})Q_0 - c((1 - atS_0)Q_0 - 1) \tag{3.23}$$

$$= (1 - tS)Q - (c - t\overline{x}S_{-1})Q_0 + c \tag{3.24}$$

and therefore we can rewrite the equation as

$$(1 - tS)Q - (c - t\overline{x}S_{-1})Q_0 = \overline{a} \tag{3.25}$$

since $1 - c = 1 - (1 - \overline{a}) = \overline{a}$. This process is rather complicated, even though it can be done for higher dimensional problems as well (as you can see in various previous approaches). Also there is an issue that we need to consider every time we are solving an equation in order to substitute quantities back into the equation. This is where the operator notation comes to the rescue. It turns out the the L operator is invertible with inverse given by

$$L^{-1} = M = 1 - c[x^0] \tag{3.26}$$

and we can apply it on the functional equation. The RHS is simply $M(1) = 1 - c = \overline{a}$ and

automatically the LHS is

$$M(1-tLES)Q = Mq - tESQ \tag{3.27}$$
$$= (q - cQ_0) - t(SQ - \overline{x}S_{-1}Q_0) \tag{3.28}$$
$$= (1-tS)Q - (c - t\overline{x}S_{-1})Q_0 \tag{3.29}$$

and we've derived the same equation but in a much neater way.

Proposition 7. *The one-dimensional random walk model with laurent step polynomial* $S = S_{-1}\overline{x} + S_0 + S_1 x$ *has a generating function* Q *satisfying the functional equation*

$$(M - tES)Q = (1-tS)Q - (c - t\overline{x}S_{-1})Q_0 = \overline{a}\,. \tag{3.30}$$

This will prove crucial later on when we derive a functional equation for higher dimensional problems. Although it can be done directly with a lot of effort it's much better to just use the operator notation. The derivation is brought down from about a dozen pages to one or two. The operator notation will also allow us to solve some multidimensional problems directly or with the help of lower dimensional problems. We can extract the x^0 coefficient from the final functional equation again and although we can do this directly, we'll use the operator notation to tidy up the process. This will show up later when dealing with higher dimensional problems

$$[x^0]M = [x^0] - [x^0]c[x^0] = [x^0] - c[x^0] = \overline{a}[x^0] \tag{3.31}$$
$$[x^0]E = [x^0] - [x^0]\overline{x}[\overline{x}] = [x^0]\,. \tag{3.32}$$

Then we extract the x^0 from the functional equation

$$[x^0]\overline{a} = \overline{a} = [x^0](M - tES)Q \tag{3.33}$$
$$= ([x^0]M - t[x^0]ES)Q \tag{3.34}$$
$$= (\overline{a}[x^0] - t[x^0]S)Q \tag{3.35}$$
$$= [x^0](\overline{a} - tS)Q\,. \tag{3.36}$$

Therefore the extraction leads to

$$[x^0](1 - atS)Q = 1\,. \tag{3.37}$$

We could extract higher coefficients from the functional equation too. Let $k \geqslant 1$. Then

$$[x^k]M = [x^k] - [x^k]c[x^0] = [x^k] \tag{3.38}$$

$$[x^k]E = [x^k] - [x^k]\overline{x}[\overline{x}] = [x^k]\,. \tag{3.39}$$

Therefore

$$[x^k]\overline{a} = 0 = [x^k](M - tES)Q \tag{3.40}$$

$$= ([x^k] - t[x^k]S)Q \tag{3.41}$$

$$= [x^k](1 - tS)Q\,. \tag{3.42}$$

That is

$$[x^k](1 - tS)Q = 0 \tag{3.43}$$

and we can readily extract whichever coefficients we want. The first equation leads to

$$(1 - atS_0)Q_0 - atS_{-1}Q_1 = 1 \tag{3.44}$$

and the second leads to

$$Q_k - t(S_{-1}Q_{k+1} + S_0Q_k + S_1Q_{k-1}) = 0 \tag{3.45}$$

$$(1 - S_0t)Q_k - t(S_{-1}Q_{k+1} + S_1Q_{k-1} = 0\,. \tag{3.46}$$

3.2 Solving the functional equation for various S

In this section we'll solve all 8 one-dimensional problems fully. Some cases are easier to deal with than others. As we'll see later in the higher dimensional cases, some problems are easierto solve than others. This difficulty tends to scale up with the inclusion of certain steps with negative direction.

3.2.1 The $S_{-1} = 0$ case.

This condition allows us to solve explicitly for Q_0

$$Q_0 = \frac{1}{1 - aS_0t}\,. \tag{3.47}$$

We can substitute this expression in the functional equation and obtain

Proposition 8. *When $S_{-1} = 0$ we have*

$$Q(x, a; t) = \frac{\overline{a}}{1 - st}\left(1 + \frac{b}{1 - aS_0t}\right) = \frac{1 - S_0t}{(1 - st)(1 - aS_0t)}\,. \tag{3.48}$$

This takes care of half of the cases! What happens when we set $a = 1$, corresponding to setting $b = 0$ or extracting the b^0 coefficient from the equation? This corresponDS to unmarkling the random walks. We obtain

$$[b^0]Q = \widehat{Q} = \frac{1 - S_0t}{(1 - st)(1 - S_0t)}\,. \tag{3.49}$$

We'll note that

$$(1 - aS_0t)Q - (1 - S_0t)\widehat{Q} = 0\,. \tag{3.50}$$

This can also be written in operator form with $B = 1 - [b^0]$

Proposition 9. *When $S_{-1} = 0$ we have*

$$B((1 - aS_0t)Q) = 0\,. \tag{3.51}$$

This relates the marked generating function to the unmarked generating function! it says we can obtain the marked generating function from the unmarked one by the following relation

$$Q = \frac{1 - aS_0t}{1 - S_0t}\widehat{Q}\,. \tag{3.52}$$

We can also establish such as relationship for Q_0 although the derivation here will be trivial. Extracting $[x^0]$ from the equation leads to

$$B((1 - aS_0t)Q_0) = 0 \tag{3.53}$$

and

$$Q_0 = \frac{1 - aS_0t}{1 - S_0t}\widehat{Q_0} \tag{3.54}$$

The relationship between Q and $\widehat{Q}$ is no accident but it tends to happen fairly often in the higher dimensional cases that we've considered. The functional equation in the $S_{-1} = 0$ case is easier

to solve most likely because when $S_{-1} = 0$, walks don't go beyond the origin in the negative direction. Consequently the E operator actS as the identity on the generating function. and the functional equation reaDS as

$$(M - tS)Q = Q - cQ_0 - tSQ = (1 - tS)Q - cQ_0 = \overline{a}\,. \tag{3.55}$$

3.2.2 The $S_{-1} \neq 0$ case.

The functional equation in the $S_{-1} \neq 0$ is also not difficult to solve. We'll try to apply the kernel method whenever possible. It requires finding an x value that makes the kernel $1 - tS$ equal to zero in the functional equation

$$(1 - tS)Q - (c - t\overline{x})Q_0 = \overline{a}\,. \tag{3.56}$$

Such a solution will exist since $S_{-1} = 1$ and therefore the equation $1 - tS = 0$ is non-trivial. Let one such solution be $x = \sigma$ and suppose that we can plug it into the functional equation. Then the kernel vanishes, taking Q away with it and we obtain

$$-(c - t\overline{\sigma})Q_0 = \overline{a}\,. \tag{3.57}$$

We can solve for Q_0 explicitly again

$$Q_0 = \frac{\overline{a}}{\overline{\sigma}t - c} \tag{3.58}$$

We can now solve for Q

$$Q = \frac{\overline{a}}{1 - tS}\left(1 - \frac{t\overline{x} - c}{t\overline{\sigma} - c}\right) = \frac{t}{1 - tS}\left(\frac{\overline{x} - \overline{\sigma}}{b - a\overline{\sigma}t}\right)\,. \tag{3.59}$$

We note that we can apply the B operator again

$$B((a\overline{\sigma}t - b)Q) = 0\,. \tag{3.60}$$

Proposition 10. *When $S_{-1} \neq 0$ we have*

$$Q = \frac{t}{1 - tS}\left(\frac{\overline{x} - \overline{\sigma}}{b - a\overline{\sigma}t}\right) \tag{3.61}$$

$$0 = B((a\overline{\sigma}t - b)Q) \tag{3.62}$$

Written out explicitly

$$(a\overline{\sigma}t - b)Q - (\overline{\sigma}t)\widehat{Q} = 0\,. \tag{3.63}$$

This also allows us to write the marked generating function in terms of the unmarked generating function as

$$Q = \frac{\overline{\sigma}t}{a\overline{\sigma}t - b}\widehat{Q} = \frac{t}{at - b\sigma}\widehat{q}\,. \tag{3.64}$$

We can also work with Q_0 at this stage. We find

$$[x_0]B((a\overline{\sigma}t - b)Q) = B((a\overline{\sigma}t - b)Q_0) = 0 \tag{3.65}$$

too. This ends up being the same as

$$(a\overline{\sigma}t - b)Q_0 - (\overline{\sigma}t)\widehat{Q_0} = 0 \tag{3.66}$$

with solution

$$Q_0 = \frac{t}{at - b\sigma}\widehat{Q_0}\,. \tag{3.67}$$

The final task for this section will be to find what σ will be for various cases. We already know $S_{-1} \neq 0$ so the result depend on S_0 and S_1. In general the kernel is zero when

$$ts = 1 \tag{3.68}$$

$$t(S_{-1} + S_0 x + S_1 x^2) = x \tag{3.69}$$

$$tS_1 x^2 - (1 - S_0 t)x + tS_{-1} = 0\,. \tag{3.70}$$

When $S_1 \neq 0$ this is quadratic with two roots that are functions of t. We have to pick the one which has no singularity at $t = 0$. When $S_1 = 0$ we have a linear equation which may have a single root. When $S_1 = 0$ we have the equation $(1 - S_0 t)x = t$ so the single root is

$$\sigma = \frac{t}{1 - S_0 t}\,. \tag{3.71}$$

When $S_1 = 1$ we have the equation $tx^2 - (1 - S_0 t)x + t = 0$ with roots

$$\sigma = \frac{1 - S_0 t - \sqrt{(1 - S_0 t)^2 - 4t^2}}{2t}, \quad \sigma^* = \frac{1 - S_0 t + \sqrt{(1 - S_0 t)^2 - 4t^2}}{2t}\,. \tag{3.72}$$

The two roots multiply to 1 so $\overline{\sigma} = \sigma^*$. Note that in the relationship between the marked and unmarked Q_0 we have

$$Q_0 = \frac{t}{at - b\sigma}\widehat{Q_0} \tag{3.73}$$

where σ is a function containing a square root, i.e. not a polynomial. However we can go at the problem from a different angle to obtain a polynomial $f(a, t)$ for which $B(fQ_0) = c$ which is not quite zero but it's a simpler relationship. This will also illustrate the group approach together with the kernel method. The two cases that we need to work on both have $S_{-1} = S_1 = 1$ with S_0 being 0 or 1. This means that s is left invariant under sending $* : x \mapsto \overline{x}$. We can apply this on the functional equation and now have a system of two functional equations

$$(1 - tS)Q - (c - t\overline{x})Q_0 - \overline{a} = 0\,, \tag{3.74}$$

$$(1 - tS)Q^* - (c - tx)Q_0 - \overline{a} = 0\,. \tag{3.75}$$

We can multiply the first by $c - tx$, the second by $c - t\overline{x}$ and subtract the equations to eliminate the Q_0 term. We can divide everything out by $1 - tS$ and finally obtain

$$(c - tx)Q(x) - (c - t\overline{x})Q(\overline{x}) - \frac{(c - tx - c + t\overline{x})}{1 - tS}\overline{a} = 0\,, \tag{3.76}$$

$$(c - tx)Q(x) - (c - t\overline{x})Q(\overline{x}) - \overline{a}\delta = 0\,. \tag{3.77}$$

where

$$\delta = \frac{t(\overline{x} - x)}{1 - tS} \tag{3.78}$$

and crucially it doesn't depent on a. We extract the x^1 coefficient from the equation to obtain an equation relating Q_1 and Q_0

$$cQ_1 - tQ_0 - \overline{a}\delta_1 = 0\,. \tag{3.79}$$

We can set $a = 1$ here so we get

$$-\delta_1 = t\widehat{Q_0} \tag{3.80}$$

and we can bring this back to the original equation to get

$$cQ_1 = tQ_0 - \overline{a}t\widehat{Q_0} \tag{3.81}$$

$$bQ_1 = t(aQ_0 - \widehat{Q_0})\,. \tag{3.82}$$

Earlier we had the equation

$$(1 - aS_0t)Q_0 = 1 + aS_{-1}tQ_1 = 1 + atQ_1 \tag{3.83}$$

which we can solve for Q_1 this time and insert the solution into the above equation

$$b((1 - aS_0t)Q_0 - 1) = t^2a(aQ_0 - \widehat{Q_0})\,. \tag{3.84}$$

Collecting coefficients and simplifying leads us to

$$(b(1 - aS_0t) - a^2t^2)Q_0 - (-at^2)\widehat{Q_0} = b \tag{3.85}$$

$$(c(1 - aS_0t) - at^2)Q_0 - (-t^2)\widehat{Q_0} = c \tag{3.86}$$

which gives $f(a,t) = c(1 - aS_0t) - at^2$ so that

Proposition 11. *When $S_{-1} = S_1 = 1$ we have*

$$B((c(1 - aS_0t) - at^2)Q_0) = c. \tag{3.87}$$

With two different approaches we obtained slightly different relationships upon applying the operator B.

3.2.3 1-dimensional case conclusion

In this section we were able to find explicit solutions for Q_0, Q and also B-operator relationships for Q and $\widehat{Q}$. In higher dimensions this may not be quite possible but we should still be able to find B-operator relationships for the equivalent of Q_0 and $\widehat{Q_0}$. Here will be a table showing the resultS of this section. Here is a table showing detailed resuls for each of the 8 possible cases of random walks restricted to the non-negative axis.

Case	S	Q_0	Q	$f(a,t)$
$S_{-1}=0$	S_0+S_1x	$\dfrac{1}{1-aS_0t}$	$\dfrac{1-S_0t}{(1-tS)(1-atS_0)}$	$1-atS_0$
$S_{-1}=1$	$1+S_0+S_1x$	$\dfrac{\sigma}{at-b\sigma}$	$\dfrac{t(x-\sigma)}{x(1-tS)(at-b\sigma)}$	$a\overline{\sigma}t-b$

Table 3.1: One-dimensional Generating functions

Chapter 4

Two-dimensional Random Walks

4.1 Introduction

The power of the operator notation really starts to show here as it allows us to easily derive functional equations for the two-dimensional problem.

Definition 12. *Let $Q = Q(x_1, x_2, a_1, a_2; t)$ be the generating function enumerating two-dimensional quarter-plane random walks with interactions with the axes. Here the step Laurent polynomial is $S = S(x_1, x_2)$ and will be made explicit if the context requires it. The x's indicate the ending coordinates of each walk while the a's count its interactions with the axes.*

Here the operators E and L are defined as $E = E_1E_2$ where $E_k = 1 - \overline{x_k}[\overline{x_k}]$ and $L = L_1L_2$ where $L_k = 1 + (a_k - 1)[x_k^0]$. As before we start with the empty walk with monomial 1. Then to find Q we take SQ to obtain all possible extensions of walks enumerated by Q. Then we need to remove all walks stepping outside of the quarter plane. Such walks have either x_1 or x_2 to a negative power (or both) so if we take ESQ we remove all problematic walks. Now we need to mark those walks stepping on the axes so we calculate $LESQ$. Since all the walks enumerated here has one more step we need to keep track of that by multiplying with t. The only missing walk is the empty walk. This yields $Q = 1 + tLESQ$ or in operator form

$$(1 - tLES)Q = 1\,. \tag{4.1}$$

But we know that the operator L is invertible with inverse $L^{-1} = L_1^{-1}L_2^{-1}$ where $L_k^{-1} = 1 - c_k[x_k^0]$. We can multiply by this inverse to find

$$(L^{-1} - tES)Q = L^{-1}1 = \overline{a}_1\overline{a}_2 = \alpha\,. \tag{4.2}$$

Proposition 13. *The function Q satisfies the following functional equations*

$$(1 - tLES)Q = 1 \tag{4.3}$$

$$(L^{-1} - tES)Q = \alpha\,. \tag{4.4}$$

We shall expand the equation to its explicit form. We have

$$L^{-1}Q = (1 - c_1[x_1^0])(1 - c_2[x_2^0])Q = Q - c_1Q_1^0 - c_2Q_2^0 + c_1c_2Q_{1,2}^{0,0} \tag{4.5}$$

and

$$ESQ = (1 - \overline{x}_1[\overline{x}_1])(1 - \overline{x}_2[\overline{x}_2])SQ \tag{4.6}$$

$$= SQ - \overline{x}_1S_1^{-1}Q_1^0 - \overline{x}_2S_2^{-1}Q_2^0 + \overline{x}_1\overline{x}_2S_{1,2}^{-1,-1}Q_{1,2}^{0,0} \tag{4.7}$$

where in the calculation we've dropped terms like Q_i^j for $j < 0$ since the model is restricted ot the quarter plane. Combining all the terms computed so far leads to the master functional equation

$$(1 - tS)Q - (c_1 - t\overline{x_1}S_1^{-1})Q_1^0 - (c_2 - t\overline{x_2}S_2^{-1})Q_2^0 + (c_1c_2 - t\overline{x_1x_2}S_{1,2}^{-1,-1})Q_{1,2}^{0,0} = \overline{a_1a_2}\,.$$

Definition 14. *For $I \subseteq [d]$ let*

$$K_I = c_I - t\overline{x}_IS_I^{-1} \tag{4.8}$$

where if $I = \varnothing$, $K_I = K$.

This allows us to re-write the equation neatly.

Corollary 15. *The generating function function Q satisfies the master functional equation*

$$KQ - K_1Q_1^0 - K_2Q_2^0 + K_{1,2}Q_{1,2}^{0,0} = \overline{a}_1\overline{a}_2\,. \tag{4.9}$$

We can also perform coefficient extractions like in the one-dimensional case and obtain further functional equations. The derivation is quite mechanical and very similar to the one-dimensional case.

Corollary 16. *The generating function Q satisfies the additional functional equations*

$$(1 - ta_1 S_1^0)Q_1^0 - (c_2 - ta_1\overline{x_2}S_{1,2}^{0,-1})Q_{1,2}^{0,0} - a_1 t(S_1^{-1}Q_1^1 - \overline{x_2}S_{1,2}^{-1,-1}Q_{1,2}^{1,0}) = \overline{a_2}\,,$$
$$(1 - ta_2 S_2^0)Q_2^0 - (c_1 - ta_2\overline{x_1}S_{1,2}^{-1,0})Q_{1,2}^{0,0} - a_2 t(S_2^{-1}Q_2^1 - \overline{x_1}S_{1,2}^{-1,-1}Q_{1,2}^{0,1}) = \overline{a_1}\,,$$
$$(1 - a_1 a_2 t S_{1,2}^{0,0})Q_{1,2}^{0,0} - ta_1 a_2 t(S_{1,2}^{-1,-1}Q_{1,2}^{1,1} + S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{0,-1}Q_{1,2}^{0,1}) = 1$$

where $S_{1,2}^{0,0} = 1$ if we have resting steps and zero otherwise.

We should contrast this quick analysis to the one done without the operator notation. We would have to use inclusion-exclusion to determine which terms are being kept and which terms are being discarded, and then we would have to take care of double-counting. The procedure can be somewhat complicated and prone to error. Furthermore, the complexity of the equation increases exponentially with the number of extractions necessary. The operator notation allows us to derive the functional equatiosn in an elegant way that is rather easy to generalize. It also allows us to encapsulate away the unnecessary details in the marked cases, so that we can think of inverting operators, rather than do coefficient extraction and back substitution (which would be the original way). Later we'll cover the general case in more detail.

4.2 Investigating relationships between G, G_1, G_2 and $G_{1,2}$

In this work we'll focus on investigating the properties of the function $G(a_1, a_2) = Q_{1,2}^{0,0}(a_1, a_2)$. Let $G_1 = G(1, a_2)$, $G_2 = G(a_1, 1)$ and $G_{1,2} = G(1, 1)$. We could also see these as coefficient extractions of G with respect to b_1, b_2 and $b_1 b_2$, i.e. $G_{1,2} = [b_1^0 b_2^0]G$. In previous works it has been shown that there is a relationship between these quantities depending on the particular step set S. In some cases (like diagonal walks $S = (x_1 + \overline{x_1})(x_2 + \overline{x_2})$) we have a linear relationship between these quantities. In other cases (like cartesian walks $S = x_1 + \overline{x}_1 + x_2 + \overline{x}_2$) we don't have such a linear relationship but we have a rational one, that is G is a rational function of G_1, G_2 and $G_{1,2}$. The aim of the followin sections is to investigate when such relationships exist, when se have a relationship of that type and whether we have other cases.

The work can be split into several cases - walks that never leave the origin and ones that leave but never return to the origin. In the first case the relationship is trivially $G - G_1 - G_2 + G_{1,2} = 0$ because $Q = 1$ itself and so $G = 1$. In the second case Q may be more complicated but again $G = 1$ trivially since no walks end at the origin. This case satisfies the same relationship. However, once we move beyond this stage we need to consider the problem more carefully.

In any of the simple cases we can write $BG = 0$ and sometime $BQ = 0$ where $B = B_1 B_2$ from before. We'll investigate the when we'll have a relationship like $B(FQ) = 0$, $B(FG) = 0$ or

$B(FG)$ is constant for some F's.

4.3 Simple cases: Walks never leaving the origin and walks leaving the origin but never returning

Here are all the easy cases of step sets that lead to walks that either don't leave the origin or leave it and never come back. We're not allowing resting steps here. There are 80 such sets among the 256 non-resting possible cases. Of course we could use symmetries to eliminate cases but for now this was an easier thing to do.

There are $2^5 = 32$ types of walks that don't leave the origin. Those are walks with steps among the set {NW,W,SW,S,SE}. Next we look at the walks that leave the origin but never come back. There is the case of $S = \{NE\}$ that never comes back to any axis. Let's look at those walks that never come back to the x_1 axis. We have some steps that go in the positive x_2 direction, that is one of {N, NE, E} together with some steps from {NW, W, SW}. If we were to include any other steps we would come back to the x_1 axis. If we only have a N step, we have 8 new cases, if we have only a NE step we have 3 new cases and if we have both, we have 4 new cases. This gives a total of 15 new cases. There's an equivalent number of walks that leave the x_2 axis and never come back to it. The running total is $32 + 1 + 15 + 15 = 63$.

To finish off we have 17 cases of walks that leave the origin, eventually interact with both axes but never come back to the origin. For this we must only include steps in such a way that we cannot pick a selection of steps that will decrease both the x_1 and x_2 coordinates. We have to pick a subset of steps from {N, NE, E} so the only other steps allowed are NW and SE. The new cases coming from {N, NE, E}, {N,E}, {N, NE }, {NE, E}, {N}, {NE}, {E} are 4, 4, 2, 2, 1, 2, 2 respectively.

Before we consider walks that come back to the origin we'll look at the behavior Q depending on whether S factors in some nice way or not.

4.4 Simple case: Factoring S into a product

We take a moment to mention a result that falls out of the operator notation and which was covered in more generality by Steve et al. We have a special case when S factors as $S = S_1(x_1)S_2(x_2)$. We have

$$(DS)^k = (D_1D_2S_1S_2)^k = (D_1S_1)^k(D_2S_2)^k \tag{4.10}$$

since the D_1 and D_2 operators commute and the D_iS_i operators also commute since they deal with different variables. The functional equation for Q is

$$(1 - tDS)Q = 1\,. \tag{4.11}$$

We can compute Q directly via the iteration as

$$Q = \sum_{k=0}^{\infty} t^k (DS)^k = \sum_{k=0}^{\infty} t^k (D_1S_1)^k (D_2S_2)^k = Q_1(x_1, a_1, t) \odot_t Q_2(x_2, a_2, t) \tag{4.12}$$

where $\odot_t$ denotes the Hadamard product of two functions with respect to variable t. From now on we'll drop the subscript t unless it is needed. Here Q_i is given by

$$Q_i = \sum_{k=0}^{\infty} t^k (D_iS_i)^k \tag{4.13}$$

and by reversing the iteration we obtain the functional equation

$$(1 - tD_iS_i)Q_i = 1\,. \tag{4.14}$$

In other words, $Q_i(x_i, a_i, t)$ is the generating function for a 1-dimensional problem with Laurent step polynomial $S_i(x_i)$. In the case of a factoring of this sort, we are able to glue the 2-dimensional solution out of two 1-dimensional solutions. (Of course this generalizes easily to many dimensions and gives an infinite family of solutions to higher-dimensional problems where S factors into lower dimensional S_i's). If we want to look at the generating function for quarter plane walks of length k it is enough to look at the product of two generating functions for one-dimensional walks of length k. Also, we can write

$$Q = Q_1 \odot Q_2, \quad \widehat{Q} = \widehat{Q}_1 \odot Q_2 \tag{4.15}$$

$$\breve{Q} = Q_1 \odot \breve{Q}_2, \quad \breve{\widehat{Q}} = \widehat{Q}_1 \odot \breve{Q}_2 \tag{4.16}$$

Therefore

$$B_1B_2Q = Q - \widehat{Q} - \breve{Q} + \breve{\widehat{Q}} = (Q_1 - \widehat{Q}_1) \odot (Q_2 - \breve{Q}_2) \tag{4.17}$$

$$B_1B_2(Q_1 \odot Q_2) = (B_1Q_1) \odot (B_2Q_2)\,. \tag{4.18}$$

In general if we are applying operators that don't involve t to a Hadamard product with respect to t we can treat the operator as a constant with respect to t and apply it on whichever term we would like to. If it applies to both terms then we have to apply it to their mixture in a non-trivial way. For example, each of the Q_i's solves a 1-dimensional problem and therefore we have a relationship of the type $B_i(f_iQ_i) = g_i$ for some f_i and g_i. Then we can write

$$B_1B_2(f_1Q_1 \odot f_2Q_2) = (B_1f_1Q_1) \odot (B_2f_2Q_2) = g_1 \odot g_2 \tag{4.19}$$

but so far this is not a relationship of the type $B_1B_2(fQ) = g$ since taking out the functions f_1 and f_2 from within the Hadamard product is non-trivial in general. One might be tempted to say that $f = f_1 \odot f_2$ but that would be incorrect. It would be neat to see if there is a way to find f in terms of f_1 and f_2.

There is a way to see the Hadamard product solution without the iteration method. Suppose that Q_i is a solution to the functional equation $(M_i - tE_iS_i)Q_i = \overline{a}_i$ for $i = 1$ and $i = 2$. Then we claim that $Q = Q_1 \odot Q_2$ is a solution to the functional equation $(M - tES)Q = \overline{a}_1\overline{a}_2$. To this end we calculate

$$\begin{aligned}(M - tES)Q &= M(Q_1 \odot Q_2) - tES(Q_1 \odot Q_2) &(4.20)\\ &= (M_1Q_1 \odot M_2Q_2) - t((E_1S_1Q_1) \odot (E_2S_2Q_2))\,. &(4.21)\end{aligned}$$

Then we substitute $M_iQ_i = \overline{a}_i + tE_iS_iQ_i$ into the first term to obtain

$$\begin{aligned}(M_1Q_1 \odot M_2Q_2) &= (\overline{a}_1 + tE_1S_1Q_1) \odot (\overline{a}_2 + tE_2S_2Q_2) &(4.22)\\ &= \overline{a}_1 \odot \overline{a}_2 + \overline{a}_1 \odot (tE_2S_2Q_2) + \overline{a}_2 \odot (tE_1S_1Q_1) &(4.23)\\ &\quad + (tE_1S_1Q_1) \odot (tE_2S_2Q_2))\,. &(4.24)\end{aligned}$$

We have

$$\overline{a}_i \odot (tE_jS_jQ_j) = \overline{a}_i[t^0]tE_jS_jQ_j = 0 \tag{4.25}$$

since we are dealing with power series in t. Therefore

$$(M_1Q_1 \odot M_2Q_2) = \overline{a}_1\overline{a}_2 + t((E_1S_1Q_1) \odot (E_2S_2Q_2)) \tag{4.26}$$

and upon substitution we are left with $\overline{a}_1\overline{a}_2$ which is what we needed in the first place.

One may also ask whether we can do something in the case where S factors and the terms share

variables.

This factoring also generalizes to higher dimensions. If S factors into a product of k factors $S_1, \ldots, S_k$ where no two of which share any variables then the problem of finding Q boils down to solving k lower-dimensional problems for functions $Q_1, \ldots, Q_k$, naming variables appropriately and taking the Hadamard product of the functions. That is

$$Q = Q_1 \odot \cdots \odot Q_k \tag{4.27}$$

The more lower-dimensional problems that we solve, the more higher dimensional problems we can solve. Although it appears that the interesting higher dimensional problems that can be solved in this manner will have a lower and lower proportion among all higher dimensional problems.

4.5 Kernel Group Considerations

It is important to find more functional equations if we can. One way to do this is by applying a transformation on the pair (x_1, x_2) that leaves as many parts of the functional equation as we can but still adding some information in the form of a new equation. One way to do this is to cosider mapping x_1 and x_2 to some quantities separately and then consider the full group generated by such maps. Since we usually eliminate the kernel in front of Q it is useful to consider transformations that preserve the kernel. We'll call this group $G(K)$, although it won't be the full group of rational transformations preserving the kernel. For example $(x_1, x_2) \mapsto (x_2, x_1)$ will not be in the group. Any transformation on (x_1, x_2) that preserves the kernel $K = 1 - tS$ must necessarily preserve the step polynomial S. Here

$$S = \alpha\overline{x} + \beta + \gamma x \tag{4.28}$$

where $x = x_1$ or x_2 and α, β, γ change correspondingly. We consider sending $x \mapsto y$ and solve for y to determine the possible transformations we can perform. That is we have

$$\alpha\overline{x} + \gamma x = \alpha\overline{y} + \gamma X \tag{4.29}$$

$$(\alpha - \gamma xy)(y - x) = 0\,. \tag{4.30}$$

We have cases

- $\alpha = 0, \gamma = 0$. The equation is then trivially true and any transformation preserves the kernel although this is not very useful though.
- $\alpha = 0, \gamma \neq 0$. Then $y = x$ and only the identity transformation preserves the kernel.

- $\alpha \neq 0, \gamma = 0$. Here $y = x$ as well.
- $\alpha \neq 0, \gamma \neq 0$. Here we either have $y = x$ or $X \neq x$ and $y = \overline{x}\alpha/\gamma = \overline{x}S_{-1}/S_1$.

In conclusion, either every transformation works and we don't have an interesting case or we have the identity transformation plus a possible second one. The last transformation creates a non-trivial group and we can have groups of various orders. Let

$$g_1 : x_1 \mapsto \overline{x_1}\frac{S_1^{-1}}{S_1^1}, \quad g_2 : x_2 \mapsto \overline{x_2}\frac{S_2^{-1}}{S_2^1} \tag{4.31}$$

whenever the S quantities are non-zero. Note that $g_i(g_i(x_1)) = x_1$, that is they are involutions. We have three cases for each x

- 1: Any transformation can be applied.
- 2: We only have the identity transformation.
- 3: We have two transformations.

We'll combine cases for analysis in the form "case $i.j$" to mean we have case i for x_1 and case j for x_2.

Case 1.1: Here we can apply any transformation on both x_1 and x_2. From the transformation analysis this means that $S_1^{-1} = S_1^1 = S_2^{-1} = S_2^1 = 0$ and thus means that we have no steps that go away from the origin. Therefore $S = S_{1,2}^{0,0}$. Thus we have only two models - no steps or a resting step. (I should include little diagrams with arrows like before).

Case 1.2 : Here any transformation applies on x_1, so $S_1^{-1} = S_1^1 = 0$ so the model has no steps that step away from the origin in the x_1-direction. Since the only transformation on x_2 is the identity transformation we either have $S_2^{-1} = 0$ or $S_2^1 = 0$ in which case the model consists of only an up step or a down step, together with a possible resting step. Similarly, the 2.1 case involves a single x_1 or $\overline{x_1}$ step with a possible resting step.

Case 1.3: (and 3.1): This case is similar to the previous one except that now we have two transformations for x_2 and this means $S_2^{-1}, S_2^1 \neq 0$. This means we must have either both vertical steps x_2 and $\overline{x}_2$ in the 1.3 case or two horizontal steps x_1 and $\overline{x}_1$ in the 3.1 case.

Case 2.2: In this case both transformations are the identity transformations. Therefore exactly one of S_1^{-1} and S_1^1 is zero and exactly one of S_2^{-1} and S_2^1 is zero. WLOG, assume that $S_1^1 \neq 0, S_2^1 \neq 0$ (the other models can be obtained by a rotation around the origing of 90, 180 and

270 degrees). The possible models are (I should include the drawings): {N,E,NE}, {N,NE}, {E,NE}, {N,E}, {NE}. In other words, the subsets of {N,E,NE} except the ones going either only in the N direction or only in the E direction. This is a case of group $G(K)$ or order 1.

Case 2.3: In this case there is only one transformation for x_1 and two for x_2. This means $S_1^{-1} = 0$ (positive case) or $S_1^1 = 0$ (negative case). Also $S_2^{-1}, S_2^1 \neq 0$ so there must be at least one step going in the positive x_2 direction and one in the negative x_2 direction. We have 17 cases (and another 17 including a resting step). Of course in the 3.2 case we have the meaning of x_1 and x_2 reversed.This is a case of group $G(K)$ of order 2.

Case 3.3: This is the most interesting case with two transformations for x_1 and x_2 each. This means that there must be at least one step going in a positive and one going in a negative coordinate for each of x_1 and x_2. There are a lot of such cases (in the hundreds). Previous results show that the group $G(K)$ has orders 4, 6, 8 and infinity depending on the step set in question.

Later we'll see that these cases correspond to the idea of dimensionality of the model. Case 1.1 is zero-dimensional. Cases 1.2, 1.3 (and 3.1) are one-dimensional. Cases 2.2, 2.3 and 3.3 are two-dimensional. It should not be a suprise that the 0 and 1-dimensional cases will be easy while the two-dimensional cases will become progressively more difficult.

4.6 Solving basic 2D models

4.6.1 Cases 1.1, 1.2, 1.3

A quick glance at the step models involved shows that we are dealing with essentially one-dimensional problems. This means they can be solved easily explicitly. One can also note that these models are all factorable

$$S(x_1, x_2) = 1 \cdot S(x_2)\,. \tag{4.32}$$

Both Q_1 and Q_2 are functions we solved for in the one-dimensional case

$$Q_1(x_1, a_1, t) = \frac{1}{1 - a_1 t}, \quad Q_2(x_2, a_2, t) = q(x_2, a_2, t) \tag{4.33}$$

for some solution q from that section. Therefore

$$Q(x_1, x_2, a_1, a_2, t) = Q_1(x_1, a_1, t) \odot Q_2(x_2, a_2, t) \tag{4.34}$$

$$= \frac{1}{1 - a_1 t} \odot q(x_2, a_2, t) = q(x_2, a_2, a_1 t)\,. \tag{4.35}$$

Similarly in the 2.1, 3.1 cases

$$Q(x_1, x_2, a_1, a_2, t) = Q_1(x_1, a_1, t) \odot Q_2(x_2, a_2, t) \tag{4.36}$$

$$= q(x_1, a_1, t) \odot \frac{1}{1 - a_2 t} = q(x_1, a_1, a_2 t) \tag{4.37}$$

and this completes the solution. What about the relationship with the $B_1 B_2$ transformation? If $f(a, t)$ is the function that goes together with $q(x, a, t)$ then the function $F(a_1, a_2, t)$ that goes together with $Q(x_1, x_2, a_1, a_2, t)$ is given by

$$F(a_1, a_2, t) = f(a_2, a_1 t)\,. \tag{4.38}$$

Then we have $B_2(FQ) = 0$ and $B_1 B_2(FQ) = 0$.

4.6.2 Case 2.2 - group of order 1

A quick glance at the step models reveals that this case is mostly trivial. We have a lot of models that never leave the origin and those have a trivial solution

$$Q(x_1, x_2, a_1, a_2) = \frac{1}{1 - a_1 a_2 S^{0,0}_{1,2} t}\,. \tag{4.39}$$

We also have a lot of models that are essentially one-dimensional with a single step going along an axis and some steps going off-axis where the latter steps can never be taken by a random walk in this model. They are essentially cases already covered in the previous section. Suppose that the single step is taken in the positive x_1 direction. This is equivalent to the $s_{-1} = 0$ case in 1D with function

$$q(x, a, t) = \frac{1 - s_0 t}{(1 - st)(1 - a s_0 t)} = \frac{1 - s_0 t}{(1 - x_1 s_1 t)(1 - a s_0 t)}\,. \tag{4.40}$$

Then we have

$$Q(x_1, x_2, a_1, a_2, t) = Q_1(x_1, a_1, t) \odot Q_2(x_2, a_2, t) \tag{4.41}$$

$$= q(x_1, a_1, t) \odot \frac{1}{1 - a_2 t} = q(x_1, a_1, a_2 t) \tag{4.42}$$

and therefore

$$Q(x_1, x_2, a_1, a_2, t) = \frac{1 - a_2 S_{1,2}^{0,0} t}{(1 - (x_1 + S_{1,2}^{0,0}) a_2 t)(1 - a_1 a_2 S_{1,2}^{0,0} t)} \,. \tag{4.43}$$

The interesting models in case 2.2 are those with non-negative steps that we listed. Here

$$S = \alpha x_1 + \beta x_2 + \gamma x_1 x_2 + \delta \,. \tag{4.44}$$

The functional equation and its zero extractions are

$$(1 - tS)Q - c_1 Q_1^0 - c_2 Q_2^0 + c_1 c_2 Q_{1,2}^{0,0} = \overline{a_1 a_2} \,, \tag{4.45}$$

$$(1 - t a_1 (\delta + \beta x_2)) Q_1^0 - c_2 Q_{1,2}^{0,0} = \overline{a_2} \,, \tag{4.46}$$

$$(1 - t a_2 (\delta + \alpha x_1)) Q_2^0 - c_1 Q_{1,2}^{0,0} = \overline{a_1} \,, \tag{4.47}$$

$$(1 - a_1 a_2 t \delta) Q_{1,2}^{0,0} = 1 \,. \tag{4.48}$$

We solve the problem by a simple back-substitution of the variables.

$$Q_{1,2}^{0,0} = \frac{1}{1 - a_1 a_2 \delta t} \,, \tag{4.49}$$

$$Q_1^0 = \frac{\overline{a}_2}{1 - t a_2 S_1^0} \left(1 + \frac{b_2}{1 - a_1 a_2 \delta t} \right) = \frac{1 - a_1 \delta t}{(1 - t a_1 S_1^0)(1 - a_1 a_2 \delta t)} \,, \tag{4.50}$$

$$Q_2^0 = \frac{\overline{a}_1}{1 - t a_1 S_2^0} \left(1 + \frac{b_1}{1 - a_1 a_2 \delta t} \right) = \frac{1 - a_2 \delta t}{(1 - t a_2 S_2^0)(1 - a_1 a_2 \delta t)} \tag{4.51}$$

$$Q = \frac{1}{1 - tS} (\overline{a}_1 \overline{a}_2 + c_1 Q_1^0 + c_2 Q_2^0 - c_1 c_2 Q_{1,2}^{0,0}) \tag{4.52}$$

$$= \frac{1}{1 - tS} \left(\overline{a}_1 \overline{a}_2 + \frac{1}{1 - t a_1 a_2 \delta} \left(c_1 \frac{1 - a_1 \delta t}{1 - t a_1 S_1^0} + c_2 \frac{1 - a_2 \delta t}{1 - t a_2 S_1^0} - c_1 c_2 \right) \right) \tag{4.53}$$

$$= \frac{1}{1 - tS} \left(\overline{a}_1 \overline{a}_2 + \frac{1}{1 - a_1 a_2 \delta t} (c_1 f_1(a_1, t) + c_2 f_2(a_2, t) - c_1 c_2) \right) \tag{4.54}$$

where the last line explicitly shows that Q is composed of functions that depend only on one of a_1, a_2. If we want to find a relation of the type $B(FQ) = 0$ then we must first find a suitable function

F. First of all the $1 - tS$ part doesn't play a role since it doesn't depend on a_1 or a_2. Then later the variables c_1 and c_2 will vanish when extracting the coefficients of b_1^0 and b_2^0. We'll now show that the function

$$F(a_1, a_2, t) = 1 - a_1 a_2 \delta t \tag{4.55}$$

works exactly. First

$$B((1 - a_1 a_2 \delta t)\overline{a}_1\overline{a_2}) = B(\overline{a}_1\overline{a}_2 - t\delta) = B(\overline{a}_1\overline{a}_2) - B(t\delta) = \overline{a}_1\overline{a}_2 - \overline{a}_2 - \overline{a}_1 + 1 = c_1 c_2 \,. \tag{4.56}$$

Then

$$B(c_1 f_1 + c_2 f_2 - c_1 c_2) = c_1 f_1 + c_2 f_2 - c_1 c_2 - c_2 f_2 - c_1 f_1 = -c_2 c_2 \,. \tag{4.57}$$

Putting the two quantities together along with dividing by $1 - tS$ shows that $B(FQ) = 0$. Of course this works for $G = Q_{1,2}^{0,0}$ too.

4.6.3 Case 2.3 - group of order 2

There are two possibilities here - positive and negative. In the negative case $S_1^1 = 0$ and the steps have a non-positive x_1 component. Due to the fact that a lot of steps are not going to be taken by random walks in those models, all of these models reduce to the cases of S consising of a rest step and maybe a single step in a positive direction. This takes care of 34 models (or 68 if we include a rest step). The interesting models are those with a positive x_1 component, or in other words, models where $S_1^{-1} = 0$. The calculations for these models become increasingly more complicated. The functional equation and its extracted forms read as

$$(1 - tS)Q - c_1 Q_1^0 - (c_2 - t\overline{x}_2 S_2^{-1})Q_2^0 + c_1 c_2 Q_{1,2}^{0,0} = \overline{a_1 a_2} \,, \tag{4.58}$$

$$(1 - ta_1 S_1^0)Q_1^0 - (c_2 - ta_1\overline{x}_2 S_{1,2}^{0,-1})Q_{1,2}^{0,0} = \overline{a_2} \,, \tag{4.59}$$

$$(1 - ta_2 S_2^0)Q_2^0 - c_1 Q_{1,2}^{0,0} - a_2 t S_2^{-1} Q_2^1 = \overline{a_1} \,, \tag{4.60}$$

$$(1 - a_1 a_2 t S_{1,2}^{0,0})Q_{1,2}^{0,0} - ta_1 a_2 S_{1,2}^{0,-1} Q_{1,2}^{0,1} = 1 \,. \tag{4.61}$$

We can identify a subproblem that is essentially one-dimensional and therefore already solved. For this we look at the second and fourth equations with the substitution $a_i = a, c_i = c, x_2 = x, ta_1 =$

$\tau, S_1^0 = T, Q_1^0(x_2) = q(x)$. These equations become

$$(1 - \tau T)q - (c - \tau \overline{x} T_{-1})q_0 = \overline{a}\,, \tag{4.62}$$

$$(1 - a\tau T_0)q_0 - a\tau S_{1,2}^{0,-1} q_1 = 1 \tag{4.63}$$

where $q_0 = q(a, \tau)$. This is the one-dimensional problem again. The solution here is $q(x, a, \tau)$ obtained by the methods in the previous chapter so the original problem has a solution

$$Q_1^0(x_2) = q(x_2, a_2, a_1 t), \quad Q_{1,2}^{0,0} = q_0(a_2, a_1 t)\,. \tag{4.64}$$

The reason the method worked in this case is that we have steps that keep a random walk staying on the x_2 axis and steps that take it away from the x_2 but no steps that have it return to it. This is similar to the case where a random walk leaves the x_2 axis never to come back. Essentially the problem splits into a one-dimensional one regarding the x_2 axis and a further 2D one once we step off the axis. We re-write the master equation as

$$(1 - tS)Q - (c_2 - t\overline{x}_2 S_2^{-1})Q_2^0 = \overline{a_1}(\overline{a}_2 + b_1 q - b_1 c_2 q_0) = \overline{a}_1 \phi(x_2)\,. \tag{4.65}$$

Since $S_2^{-1}, S_2^1 \neq 0$ we have a root $x_2 = \sigma_2$ that allows us to remove the $(1 - tS)Q$ term and get

$$Q_2^0 = -\frac{\overline{a}_1 \phi(\sigma_2)}{c_2 - t\overline{\sigma}_2 S_2^{-1}} \tag{4.66}$$

so this allows us to write Q as

$$Q = \frac{1}{1 - tS}\left(\overline{a}_1 \phi(x_2) - \frac{c_2 - t\overline{x}_2 S_2^{-1}}{c_2 - t\overline{\sigma}_2 S_2^{-1}} \overline{a}_1 \phi(\sigma_2)\right)\,. \tag{4.67}$$

Of course, this can be expanded and simplified based on what q and q_0 are from the one-dimensional setting but the important thing is that we can obtain the solution in this case via one-dimensional equations.

4.6.4 Summary of cases

We can give a short summary of the result via a table. Here q stands for a one-dimensional solution and f to its associated function such that $B(fq) = 0$.

Case	Q	$F(a_1, a_2, t)$
1.1	1	1
1.2, 1.3	$q(x_2, a_2, a_1 t)$	$f(a_2, a_1 t)$
2.1, 3.1	$q(x_1, a_1, a_2 t)$	$f(a_1, a_2 t)$
2.2	$\frac{\overline{a}_1 \overline{a}_2 + \frac{c_1 f_1 + c_2 f_2 - c_1 c_2}{1 - a_1 a_2 \delta t}}{1 - tS}$	
2.3		

Table 4.1: Easy 2D cases

Chapter 5

Hexagonal Walks

5.1 Introduction

We'll tackle the problem of random walks on a honey-comb lattice. This is an example of a problem set on a different topology. It's our first problem dealing with a random walk model where the direction in which you go depends on where you are on the lattice. It can be interpreted as a random walk model on the plane where we have two distinct step polynomials depending on where we are. Because of this the flavour of the problem will be quite different. However it shall serve as a good example of the use of operators. We start by embedding the honeycomb lattice in $\mathbb{Z}^2$ where the first hexagon has coordinates

$$\{(0,0),(1,1),(1,2),(0,0),(-1,2),(-1,1)\} \tag{5.1}$$

and the rest of the lattice is obtained by tiling the plane with this hexagon. As before we let Q be the generating function for the random walks on the lattice where x, y keep track of the endpoint of a walk and t keeps track of its length. Of course, off the lattice the terms will have zero coefficients. Out of thin air we create two operators C_1 and C_2, acting on Q, which pick out all the bottom and top corners of the hexagon tiles respectively. By picking out we mean that C_1 results in the partial generating function corresponding the the coordinates given in the two situations. The condition is that $C_1 + C_2 = 1$ the identity. These two locations give two distinct possibilities for step sets so we'll give them their own step polynomials. Let's call their respective step polynomials S_1 and S_2 given by

$$S_1 = (x + \overline{x})y + \overline{y}, \quad S_2 = (x + \overline{x})\overline{y} + y\,. \tag{5.2}$$

Now we can continue to create a functional equation for the hexagonal model. We perform an iteration. We start with the empty walk with generating function $Q = 1$. Then for every iterate of Q we create the next iterate by taking the walks enumerated in Q and extending them by one step. Since we have two situations we need to extend by different steps in each case. That is we start with Q, extract the S_1-centric walks with C_1Q and then extend them by multiplying with S_1. The result is that the new Q is given by

$$Q = 1 + t(S_1C_1 + S_2C_2)Q = 1 + t\Sigma Q\,. \tag{5.3}$$

We can also write this in the operator format as

$$(1 - t\Sigma)Q = 1\,. \tag{5.4}$$

We may wish to restrict the random walks to the quarter plane or in some other way. To do this we need to also use an extraction operator E that removes those walks that stray out of the boundaries. Since each step is of size at most 1 we only need to remove those walks that step 1 unit outside of the boundaries and E is then easy to write. The iteration and functional equation are given by

$$Q = 1 + tE(S_1C_1 + S_2C_2)Q = 1 + tE\Sigma Q\,, \tag{5.5}$$

$$(1 - tE\Sigma)Q = 1\,. \tag{5.6}$$

Of course we can also label the walks that interact with specific locations. We repeate the work with the extra operator L. In general we can use an operator Γ to encapsulate everything extra that we do. It remains to find the actual operators C_1 and C_2.

5.2 Operators

Before we find the operators we need to find which points they need to act on. We can find all the lower vertices by starting with the hexagon with lower vertex $(0, 0)$ and translating it along the lattice. It's not difficult to see that the lower vertices of these hexagons are those with coordinates (m, n) where

- $m \equiv 0 \pmod 2$, $n \equiv 0 \pmod 4$,
- $m \equiv 1 \pmod 2$, $n \equiv 2 \pmod 4$.

The upper vertices of these hexagons are the above coordinates translated by $(0, 3)$ in the vertical direction. Those are the coordinates (m, n) where

- $m \equiv 0 \pmod 2$, $n \equiv 3 \pmod 4$,
- $m \equiv 1 \pmod 2$, $n \equiv 1 \pmod 4$.

This allows us to write the operators as

$$C_1 = \sum_{m,n} x^{2m} y^{4n} [x^{2m} y^{4n}] + \sum_{m,n} x^{2m+1} y^{4n+2} [x^{2m+1} y^{4n+2}] \,, \tag{5.7}$$

$$C_2 = \sum_{m,n} x^{2m+1} y^{4n+1} [x^{2m+1} y^{4n+1}] + \sum_{m,n} x^{2m} y^{4n+3} [x^{2m} y^{4n+3}] \,. \tag{5.8}$$

$$\tag{5.9}$$

We can write these operators more succinctly by using the following conventions

$$d_{j,k}(z) = \sum_m z^{jm+k} [z^{jm+k}] \,, \tag{5.10}$$

$$D_{ij} = d_{2,i}(x) d_{4,j}(y) \,, \tag{5.11}$$

$$D = D_{00} \,. \tag{5.12}$$

Note that

$$D_{ij} = x^i y^j D\overline{x}^i \overline{y}^j \,. \tag{5.13}$$

This allows us to write the operators C_1 and C_2 succinctly as

$$C_1 = D_{00} + D_{12} = D + xy^2 D\overline{xy}^2 \,, \tag{5.14}$$

$$C_2 = D_{03} + D_{11} = y^3 D\overline{y}^3 + xy D\overline{xy} \,. \tag{5.15}$$

The functional equation becomes

$$(1 - t(S_1(D_{00} + D_{12}) + S_2(D_{03} + D_{11}))Q = 1 \tag{5.16}$$

and analogously we can append an operator Γ to to the expression.

5.3 Group Structure

There are nicer ways to express these extractions without performing the extractions themselves. Let k be a positive integer and $w_k = e^{2\pi i/k}$. Then it is a somewhat well-known result that we can extract the coefficient $0 \pmod k$ of $f(x)$ with the following summation

$$d_{k,0}(x)f(x) = \frac{1}{k}\sum_{j=0}^{k-1} f(w_k^j x)\,. \tag{5.17}$$

This establishes the concrete operator representations

$$d_{2,0}(x)f(x) = \frac{1}{2}(f(x) + f(-x))\,, \tag{5.18}$$

$$d_{4,0}(y)f(y) = \frac{1}{4}(f(y) + f(iy) + f(-y) + f(-iy))\,. \tag{5.19}$$

Using this definition in the functional equation would lead to a functional equation relating $Q(x, y)$ to functions Q evaluated at different arguments like $(-x, y)$, (x, iy), etc. This allows us to explore the group structure of the functional equation. To this end let $g : x \mapsto -x$ and $h : y \mapsto iy$ be the two commuting operations and the group of symmetries be

$$G \cong \langle g\rangle \times \langle h\rangle \cong \langle -1\rangle \times \langle i\rangle\,. \tag{5.20}$$

It remains to see how these group actions affect the various quantities in the problem. First we deal with S_1 and S_2 and then with the operators D_{jk}. This will allow us to find how the actions affect Σ and the functional equation as a whole. First we have the actions of g

$$gS_1 = S_1^* = -(x + \overline{x})y + \overline{y}\,, \tag{5.21}$$

$$gS_2 = S_2^* = -(x + \overline{x})\overline{y} + y\,. \tag{5.22}$$

Then the actions of h

$$hS_1 = i(x + \overline{x})y - i\overline{y} = -igS_1 = -iS_1^*\,, \tag{5.23}$$

$$hS_2 = -i(x + \overline{x})\overline{y} + iy = igS_2 = iS_2^*\,. \tag{5.24}$$

These relatons allows us to complete the rest of the relations iteratively (maybe a multiplication table would be good here)

$$hS_1 = -iS_1^*, \quad h^2S_1 = -S_1, \quad h^3S_1 = iS_1^*\,, \tag{5.25}$$

$$ghS_1 = -iS_1, \quad gh^2S_1 = -S_1^*, \quad gh^3S_1 = -iS_1\,, \tag{5.26}$$

$$hS_2 = iS_2^*, \quad h^2S_2 = -S_2, \quad h^3S_2 = iS_2^*\,, \tag{5.27}$$

$$ghS_2 = iS_2, \quad gh^2S_2 = -S_2^*, \quad gh^3S_2 = -iS_2\,. \tag{5.28}$$

$$\tag{5.29}$$

Now g acts on D_{jk} in the following way

$$g(D_{jk}f(x,y)) = g\sum_{m,n} x^{2m+j}y^{4m+k}[x^{2m+j}y^{4m+k}]f(x,y) \tag{5.30}$$

$$= \sum_{m,n}(-x)^{2m+j}y^{4m+k}f_{2m+j,4m+k} \tag{5.31}$$

$$= (-1)^j D_{jk}f(x,y)\,. \tag{5.32}$$

Similarly h acts on D_{jk} in the following way

$$h(D_jkf(x,y)) = i^k D_{jk}f(x,y)\,. \tag{5.33}$$

Or specifically

$$gD_{00} = D_{00}, \quad gD_{11} = -D_{11}, \quad gD_{03} = D_{03}, \quad gD_{12} = -D_{12}\,, \tag{5.34}$$

$$hD_{00} = D_{00}, \quad hD_{11} = iD_{11}, \quad hD_{03} = -iD_{03}, \quad hD_{12} = -D_{12}\,. \tag{5.35}$$

$$\tag{5.36}$$

We can now move on to finding the actions on the operator Σ. To spare some of the work it is enough to find the actions $1, gh, h^2$ and gh^3 since we'll have only 4 independent relations on the operators. Let

$$A = S_1(D_{00} + D_{12}), \quad B = S_2(D_{11} + D_{03})\,. \tag{5.37}$$

Then

$$\Sigma = B + A, \quad gh\Sigma = B - iA\,, \tag{5.38}$$

$$h^2\Sigma = B - A, \quad gh^3\Sigma = B + iA\,. \tag{5.39}$$

We have

$$(1 + h^2)\Sigma = gh(1 + h^2)\Sigma = 2B \tag{5.40}$$

and

$$(1 - h^2)\Sigma = igh(1 - h^2) = 2A\,. \tag{5.41}$$

By using these eliminations we can write new forms of the functional equation. We have

$$\frac{1}{2}(1 + h^2)(1 - t\Sigma)Q = \left(\frac{1}{2}(1 + h^2) - tB\right) Q \tag{5.42}$$

and

$$\frac{1}{2}(1 + h^2)(1) = \frac{1}{2}2 = 1 \tag{5.43}$$

so the functional equation is

$$\left(\frac{1}{2}(1 + h^2) - tS_2(D_{11} + D_{03})\right) Q = 1\,. \tag{5.44}$$

Repeating the analysis with the other relations yields

$$\left(\frac{1}{2}(1 - h^2) - tS_1(D_{00} + D_{12})\right) Q = 0\,. \tag{5.45}$$

The relations allow us to write two different representations of Σ

$$2\Sigma = ((1 + h^2) + igh(1 - h^2))\Sigma = ((1 - h^2) + gh(1 + h^2))\Sigma\,. \tag{5.46}$$

Or in other words

$$(1 - h^2 - igh(1 - h^2))\Sigma = (1 - h^2)(1 - igh)\Sigma = 0\,, \tag{5.47}$$

$$(1 + h^2 - gh(1 + h^2))\Sigma = (1 + h^2)(1 - gh)\Sigma = 0\,. \tag{5.48}$$

We can use these on the functional equation to eliminate Σ

$$(1-h^2)(1-igh)(1-t\Sigma)Q = (1-h^2)(1-igh)Q = 0 = (1-h^2)(1-igh)1\,, \tag{5.49}$$

$$(1+h^2)(1-gh)(1-t\Sigma)Q = (1+h^2)(1-gh)Q = 0 = (1+h^2)(1-gh)1\,. \tag{5.50}$$

Proposition 17. *The function Q satisfies the functional equation*

$$(1-h^2)(1-igh)Q = (1+h^2)(1-gh)Q = 0 \tag{5.51}$$

where we need both factors to eliminate Q.

If we use the series definition for Q here we find that these relations are saying nothing more than the fact that the terms of Q corresponding to the "holes" inside the hexagons are zero.

5.4 Finding the generating function Q

Note that

$$\frac{1}{2}(1+h^2)Q(x,y) = \frac{1}{2}(Q(x,y)+Q(x,-y)) = C_1Q\,, \tag{5.52}$$

$$\frac{1}{2}(1-h^2)Q(x,y) = \frac{1}{2}(Q(x,y)-Q(x,-y)) = C_2Q\,. \tag{5.53}$$

where the latter equalities follow since we are dealing with the even and odd y parts of Q respectively, which cover conditions C_1 and C_2. We can re-write the two functional equations from the previous section as

$$C_1Q - tS_2C_2Q = 1\,, \tag{5.54}$$

$$C_2Q - tS_1C_1Q = 0\,. \tag{5.55}$$

This is a linear system in the quantities C_1Q and C_2Q. The associated determinant is $1-t^2S_1S_2 \neq 0$ so we find the unique solution to be

$$C_1Q = \frac{1}{1-t^2S_1S_2} = \frac{x^2y^2}{x^2y^2 - t^2(x+y^2+x^2y^2)(1+x^2+xy^2)} \tag{5.56}$$

$$= 1 + (3+\overline{x}^2+x^2+\overline{x}\overline{y}^2+x\overline{y}^2+y^2\overline{x}+xy^2)t^2+\cdots\,, \tag{5.57}$$

$$C_2Q = \frac{tS_1}{1-t^2S_1S_2} = \frac{txy(x+(1+x^2)y^2)}{x^2y^2 - t^2(x+y^2+x^2y^2)(1+x^2+xy^2)} \tag{5.58}$$

$$= (\overline{y}+y\overline{x}+xy)t+\cdots\,. \tag{5.59}$$

Finally

$$Q = C_1 Q + C_2 Q = \frac{1 + tS_1}{1 - t^2 S_1 S_2} = \sum_{k=0}^{\infty} t^{2k} S_1^k S_2^k + \sum_{k=0}^{\infty} t^{2k+1} S_1^{k+1} S_2^k \,. \tag{5.60}$$

Proposition 18. *The generating function Q for unrestricted random walks on the honeycomb lattice is given by*

$$Q(x, y; t) = \frac{x^2 y^2 + txy(x + y^2 + x^2 y^2)}{x^2 y^2 - t^2 (1 + x^2 + xy^2)(x + y^2 + x^2 y^2)} \tag{5.61}$$

$$= 1 + (\overline{y} + y\overline{x} + xy)t + (3 + \overline{x}^2 + x^2 + \overline{xy}^2 + x\overline{y}^2 + y^2\overline{x} + xy^2)t^2 + \cdots \,. \tag{5.62}$$

Chapter 6

Higher-dimensional walks

6.1 Using Operators to Derive the Functional Equation

As mentioned previously, the generalization to higher-dimensions is straight-forward. Here we have an operators of the form $D = D_1 \dots D_d$. We take the generating function Q enumerating all random walks from model S with interactions. We extend all walks by one step via the multiplication SQ. Then we remove all walks venturing outside of the positive orthant by applying E, that is we have ESQ. We mark all walks that have ended up on the zero hyperplanes via L. The result is $LESQ$. We need to multiply by t to track the extension of the walks. The resulting expression is almost the same as Q itself. The only walk missing is the empty walk with monomial 1. This allows us to write the functional equation for Q as

$$Q = 1 + tLESQ \tag{6.1}$$

which we can re-write as

$$(1 - tLES)Q = 1\,. \tag{6.2}$$

We can apply the operator L^{-1} and obtain the equation

$$L^{-1}(1) = \sum_{I \subseteq [d]} c_I = \overline{a}_{[d]} \tag{6.3}$$

which allows us to write down the following result.

Proposition 19. *The $d-$dimensional random walk model with small steps in the positive orthant and*

marking of hyperplane $x_k = 0$ interactions has a generating function Q satisfying the functional equations

$$(1 - tLES)Q = 1\,, \tag{6.4}$$

$$(L^{-1} - tES)Q = \overline{a}_{[d]} \tag{6.5}$$

where L, E, L^{-1} are of the form $D = D_1 \cdots D_d$.

The second equation contains many less terms. To see this we take a look at the $L_k E_k$ operator

$$L_k E_k = 1 + b_k[x_k^0] - \overline{x}_k[\overline{x}_k] \tag{6.6}$$

which increases the number of terms in the equation by a factor of 3. We have 3^d terms relating to SQ by the end of applying LE. On the other hand once we apply the operator L^{-1} we introduce 2^d terms relating to Q but then since we only have E applied on SQ we'd only have at most 2^d terms from this. For the case of walks of step size at most 1 and labelling the hyperplane interactions we have exactly 2^d terms $Q_I^{\mathbb{O}}$, one for each subset $I \subseteq [d]$. In this case we can write out the functional equation explicitly. Firstly,

$$L^{-1}Q = (1 - c_1[x_1^0]) \dots (1 - c_d[x_d^0])Q \tag{6.7}$$

$$= \sum_{I \subseteq [d]} (-1)^{|I|} c_I Q_I^{\mathbb{O}} \tag{6.8}$$

and

$$ESQ = (1 - \overline{x}_1[\overline{x}_1]) \dots (1 - \overline{x}_d[\overline{x_d}])(SQ) \tag{6.9}$$

$$= \sum_{I \subseteq [d]} (-1)^{|I|} \overline{x}_I [x_I^{-\mathbb{I}}](SQ) \tag{6.10}$$

$$= \sum_{I \subseteq [d]} (-1)^{|I|} \overline{x}_I S_I^{-\mathbb{I}} Q_I^{\mathbb{O}} \tag{6.11}$$

where the last line follows from the fact that there are no negative powers of x_k in Q for any k. Therefore the functional equation is

$$\sum_{I \subseteq [d]} (-1)^{|I|} (c_I - t\overline{x_I} S_I^{-\mathbb{I}}) Q_I^{\mathbb{O}} = \overline{a_{[d]}}\,. \tag{6.12}$$

As in the two-dimensional case we let $K_I = c_I - t\overline{x_I} S_I^{-\mathbb{I}}$ where if $I = \varnothing$ we have the empty

products and empty extractions leaving us with $K_\varnothing = K$. Finally we can write the master equation in d-dimensions.

Theorem 20. *The full generating function Q for random walks with interactions satisfies the functional equation*

$$\sum_{I\subseteq[d]} (-1)^{|I|} K_I Q_I^{\mathbb{O}} = \overline{a}_{[d]} \tag{6.13}$$

where $K_I = (c_I - t\overline{x_I} S_I^{-\mathbb{I}})$.

As in the two-dimensional case we can now perform coefficient extractions with $[x_I^{\mathbb{O}}]$ to obtain further functional equations relating $Q_I^{\mathbb{O}}$ to other Q terms of the same variables or fewer variables.

6.2 Coefficient Extraction

We can perform coefficient extraction like in the 1-dimensional case. The operator $[x_i^k]$ will commute with all M_j, E_j with $j \neq i$ so that we only need to consider the action on the operators $M_i = L_i^{-1}$ and E_i. We've already done so in the 1-dimensional case. The zeroth power extraction yields

$$[x_i^0]M_i = [x_i^0] - [x_i^0]c_i[x_i^0] = [x_i^0] - c_i[x_i^0] = \overline{a_i}[x_i^0]\,, \tag{6.14}$$

$$[x_i^0]E_i = [x_i^0] - [x_i^0]\overline{x_i}[\overline{x_i}] = [x_i^0] \tag{6.15}$$

and the k-th power extraction for $k \geqslant 1$ yields

$$[x_i^k]M_i = [x_i^k] - [x_i^k]c_i[x_i^0] = [x_i^k]\,, \tag{6.16}$$

$$[x_i^k]E_i = [x_i^k] - [x_i^k]\overline{x_i}[\overline{x_i}] = [x_i^k]\,. \tag{6.17}$$

So, for $k \geqslant 0$ the operator $[x_i^k]$ eliminates both M_i and E_i from the picture while for $k = 0$ it removes E_i but leaves a residual $\overline{a_i}$ behind for M_i. Therefore if we extract $[x_I^{\mathbb{O}}]$ from the functional equation we have

$$[x_I^{\mathbb{O}}]a_{[d]} = \overline{a}_{[d]} = [x_I^{\mathbb{O}}](M_{[d]} - tE_{[d]}S)Q \tag{6.18}$$

$$= \left(\overline{a}_I M_{[d]\setminus I}[x_I^{\mathbb{O}}] - tE_{[d]\setminus I}[x_I^{\mathbb{O}}]S\right) Q \tag{6.19}$$

$$= [x_I^{\mathbb{O}}](\overline{a}_I M_{[d]\setminus I} - tE_{[d]\setminus I}S)Q \tag{6.20}$$

so that

$$\overline{a}_{[d]\setminus I} = [x_I^{\mathbb{O}}](M_{[d]\setminus I} - a_I t E_{[d]\setminus I} S)Q\,. \tag{6.21}$$

Written in another way this becomes

$$\overline{a}_I = [x_{[d]\setminus I}^{\mathbb{O}}](M_I - a_{[d]\setminus I} t E_I S)Q \tag{6.22}$$

and we can use whichever notation is more useful at the time. The latter notation underscores the idea that when we extract coefficients for walks that end at some hyperplane intersection we are having a reduced version of the functional equation at the cost of an added a term. When we extract a general term we can use the operator $[x_I^{\mathbb{O}} x_J^P]$ where P is a set of powers greater than 0 and $I \cap J = \varnothing$, $|P| \geqslant 1$ and $I \cup J = K$. We find

$$[x_I^{\mathbb{O}} x_J^P] a_{[d]} = 0 = [x_I^{\mathbb{O}} x_J^P](M_{[d]} - tE_{[d]}S)Q \tag{6.23}$$

$$= [x_I^{\mathbb{O}} x_J^P](\overline{a}_I M_{[d]\setminus K} - tE_{[d]\setminus K}S)Q \tag{6.24}$$

which we can write as

$$[x_I^{\mathbb{O}} x_J^P](M_{[d]\setminus K} - a_I t E_{[d]\setminus K} S)Q = 0\,. \tag{6.25}$$

If I is empty then

$$[x_J^P](M_{[d]\setminus J} - tE_{[d]\setminus J}S)Q = 0\,. \tag{6.26}$$

The main role of these calculations is to provide a shortcut for some of the manual calculations.

6.3 Further functional equations

We can try going a little further in the coefficient extraction to come up with an equation that looks like the original functional equation. We know that

$$[x_I^{\mathbb{O}}]SQ = S_I^{\mathbb{O}} Q_I^{\mathbb{O}} + R_I \tag{6.27}$$

where R_I is a collection of $S_I^{-P}Q_I^P$ terms that involves sets P whose terms are not all zero. We can use this inside the functional equation

$$\begin{aligned}\overline{a}_{[d]\setminus I} &= M_{[d]\setminus I}[x_I^{\mathbb{O}}]Q - a_I t E_{[d]\setminus I}[x_I^{\mathbb{O}}](SQ) && (6.28)\\ &= M_{[d]\setminus I}Q_I^{\mathbb{O}} - a_I t E_{[d]\setminus I}(S_I^{\mathbb{O}}Q_I^{\mathbb{O}} + R_I) && (6.29)\\ &= (M_{[d]\setminus I} - a_I t E_{[d]\setminus I}S_I^{\mathbb{O}})Q_I^{\mathbb{O}} - a_I t E_{[d]\setminus I}R_I && (6.30)\end{aligned}$$

so we can re-write the functional equation as

$$(M_{[d]\setminus I} - a_I t E_{[d]\setminus I}S_I^{\mathbb{O}})Q_I^{\mathbb{O}} = a_I(\overline{a}_{[d]} + tR_I') \tag{6.31}$$

where $R_I' = E_{[d]\setminus I}R_I$. We can calculate the following terms

$$M_{[d]\setminus I}Q_I^{\mathbb{O}} = \sum_{J\subseteq[d]\setminus I}(-1)^{|J|}c_J Q_{I,J}^{\mathbb{O},\mathbb{O}} \tag{6.32}$$

and

$$a_I t E_{[d]\setminus I}S_I^{\mathbb{O}}Q_I^{\mathbb{O}} = \sum_{J\subseteq[d]\setminus I}(-1)^{|J|}a_I t\overline{x}_J S_{I,J}^{\mathbb{O},-\mathbb{I}}Q_{I,J}^{\mathbb{O},\mathbb{O}} \tag{6.33}$$

so the functional equation can be written as

$$\sum_{J\subseteq[d]\setminus I}(-1)^{|J|}(c_J - a_I t\overline{x}_J S_{I,J}^{\mathbb{O},-\mathbb{I}})Q_{I,J}^{\mathbb{O},\mathbb{O}} = a_I(\overline{a}_{[d]} + tR_I')\,. \tag{6.34}$$

This suggests the notation

Definition 21. *For sets I and J we write*

$$K_J(I) = c_J - a_I t\overline{x}_J S_{I,J}^{\mathbb{O},-\mathbb{I}} \tag{6.35}$$

where when $I = \varnothing$, $K_J(I) = K_J$.

This notation allows us to write the 2^d functional equations in a simplified form.

Corollary 22. *For $I \subseteq [d]$ the function Q satisfies the functional equation*

$$\sum_{J\subseteq[d]\setminus I}(-1)^{|J|}K_J(I)Q_{I,J}^{\mathbb{O},\mathbb{O}} = \overline{a}_{[d]\setminus I} + a_I tR_I' \tag{6.36}$$

where the R'_I term involves only higher order coefficient extractions from Q, that is not all the powers are zero.

Now, to calculate the R'_I terms we first determine what R_I is. It is a sum of terms $S_I^{-P}Q_I^P$ for various nonnegative P with size $|I|$ whose elements are not all 0. To calculate R'_I we apply the operator $E_{[d]\setminus I}$ on R_I. We can try calculating some of these quantities. A key observation is that the terms Q_I^P are power series in the remaining variables so they don't have non-negative coefficients. Let $sq = S_I^{-P}Q_I^P$. Then

$$\begin{aligned} E_j sq &= sq - \overline{x}_j (sq)_j^{-1} && (6.37)\\ &= sq - \overline{x}_j (s_j^{-1}q_j^0 + s_j^0 q_j^{-1} + s_j^1 q_j^{-2}) && (6.38)\\ &= sq - \overline{x}_j s_j^{-1} q_j^0 \,. && (6.39) \end{aligned}$$

Hence

$$E_K sq = \sum_{J\subseteq K} (-1)^{|J|} \overline{x}_J s_J^{-\mathbb{I}} q_J^{\mathbb{O}} \,. \tag{6.40}$$

Finally we can write

$$\begin{aligned} R'_I &= E_{[d]\setminus I} R_I && (6.41)\\ &= \sum_{P>\mathbb{O}} E_{[d]\setminus I} S_I^{-P} Q_I^P && (6.42)\\ &= \sum_{P>\mathbb{O}} \sum_{J\subseteq[d]\setminus I} (-1)^{|J|} S_{I,J}^{-P,-\mathbb{I}} Q_{I,J}^{P,\mathbb{O}} && (6.43)\\ &= \sum_{J\subseteq[d]\setminus I} (-1)^{|J|} \sum_{P>\mathbb{O}} S_{I,J}^{-P,-\mathbb{I}} Q_{I,J}^{P,\mathbb{O}} \,. && (6.44) \end{aligned}$$

Proposition 23. *The functions R'_I are given by*

$$R'_I = \sum_{J\subseteq[d]\setminus I} (-1)^{|J|} \sum_{P>\mathbb{O}} S_{I,J}^{-P,-\mathbb{I}} Q_{I,J}^{P,\mathbb{O}} \,. \tag{6.45}$$

This formula is mostly of theoretical value although it can be used to speed up manual calculations.

6.4 Using the formal inverse for iteration

In a previous subsection we obtained the equation

$$(1 - tLES)Q = 1\,. \tag{6.46}$$

Let $D = LE$. Then we can think of the formal inverse of this operator given as Neumann series as

$$(1 - tDS)^{-1} = \frac{1}{1 - tDS} = \sum_{k=0}^{\infty} t^k (DS)^k\,. \tag{6.47}$$

We can write the series solution

$$Q = \sum_{k=0}^{\infty} t^k (DS)^k(1) \tag{6.48}$$

where $(DS)^k(1) = DS(\ldots(DS(1))\ldots)$. This allows us to compute the series expansion for Q directly using an iteration. The next coefficient is the previous one with DS applied to it. It can be easily implemented in symbolic calculation software like Maple or Mathematica. The iteration is a formalized way of finding the first few terms of the series in the naive way - by taking a set of walks of length n, extending them by 1 step, removing the ones that end outside of the quarter-plane and finally mark the ones that interact with the hyperplanes $x_i = 0$ at their endpoint. We haven't been able to find closed forms for the repeated operator applications in general. Here's an example that illustrates the use of the operators.

Example 24. Let $S = x_1 + \overline{x}_1 + x_2 + \overline{x}_2 = S_1(x_1) + S_2(x_2)$. We start with $DS^0(1) = 1$. Then

$$\begin{aligned}
DS^1(1) &= LE(S_1 + S_2) = L(x_1 + x_2) = a_2x_1 + a_1x_2\,, && (6.49)\\
DS^2(1) &= LE(S_1 + S_2)(a_2x_1 + a_1x_2) && (6.50)\\
&= LE(a_2S_1x_1 + a_1x_2S_2 + a_2x_1S_2 + a_1x_2S_1) && (6.51)\\
&= L(a_2S_1x_1 + a_1x_2S_2 + (a_1 + a_2)x_1x_2) && (6.52)\\
&= a_1a_2^2 + a_2^2x_1^2 + a_1a_2^2 + a_1^2x_2^2 + (a_1 + a_2)x_1x_2 && (6.53)
\end{aligned}$$

which allows us to write

$$Q = 1 + (a_2x_1 + a_1x_2)t + \left(a_1a_2(a_1 + a_2) + (a_2^2x_1^2 + a_1^2x_2^2) + (a_1 + a_2)x_1x_2\right)t^2 + \cdots\,. \tag{6.54}$$

6.5 Kernel Form

We can write out the functional equations in a way that highlights the kernel K in the equation. This confirms that K is indeed special in some way. It's not so surprising that a lot of the tools in the analysis of random walk models use K. We start out as follows

$$1 = Q - tLESQ = Q + LE(-tS + 1 - 1)Q \tag{6.55}$$

$$= Q + LEKQ - LEQ = (1 - L)Q + LEKQ \tag{6.56}$$

where we have $EQ = Q$ since Q only has non-negative powers of x_k. Next we can write

$$(1 - L)Q + LEKQ = (1 - L)Q + (LE - 1)KQ + KQ \tag{6.57}$$

and finish off the calculation with

$$KQ + (1 - L)Q - (1 - LE)KQ = 1\,. \tag{6.58}$$

We can also start with the reduced form

$$\alpha = MQ - tESQ = MQ - EQ + EKQ = (M - 1)Q + EKQ \tag{6.59}$$

where we can expand $E = E - 1 + 1$ to find

$$KQ = \alpha + (1 - M)Q + (1 - E)KQ = \alpha + (1 - M)Q - t(1 - E)SQ \tag{6.60}$$

where the last step follows from

$$(1 - E)KQ = (1 - E)(Q - tSQ) = -t(1 - E)SQ\,. \tag{6.61}$$

The operators $1 - M$ and $1 - E$ are purely extractions . The last kernel form shows more clearly where the K_I's come from in the functional equation above.

Proposition 25. *The first and second kernel forms are*

$$KQ = 1 - (1 - L)Q + (1 - LE)KQ\,, \tag{6.62}$$

$$KQ = \alpha + (1 - L^{-1})Q + (1 - E)KQ\,. \tag{6.63}$$

6.6 Walks without interactions

The Kernel forms above suggest an easier route for dealing with the models without interactions. Those models are equivalent to setting all the $a_k = 1$ which removes any marking or unmarking operators. The first kernel form reduces to the simple functional equation

$$E(KQ) = 1\,. \tag{6.64}$$

This suggests using the following substitution

$$Q = \frac{W}{x_{[d]}K} \tag{6.65}$$

so we have

$$x_{[d]} = x_{[d]}E(KQ) = x_{[d]}E\overline{x}_{[d]}W = \Gamma W \tag{6.66}$$

where $\Gamma = \Gamma_1 \dots \Gamma_d$ and

$$\Gamma_k = x_k E_k \overline{x}_k = x_k(1 - \overline{x}_k[\overline{x}_k])\overline{x}_k = 1 - [x_k]^0\,. \tag{6.67}$$

Therefore we have the functional equation for W as

Proposition 26. *Let $W = x_{[d]}KQ$ where Q is the generating function for walks without interactions in the positive orthant*

$$\Gamma W = \sum_{I \subseteq [d]} (-1)^{|I|} W_I^{\mathbb{O}} = x_{[d]}\,. \tag{6.68}$$

For example in two dimensions we have

$$W(x_1, x_2) - W_1^0(x_2) - W_2^0(x_1) + W_{1,2}^{0,0} = x_1 x_2\,. \tag{6.69}$$

Thus we can write the solution to the problem as

$$Q = \frac{1}{x_{[d]}K}\left(x_{[d]} - \sum_{\varnothing \neq I \subseteq [d]} (-1)^{|I|} W_I^{\mathbb{O}}\right). \tag{6.70}$$

In two dimensions this becomes

$$Q(x_1, x_2) = \frac{x_1 x_2 + W_1^0(x_2) + W_2^0(x_1) - W_{1,2}^{0,0}}{x_1 x_2 (1 - tS)} \,. \tag{6.71}$$

Of course each $W_I^{\mathbb{O}}$ depends on $Q_I^{\mathbb{O}}$ so we could have seen this relationship from the master functional equation for Q too. Interestingly enough this calculation points to a differential equation. Clearly, W satisfies the differential equations

$$\partial_1 \partial_2 W = 1 \tag{6.72}$$

Proposition 27. *The function Q for walks without interactions is D-finite in the $x_1, \ldots, x_d$ variables since it satisfies the inhomogeneous PDE*

$$\partial_1 \ldots \partial_d W = \partial_1 \ldots \partial_d (x_1 \ldots x_d K Q) = 1 \,. \tag{6.73}$$

We'll see a more general relationship in a later chapter.

Chapter 7

Operators

In this chapter we'll explore some of the properties of the operators that interest us, showing when the commute, if they have inverses, what their eigenvalues and eigenvectors are and so on.

7.1 Homogeneous Operators

7.1.1 Single 1D Operators

We start off with the observation that an extraction operator E is just a marking operator L with mark $a = 0$. In other words, we can combine the discussion for E and L by combining the operators into one definition.

Definition 28. *For integer n and a number a let*

$$\Gamma_n(a) = 1 + (a-1)x^n[x^n] = 1 + bx^n[x^n]\,. \tag{7.1}$$

Lemma 29. *The operators $\Gamma_m(a_1)$ and $\Gamma_n(a_2)$ commute.*

Proof. Whether the operators mark the same monomial or not, since they act by multiplication by a_i it doesn't matter in which order we apply them. We can also see this computationally. If $m \neq n$

$$\Gamma_n(a_1)\Gamma_m(a_2) = 1 + b_1x^n[x^n] + b_2x^m[x^m] + b_1b_2x^n[x^n]x^m[x_m] \tag{7.2}$$

$$= 1 + b_1x^n[x^n] + b_2x^m[x^m] \tag{7.3}$$

$$= \Gamma_m(a_2)\Gamma_n(a_1)\,. \tag{7.4}$$

The case for $m = n$ is analogous. □

Because of the commutativity we can write a product of such operators in any order we like. We'll usually write them in alphabetical order. Next, we'll find the eigenvalues and eigenfunctions for single operators.

We note further properties of $\Gamma_n(a)$. The first has to do with its eigenvalues and eigenfunctions. If $a = 1$ the operator acts as the identity so every function is an eigenfunction with eigenvalue 1. When $a \neq 1$ we see that the operator keeps all monomials the same except for the one it marks x^n which will be an eigenfunction with eigenvalue a. It's worth seeing this proved in a linear algebra style both to illustrate the method and to act as a consistency check.

Proposition 30. *The operator $\Gamma_n(a)$ has eigenvalues a and 1 with corresponding eigenfunctions x^n for a and x^k, $k \neq 1$ for 1.*

Proof. Suppose that for some function $u \neq 0$ and $a \neq 1$

$$\Gamma_n(a)u = \lambda u\,. \tag{7.5}$$

We can write $u = u_n x^n + v$ where v has no x^n term. Therefore we find

$$0 = \Gamma_n(a)u - \lambda u = a u_n x^n + v - \lambda u_n x^n - \lambda v = (a-\lambda)u_n x^n + (1-\lambda)v\,. \tag{7.6}$$

We have two cases.

- If $v = 0$ then $u_n \neq 0$ and $\lambda = a$. The eigenfunction is x^n.
- If $v \neq 0$ then $\lambda = 1$ and $u_n = 0$. Here every x^k for $k \neq n$ is an eigenfunction.

□

The second interesting property has to do with taking powers of $\Gamma_n(a)$, including inverses.

Proposition 31. *We have*

$$\Gamma_n(a)^m = \Gamma_n(a^m) \tag{7.7}$$

which holds true for all integers m when $a \neq 0$ and all non-negative integers when $a = 0$. In particular $\Gamma_n(0)^m = \Gamma_n(0)$.

When $a = 0$ the operator has a non-trivial kernel generated by x^n. We can think of minimal polynomials too. For example $\Gamma_n(0)$ is non-trivial so its minimal polynomial is not X or $X - 1$. However since it's idempotent, its minimal polynomial must divide $X(X-1)$ and since it's neither of the factors it must be the minimal polynomial. On the other hand $\Gamma_n(a)$ for $a \neq 0$ does not have a minimal polynomial because the powers of a are independent from each other.

7.1.2 Compounded 1D Operators

Naturally we are led to consider compounded operators and their properties. Let $\Gamma = \Gamma_{n_1}(a_1)\dots\Gamma_{n_m}(a_m)$ where the powers n_i are distinct; otherwise treat any repetitions using commutativity and the identity $\Gamma_n(a)^m = \Gamma_n(a^m)$. Using induction with the identity above gives

$$\Gamma = 1 + b_1 x^{n_1}[x^{n_1}] + \cdots + b_m x^{n_m}[x^m]\,. \tag{7.8}$$

Letting $u = u_{n_1}x^{n_1} + \cdots + u_{n_m}x^{n_m} + v$ in the same way as the single operator case and going through the calculation shows that the eigenvalues of Γ are

- $\lambda = a_i$ for $1 \leqslant i \leqslant m$ with eigenfunction x^{n_i},
- $\lambda = 1$ with eigenfunctions x^m where $m \neq n_i$ for any i.

When one or more of the a_i's is zero we have a non-trivial kernel

$$\ker(\Gamma) = \langle x^m | m = n_i, a_i = 0\rangle\,. \tag{7.9}$$

7.1.3 Multidimensional Operators

Multidimensional operators are ones composed of one-dimensional operators (whether single or compounded) and they can also be considered compounded. For this section we'll use superscript notation

$$\Gamma_n^{(k)}(a) = 1 + b x_k^n[x_k^n]\,. \tag{7.10}$$

Operators working on different dimensions commute with each other

$$\begin{aligned}\Gamma_n^{(i)}(a_1)\Gamma_m^{(j)}(a_2) &= 1 + b_1 x_i^n[x_i^n] + a_2 x_j^m[x_j^m] + a_1 a_2 x_i^n x_j^m[x_i^n x_j^m] &(7.11)\\ &= \Gamma_m^{(j)}(a_2)\Gamma_n^{(i)}(a_1)\,. &(7.12)\end{aligned}$$

We can write $\Gamma^{(k)} = \Gamma_{n_1}(a_{k,n_1})\dots\Gamma_{n_r}(a_{k,n_r})$ as before. This allows us to write a multidimensional operator in terms of its constituent pieces in whichever order we prefer. We are of course interested in finding eigenvalues and eigenfunctions for such operators. We could either find those with the same method as before or we could find it via induction.

Proposition 32. *All monomials $x_{[d]}^P$ are eigenfunctions of $\Gamma = \Gamma^{(1)}\dots\Gamma^{(d)}$. Their eigenvalues are determined by the action of $\Gamma^{(k)}$ on $x_k^{P_k}$. That is, the eigenvalues are products of the eigenvalues produced by the single-dimensional operators.*

This helps us characterize the kernel of Γ.

Corollary 33. *The kernel of Γ is generated by those monomials $x^P_{[d]}$ for which $\Gamma^{(k)}x^{P_k}_k = 0$ for some k.*

7.2 Inhomogeneous Operators

We'll call an operator inhomogeneous if it consists of a homogeneous operator Γ and a laurent polynomial (possibly infinite) S and and acts in one of the following ways

$$\Gamma S(u) = \Gamma(Su) \quad \text{or} \quad S\Gamma u = S \cdot \Gamma u\,. \tag{7.13}$$

This kind of operator is important because it features quite a bit in the funcitonal equations of random walk models. We seek to find the eigenpairs for inhomogeneous operators. Our first task is to show that these operators are equivalent.

Proposition 34. *The two operators have the same eigenvalues with eigenfunctions differing by a multiple of S. That is (λ, u) is an eigenpair for ΓS if and only of (λ, Su) is an eigenpair for $S\Gamma$. In particular, if $\lambda = 0$ we have*

$$\ker(\Gamma S) = \overline{S}\ker(\Gamma)\,. \tag{7.14}$$

Proof. Note that

$$\Gamma S u = \lambda u \quad \iff \quad S\Gamma(Su) = \lambda(Su)\,. \tag{7.15}$$

Therefore the operators have the same eigenvalues and they have the eigenfunctions u and Su respectively. Now if we pick $\lambda = 0$ to find the kernels of the operators we find that $u \in \ker(\Gamma S)$ if and only if $Su \in \ker(S\Gamma) = \ker(\Gamma)$ which completes the proof. □

7.2.1 Single 1D Operators

Let G and u be given as

$$S = \sum_k S_k x^k, \quad u = \sum_k u_k x^k \tag{7.16}$$

where u is an eigenfunction of ΓS. We have

$$0 = (\Gamma S - \lambda I)u \tag{7.17}$$

$$= \sum_{k \neq n} x^k \left(\sum_i S_i u_{k-i} - \lambda u_i \right) + x^n \left(a \sum_i S_i u_{n-i} - \lambda u_n \right) \tag{7.18}$$

which leads to the system of equations

$$\lambda u_n = a \sum_i S_i u_{n-i} \,, \tag{7.19}$$

$$\lambda u_k = \sum_i S_i u_{k-i}, \quad \text{if } k \neq n \,. \tag{7.20}$$

The analysis is non-trivial even for a simple function S like $S = x$. We have

$$\lambda u_n = a u_{n-1} \,, \tag{7.21}$$

$$\lambda u_k = u_{k-1}, \quad \text{if } k \neq n \,. \tag{7.22}$$

We have a doubly-infinite recursion so we can pick an arbitrary initial condition. If $u_n = 0$ the rest of the terms will be zero as well so we must have $u_n \neq 0$. Set $u_n = 1$. Then via induction we can find

$$u_{n+k} = 1/\lambda^k, \quad \text{for } k \geqslant 0 \,, \tag{7.23}$$

$$u_{n-k} = \lambda^k / a^k, \quad \text{for } k \geqslant 1 \,. \tag{7.24}$$

Computing $u(x)$ formally leads to

$$u(x) = \sum_{k=1}^{\infty} u_{n-k} x^{n-k} + \sum_{k=0}^{\infty} u_{n+k} x^{n+k} \tag{7.25}$$

$$= x^n \sum_{k=1}^{\infty} \left(\frac{\lambda}{ax} \right)^k + x^n \sum_{k=0}^{\infty} \left(\frac{x}{\lambda} \right)^k \tag{7.26}$$

$$= \frac{x^{n-1} x}{ax - \lambda} - \frac{x^n \lambda}{x - \lambda} \tag{7.27}$$

$$= \lambda x^n \left(\frac{1}{\lambda - x} - \frac{1}{\lambda - ax} \right) \tag{7.28}$$

$$= -\frac{\lambda b x^{n+1}}{(\lambda - x)(\lambda - ax)} \,. \tag{7.29}$$

We could also try using the other definition. Let $u = u_n x^n + v$ where u is now an eigenfunction of $S\Gamma$. Let $S = S_0 + S'$. Then we have

$$S\Gamma u = (S_0 + S')(au_n x^n + v) = \lambda(u_n x^n + v) \tag{7.30}$$

$$0 = (au_n S_0 - \lambda u_n + \sum_{k \neq n} S_k u_{n-k})x^n + au_n S' x^n + S'v - \lambda v\,. \tag{7.31}$$

We find that for example

$$(\lambda - aS_0)u_n = \sum_{k \neq n} S_k u_{n-k} \tag{7.32}$$

which doesn't seem to help much.

7.3 Random Walks Operator

In this section we explore the operator $\mathcal{L} = 1 - t\Gamma S$ for a step Laurent polynomial S and some random walk operator Γ. We haven't made it rigorous yet but $\mathcal{L}$ is invertible with inverse

$$\mathcal{L}^{-1} = \sum_{k=0}^{\infty} t^k (\Gamma S)^k \tag{7.33}$$

where $(\Gamma S)^k$ is an iteration of the inhomogeneous operator with $(\Gamma S)^0 = 1$. However this inverse is not easy to use as the iteration become quite complicated due to the nature of the random walks. Let (u, λ) be an eigenpair for $\mathcal{L}$.Then we calculate

$$(\mathcal{L} - \lambda I)u = u - \lambda u - t\Gamma S u \tag{7.34}$$

$$t\Gamma S u = (1 - \lambda)u = \mu u\,. \tag{7.35}$$

If $\lambda = 1$ we seek to find the kernel of ΓS which we did in the previous section. The eigenfunctions of $\mathcal{L}$ for eigenvalue $\lambda = 1$ are the functions $u \in \overline{S} \ker \Gamma$.

Next let u be given by a finite order Laurent series

$$u = u_n t^n + u_{n+1} t^{n+1} + \dots \tag{7.36}$$

where $u_n \neq 0$. Then we can re-write the functional equation as

$$t^{-n}(1 - t\Gamma S)u = \lambda t^{-n} u\,. \tag{7.37}$$

Letting $t \to 0$ in this equation yields

$$(\lambda - 1)u_n = -\lim_{t\to 0} t^{1-n}\Gamma S u = -\Gamma S \lim_{t\to 0}(t^{1-n}u) = 0\,. \tag{7.38}$$

Since $u_n \neq 0$ this implies that $\lambda = 1$ and again we see that $u \in \overline{S}\ker\Gamma$. Since S is a Laurent polynomial, the resulting functions will indeed be finite order Laurent series. For the remainder of the analysis we assume that $\lambda \neq 1$. This excludes finite order Laurent series so u is necessarily given by an infinite order Laurent series.

7.4 Interaction with ∂_t

Here we'll investigate ∂_t interacts with $\mathcal{L}$.

$$\partial_t\mathcal{L} = \partial_t - \partial_t(t\Gamma S) = \partial_t - \Gamma S - t\Gamma S\partial_t = \mathcal{L}\partial_t - \Gamma S\,, \tag{7.39}$$

$$\partial_t^2\mathcal{L} = \mathcal{L}\partial_t^2 - 2\partial_t\Gamma S\,, \tag{7.40}$$

$$\partial_t^n\mathcal{L} = \mathcal{L}\partial_t^n - n\partial_t^{n-1}\Gamma S \tag{7.41}$$

for integer n. Another way to write these is with bracket notation

$$[\mathcal{L}, \partial_t^n] = \mathcal{L}\partial_t^n - \partial_t^n\mathcal{L} = n\partial_t^{n-1}\Gamma S\,. \tag{7.42}$$

7.5 Interaction with ∂_i

Let's take the operator $\Gamma_n(a) = 1 + bx^n[x^n]$ again. Let $\partial_k = \partial/\partial_k$ and in this case just ∂. We have

$$\partial\Gamma_n(a) = \partial + b\partial(x^n[x^n]) = \partial + bnx^{n-1}[x^n]\,. \tag{7.43}$$

Now if $n \geqslant 0$ we have

$$\partial^{n+1}\Gamma_n(a) = \partial^{n+1} + b\partial^{n+1}(x^n)[x^n] = \partial^{n+1}\,. \tag{7.44}$$

If $n \leqslant 0$ taking derivatives will yield ever greater powers of $\overline{x}$ so we want to pre-multiply by a power of x first.

$$\partial x^{-n}\Gamma_n(a) = \partial x^{-n} + b\partial[x^n] = \partial x^{-n} = -nx^{-n-1} + x^{-n}\partial\,. \tag{7.45}$$

If we have more terms in an operator then we need to multiply by a power of x to clear out any negative powers of x. Then we take the lowest order derivative in order to clear out all the extraction terms. Suppose that

$$\Gamma = 1 + \sum_{i=0}^{m} a_i x^i [x^i] - \sum_{i=1}^{n} b_i \overline{x}^i [\overline{x}^i] \tag{7.46}$$

where m and n are non-negative. We need to pre-multiply by x^n to clear out all the negative powers of x. Then we have

$$x^n \Gamma = x^n + \sum_{i=0}^{m} a_i x^{i+n} [x^{i+n}] - \sum_{i=1}^{n} b_i x^{n-i} [x^{n-i}] \,. \tag{7.47}$$

This operator is cleared by ∂^{m+n+1}.

Chapter 8

Discussion on Random Walk Model Generalizations

8.1 Introduction

In this chapter we'll push the operator notation further to see what kinds of random walk models have a relatively easy functional equation to write. We start with the following definition for a random walk model in $\mathbb{Z}^d$.

Definition 35. *Let $\mathcal{D}_0 \subseteq \mathcal{D}$ be regions in $\mathbb{Z}^d$ and S be a set of d-tuples in $\mathbb{Z}^d$. A triple $\mathcal{M} = (\mathcal{S}, \mathcal{D}, \mathcal{D}_0)$ is a random walk model $\mathcal{M}$ with steps $\mathcal{S}$, living in the region $\mathcal{D}$ and interacting with region $\mathcal{D}_0$. We may impose a starting location later on.*

In other words, we are given a set of steps $\mathcal{S}$ and a starting location in $\mathcal{D}$. Then we want to evolve the random walks restricted to the region $\mathcal{D}$. We also have some special locations $\mathcal{D}_0$ within $\mathcal{D}$ and we wish to count the interactions of the random walks with points in this region. The interactions may be further specified later on.

Definition 36. *Let $\mathcal{M}$ be a random walk model and Q its associated generating function counting length with t, ending locations with $x_1, \ldots, x_d$ and interactions with $\mathcal{D}_0$ with some multi-index variable a. We write*

$$Q = Q(x_1, \ldots, x_d, a; t) = \sum_{k=0}^{\infty} q_k(x_1, \ldots, x_d, a) t^k \tag{8.1}$$

where q_k are Laurent poynomials in the x and a variables.

Definition 37. *Suppose D is an operator involving the x and a variables but not t. We shall apply D on power series $F = \sum_{k=0}^{\infty} f_k t^k$ in the following linearized way*

$$DF = \sum_{k=0}^{\infty} D(f_k) t^k \tag{8.2}$$

to avoid issues of expanding F in Laurent series in some of the x variables.

Definition 38. *Let $\mathcal{M}$ be a given model. We'll denote with S the Laurent polynomial (or series) keeping inventory of the stes $\mathcal{S}$. Let $K = 1 - tS$ be the Kernel of $\mathcal{M}$. Let E be an operator that eliminates monomials $x_{[d]}^P$ when P is not in $\mathcal{D}$. Let L be an operator that marks (or labels) monomials $x_{[d]}^P$ for $P \in \mathcal{D}_0$ in a specified way in the model. Note that L is invertible with inverse L^{-1} that is given by marking with $1/a$ instead.*

This operator notation allows us to write down functional equations for Q really quickly. We have

Theorem 39. *Let $\mathcal{M}$ be a random walk model and Q its associated generating function with an initial term $q_0 = q$. Then Q satisfies the following functional equations*

$$(1 - tLES)Q = q \tag{8.3}$$

$$(L^{-1} - tES)Q = L^{-1}(q) = \alpha \tag{8.4}$$

Proof. The second functional equation is obtained by applying L^{-1} on the first one. It only remains to show that the first functional equation holds. We do this via iteration. We start with Q and multiply it with tS. This gives tSQ or an enumeration of all walks extended by a single step. Here the power of t is raised by one to keep track of this extension. Then we apply E to this result to eliminate all random walks that venture outside of the region $\mathcal{D}$. Finally we mark those walks that have arrived at locations in $\mathcal{D}_0$ by applying L to $E(tSQ)$. The resulting expression is almost Q but with one term missing, namely $q_0 = q$. That is we have transformed Q to itself in a non-trivial way giving us the functional equation

$$Q = q + LE(tSQ) = q + tLE(SQ) \tag{8.5}$$

which we re-write in the form at the start of the theorem. □

Corollary 40. *We have the following kernel forms*

$$KQ = q - t(1 - LE)SQ \tag{8.6}$$

$$KQ = \alpha + (1 - L^{-1})Q - t(1 - E)SQ. \tag{8.7}$$

If there is no extraction involved we have the kernel forms

$$KQ = q - t(1 - L)SQ \tag{8.8}$$

$$KQ = \alpha + (1 - L^{-1})Q. \tag{8.9}$$

If there is no marking involved then we have the simple kernel form

$$E(KQ) = q. \tag{8.10}$$

The kernel forms are obtained as before since it was mostly a symbollic calculation. We use the fact that $EQ = Q$ since Q is already composed of terms living in $\mathcal{D}$. We won't consider more general functional equation statements in this work. It is possible to also consider step sets $\mathcal{S}$ varrying with location as we saw in the honey-comb lattice chapter but writing a functional equation for this type of problem will be left as a (rather long) exercise to the reader.

The second form of the functional equation involving L^{-1} should have less terms than the first one by virtue of separating L and E from one another. Using extraction operators we can constrain random walks to lie within a finite region, semi-infinite strip, a semi-infinite sector or any other discrete shape (with diminishing returns for more complicated shapes of course). Furthermore using marking operators we can easily track the behavior of random walks as they cross or interact with boundaries or interact with regions within. For example we may consider walks interacting with the boundary of a box within some region in space.

Example 41. We can also consider a different approach where instead of removing walks that venture outside of $\mathcal{D}$ we just mark them with some variable z for later deletion. Suppose we are given some $\mathcal{M}$ and we want to frame in this new way. We have a new random walk model $\widetilde{\mathcal{M}}$ with $\widetilde{\mathcal{D}} = \mathbb{Z}^d$ and $\widetilde{\mathcal{D}}_0 = \mathcal{D}_0 \cup \mathcal{D}^c$ where we mark monomials in $\mathcal{D}^c$ with z and those in $\mathcal{D}_0$ as in $\mathcal{M}$. Here $\widetilde{E}$ is the identity since we don't remove any walks. We have the functional equations

$$(1 - t\widetilde{L}S)\widetilde{Q} = q \tag{8.11}$$

$$(\widetilde{L}^{-1} - tS)Q = \widetilde{\alpha} \tag{8.12}$$

and kernel forms

$$K\widetilde{Q} = q - t(1 - \widetilde{L})S\widetilde{Q} \tag{8.13}$$

$$K\widetilde{Q} = \widetilde{\alpha} + (1 - \widetilde{L}^{-1})\widetilde{Q}. \tag{8.14}$$

where $\widetilde{Q} = Q + O(z)$ wher Q is the generating function associated with model $\mathcal{M}$. We just need to extract $[z^0]$ from $\widetilde{Q}$ to obtain Q.

We can talk about some of the abstract properties of the operators L and E. The operator E has the eigenvalues 1 or 0 for monomials $x^P_{[d]}$ where $P \in \mathcal{D}$ or $P \in \mathcal{D}^c$ respectively. The latter characterizes the monomials contained in $\ker E$. Similarly L has eigenvalues 1 for when $P \in \mathcal{D}^c_0$ and the marking monomial a for when $P \in \mathcal{D}_0$. Here a will varry from point to point. Also L^{-1} will have the reciprocal eigenvalues of L.

The task for the rest of the chapter is to come up with representations for the operators involved. In general this should not be an easy task because if we could express arbitrary operators L and E with a small number of coefficient extractions we'd also be able to use this scheme to compress data without loss, which is impossible in general.

8.2 Marking Operators

The example in the previous section shows that we can talk about marking operators on their own without considering operators E. Alternatively, the operators E are just marking operators which apply a label $a = 0$. Therefore we can combine the discussions for operators L and E into one holistic approach. Whenever invertibility is concerned the discussion will apply to operators L only. Suppose that we wish to mark points $p \in P \subseteq \mathbb{Z}^d$ with labels a_p via an operator Γ_P. It can be written as

$$\Gamma_P = 1 + \sum_{p \in P} (a_p - 1) x^p_{[d]} [x^p_{[d]}] \tag{8.15}$$

The form $a_p - 1$ follows from the fact that we have the original term and we need to remove it before marking with a_p. In the case where we mark points for deletion $P = \mathcal{D}^c$ and $a_p = a$. The good thing is that in a lot of cases we don't need all the terms in the operator and in some cases we even find the operator has a portion that factors. We just need to ensure that we mark all the walks that touch the boundaries or go within the region marked for deletion. For example we can take the walks in one dimensions on the non-negative axis with steps of size at most n in the negative

direction. Then we can demonstrate the truncation with

$$\Gamma = 1 + (a-1)\sum_{k=1}^{\infty} \overline{x}^k[\overline{x}^k] = 1 + (a-1)\sum_{k=1}^{n} \overline{x}^k[\overline{x}^k] \tag{8.16}$$

We only need to consider how far a walk can step outside of the region $\mathcal{D}$ via the step set in S. In two dimensions we have

$$\Gamma = 1 + (a-1)\sum_{m \text{ or } n<0}^{\infty} \overline{x}_1^m \overline{x}_2^n[\overline{x}_1^m \overline{x}_2^n] \tag{8.17}$$

Without a this reduces to

$$\Gamma = 1 - \overline{x}_1[\overline{x}_1] - \overline{x}_2[\overline{x}_2] + \overline{x}_1\overline{x}_2[\overline{x}_1\overline{x}_2] = (1 - x[x]_1)(1 - x[x]_2) \tag{8.18}$$

in which case we also see that the operator can factor into two or more operators sometimes. When the boundary is complicated we may not find an a finite form for Γ let alone see any factoring. However, there are some nice special cases where we can say a lot about the operator Γ. We can build the theory. We have only finitely many steps (and sizes of steps) in S. We say Γ is finite if it can be represented by finitely many coefficient extraction terms. Suppose that $\mathcal{D}$ is a finite box in $\mathbb{Z}^d$. Then Γ is finite. To see this suppose that B has x_k coordinates in the interval $[a_k, b_k]$ and that stepping outside from B from within B via step polynomial S leads to x_k coordinates within the interval $[m_k, n_k]$. Then Γ can be written finitely with extraction part

$$\Gamma_{B^c} = \sum_{k=1}^{d} \left(\sum_{i=m_k}^{a_k-1} x_k^i[x_k^i] + \sum_{i=b_k+1}^{n_k} x_k^i[x_k^i] \right) \tag{8.19}$$

The discussion continues for semi-infinite and infinite boxes with finite marked regions within. The analysis is the same as above except that now one or more of a_k, b_k may be infinite, in which case we actually lose parts of Γ_{B^c} and the operator is still finite. We should see if we can do regions whose complements are boxes and then build up to finite regions and intersections of finite regions with boxes.

Definition 42. *A coefficient extraction operator Γ in the variables $x_1, \ldots, x_d$ is of finite type if it can be written in a form with only finitely man terms like*

$$\Gamma = \sum_{\substack{I \subseteq [d] \\ J \subseteq \mathbb{Z}^d}} a_{I,J} x_I^J[x_I^J] \tag{8.20}$$

where each $|J| < \infty$. Then Γ is of infinite type if it's not of finite type. We'll just say Γ is finite or infinite.

Definition 43. *A random walk model with finitely many steps is of finite type if its extraction and marking operator Γ is of finite type and its starting condition q is of finite degree. If not, the model is of infinite type.*

Not surprisingly all the models so far has been of finite type. In the next section we'll see the model of self-avoiding walks is most likely of infinite type.

Theorem 44. *The generating function Q for a random walk model of finite type is D-finite in the variables $x_1, \ldots, x_d$.*

Proof. We'll use the kernel form for the functional equation

$$KQ + t(1 - \Gamma)SQ = q. \tag{8.21}$$

Here $\Gamma' = 1 - \Gamma$ is purely a coefficient extractor and is of finite type. We'll prove this by finding the differential equation explicitly. Since the model is of finite type we can bound the degree of x_k that we see on either side of the functional equation. In one scheme we let A_k be the largest power of x_k that we need to clear a negative power of x_k in Γ' and in q. Then we let B_k be the largest power of x_k that we see on either side. Once we multiply by $x_k^{A_k}$ we see that the degree of x_k in the equation is $A_k + B_k$. We take the functional equation and multiply both sides by $x_{[d]}^A$

$$x_{[d]}^A(KQ + t\Gamma' SQ) = x_{[d]}^A q \tag{8.22}$$

and we obtain a polynomial equation in the variables $x_1, \ldots, x_d$. We take the derivative operator

$$D = \partial_{[d]}^{A+B+\mathbb{I}} = \partial_1^{A_1+B_1+1} \ldots \partial_d^{A_d+B_d+1} \tag{8.23}$$

and apply it to both sides of the equation to get

$$D(x_{[d]}^A(KQ + t\Gamma' SQ) = D(x_{[d]}KQ) = D(x_{[d]}^A q) = 0 \tag{8.24}$$

which follows since we're taking more derivatives than the degrees of the monomials which clears all of them. The other scheme lets us obtain a potentially lower degree differential equation at the cost of it beind inhomogeneous. Let A_k be the power of x_k that we need to clear denominators on the LHS of the functional equation and B_k be the largest power of x_k on that side. Then we have

D as before but this time it may have lower A and B values since we are not taking q into account. We have

$$D(x^A_{[d]}(KQ + t\Gamma' SQ) = D(x_{[d]}KQ) = D(x^A_{[d]}q) \tag{8.25}$$

□

For example let the powers of x_k be $\{-3, -1, 4\}$ and $\{-2, 0, 6\}$ on the LHS and RHS respectively. We see that $A_k = 3$ and $B_k = 6$. Then we're taking the $A_k + B_k + 1 = 10$-th derivative of both sides of the functional equation. In The second scheme we take $A_k = 3$ still but $B_k = 4$ and we are taking the 8-th derivative w.r.t. x_k this time. Even in the quarter-plane we have models that have a generating function that is not D-finite (with respect to the t variable) but it turns out that every model so far considered is D-finite in the $x_1, \ldots, x_d$ variables.

In order to solve the differential equation we need to investigate the kernel of the differential operator.

Corollary 45. *The generating function Q of a finite random model is given by*

$$x^A_{[d]}KQ(x_1, \ldots, x_d; t) = \sum_{i=1}^{d} \sum_{j=0}^{B_i} Aij(x_1, \ldots, x_{i-1}, x_i, \ldots, x_d, t)x_i^j \tag{8.26}$$

where A_{ij} are functions that need to be determined.

We could have also obtained this kind of solution from going through the calculations in the functional equation and extracted coefficients. However, this method is more efficient. If we had used the second scheme in the proof we would have ended up with a slightly different form but maybe it would have had a slightly lower degree. It would involve an indefinite integral of q and elements in the kernel of the differential operator.

8.3 Self-avoiding walks

We can handle self-avoiding random walks with this theory. If we pick a marking operator Γ that marks every location on the plane this will allow us to track the random walks' history. All we'll need to do in the end is remove those walks which have stepped at some location more than once. However we'll do this problem in a bit more detail as it serves as an example of the theory. To use the generating function iteration properly we need a sort of limiting argument. For the first n steps we "only" require an operator covering the square region of $(2n + 1)^2$ lattice points. The overall operator is defined through a limiting process of applying the full operator (or truncated version) on

a finite Laurent polynomial. Let

$$\Gamma_{i,j} = 1 + b_{i,j} x_1^i x_2^j [x_1^i x_2^j] \tag{8.27}$$

where as usual $b_{i,j} = a_{i,j} - 1$. This operator takes a generating function and marks the monomial $x_1^i x_2^j$. We'll be using a lot of these together so we let

$$\Gamma_n = \prod_{-n \leqslant i,j \leqslant n} \Gamma_{i,j} \tag{8.28}$$

To find a functional equation for SAWs on the plane we perform an iteration. Let Q be the generating function for walks with history, that is they have a mark for each coordinate they pass through and let $Q^{\leqslant n}$ be the truncated version of this generating function. It's given by

$$Q^{\leqslant n} = \sum_{k=0}^{n} Q_k t^k \tag{8.29}$$

where the initial condition is $Q_0 = a_{0,0}$. Starting from Q_n we can find Q_{n+1} by iterating with the step set S, marking with Γ_{n+1} as follows

$$Q_{n+1} = \Gamma_{n+1} S Q_n \tag{8.30}$$

Using a computer we can easily iterate this to get more and more steps. We can also get $Q^{\leqslant n+1}$ from $Q^{\leqslant n}$ since

$$t\Gamma_{n+1} S Q^{\leqslant n} = \sum_{k=0}^{n} \Gamma_{n+1} S Q_k t^{k+1} \tag{8.31}$$

$$= \sum_{k=0}^{n} \Gamma_{k+1} S Q_k t^{k+1} \tag{8.32}$$

$$= \sum_{k=0}^{n} Q_{k+1} t^{k+1} \tag{8.33}$$

$$= Q^{\leqslant n+1} - Q_0 \tag{8.34}$$

or in other words

$$Q^{\leqslant n+1} = a_{0,0} + t\Gamma_{n+1} S Q^{\leqslant n} \tag{8.35}$$

In the interest of finding a functional equation we would like to take the limiit of this as $n \to \infty$. For this we define Γ to be the operator marking all monomials $x_1^i x_2^j$ and acting in the following way

$$\Gamma x^i y^j = \Gamma_{\max(|i|,|j|)} x^i y^j \tag{8.36}$$

and extended by linearity. The limiting process is

$$Q = a_{0,0} + t\Gamma S Q \tag{8.37}$$

We finally have the functional equation for random walks with full history

$$(1 - t\Gamma S)Q = a_{0,0}. \tag{8.38}$$

From this we can obtain the generating function W for the SAWs by removing all walks with power of mark $a_{i,j}$ greater than 1. For this we need a removal operator. Let

$$R_{i,j} = a^0_{i,j}[a^0_{i,j}] + a_{i,j}[a_{i,j}] \tag{8.39}$$

$$R_n = \prod_{-n \leqslant i,j \leqslant n} R_{i,j} \tag{8.40}$$

$$R = \lim_{n \to \infty} R_n \tag{8.41}$$

where the limit is understood as in the example of Γ above. The solution for Q can also be restated as

$$Q = \sum_{k=0}^{\infty} t^k (\Gamma S)^k (a_{0,0}) = \sum_{k=0}^{\infty} t^k (\Gamma_k S)^k (a_{0,0}) \tag{8.42}$$

We can either redo the analysis applying R at every iteration or we can apply R to Q and get

$$W = RQ = \sum_{k=0}^{\infty} t^k R(\Gamma S)^k (a_{0,0}) = \sum_{k=0}^{\infty} t^k (R\Gamma S)^k (a_{0,0}) \tag{8.43}$$

We can write this as a functional equation

$$\mathcal{L}W = (1 - t\Sigma)W = (1 - tR\Gamma S)W = a_{0,0} \tag{8.44}$$

8.3.1 Eigenpairs

We can try finding eigenvalues and eigenfunctions of $\mathcal{L}$. Let $\mathcal{L}u = \lambda u$. Writing it out leaves us with

$$t\Sigma u = (1 - \lambda)u \tag{8.45}$$

So if $\lambda = 1$, $u \in \ker(\Sigma)$. That is $R\Gamma Su = 0$. In other words ΓSu has only terms that involve powers greater than 1 of $a_{i,j}$ for various i, j. Suppose that

$$\Gamma Su = a_{i,j}^2 f(x_1, x_2) \tag{8.46}$$

Then

$$u = \frac{a_{i,j}^2}{S}\Gamma^{-1}(f) \tag{8.47}$$

So we here we can let

$$f = a_{k,l}^m x_1^r x_2^s t^n \tag{8.48}$$

and we find that

$$u = \frac{a_{i,j}^2 a_{k,l}^m}{a_{r,s} S} x_1^r x_2^s t^n \tag{8.49}$$

Thus the kernel of Σ is generated by these functions for all coefficients $i, j, k, l, r, s, n \in \mathbb{Z}$ and $m \geqslant 0$. This may perhaps prove useful later on. Now, if we look at eigenvalues $\lambda \neq 1$ we have to solve the equation

$$t\Sigma u = (1 - \lambda)u \tag{8.50}$$

Because of the t, if u is a finite order laurent series in t with first non-zero coeficient u_n. We find that since Σ does not affect the t monomials

$$\Sigma u_n = \mu u_{n-1} = 0 \tag{8.51}$$

and this forces all u_k coefficients to be zero. Therefore Σ has no other eigenvalues except 1 among the finite order power series in t.

8.3.2 Generalizations

The discussion lends itself nicely to generalization where now we have d variables and operators $\Gamma_{i_1,\ldots,l_d}$, $R_{i_1,\ldots,i_d}$, etc defined analogously. The eigenpair discussion works out in the same way with eigenfunctions for $\lambda = 1$

$$u = \frac{a^2_{i_1,\ldots,i_d} a^m_{j_1,\ldots,j_d}}{a_{k_1,\ldots,k_d} S} x_1^{r_1} \ldots x_d^{r_d} t^n \tag{8.52}$$

We can also constrain the SAWs on a part of $\mathbb{Z}^d$ with an extraction operator E and mark special locations with an operator L. The functional equation is

$$(1 - tLE\Sigma)W = a_{0,\ldots,0} \tag{8.53}$$

for arbitrary number of dimensions, with region constraints and marking of special locations.

Chapter 9

Algebraic Kernel method

9.1 Groups for walks with small steps

Suppose that the random walk model has a step polynomial S in the variables $x_1, \ldots, x_d$. We are interested in the bi-rational transformations that leave the kernel $K = 1 - tS$ intact. For some $x = x_k$ we have bi-rational transformations sending $g : x \mapsto y$ that preserse K. That is $K(y) = K(x)$. Expanding the definition we have

$$1 - tS(y) = 1 - tS(x) \tag{9.1}$$

$$S(y) = S(x) \tag{9.2}$$

$$S_{-1}\overline{x} + S_1 x = S_{-1}\overline{y} + S_1 y \tag{9.3}$$

Because of this equation it's equally valid to speak of the birational transformations preserving S. Collecting terms and simplifying yields

$$(S_{-1} - S_1 xy)(x - y) = 0 \tag{9.4}$$

There are cases here. If both S_{-1}, S_1 are zero the equation is trivially valid and g can be anything. When exactly one is zero then the equation only has the trivial solution $y = x$. When neither is zero we have the two solutions $y = x$ and $y = \overline{x}S_{-1}/S_1$ in which case we say that the transformation is non-trivial. That is we define $g_k = \overline{x}_k S_k^{-1}/S_k^1$ whenever we have a non-trivial transformation. Note that

$$g_k^2(x_k) = g_k\left(S_k^{-1} x_k S_k^1\right) = x_k \tag{9.5}$$

so these transformations are involutions. We can define a group G_S of the non-trivial transformations.

Definition 46. *Let $G = G_S$ be the group generated by the non-trivial transformations g_k.*

In the case where the transformation on x is trivial there is something we can say about the random walk model. We use the operator $E = 1 - \overline{x}[\overline{x}]$ to gain information on the generating function Q. Since Q only has non-negative powers of x in its expansion $EQ = Q$. However in the master functional equation we have

$$ESQ = SQ - \overline{x}(S_{-1}Q_1 + S_0Q_0 + S_1Q_{-1}) = SQ - \overline{x}(S_{-1}Q_1 + S_0Q_0) \tag{9.6}$$

We have several cases

1) If $S_{-1} = 0$ and $S_1 = 0$, $S = S_0$ and therefore $ESQ = S_0EQ = S_0Q$ so the functional equation doesn't "see" the variable x.

2) If $S_{-1} = 0$ and $S_1 \neq 0$ then

$$ESQ = SQ - \overline{x}(S_{-1}Q_0 + S_0Q_{-1}) = SQ \tag{9.7}$$

since $Q_{-1} = 0$ so the extraction doesn't see the variable x. This is equivalent to saying that even though we are working in the positive orthant we can relax the constraint for the x to the whole space since there are no steps going back in that direction. The variable x is still present in the functional equation. If we have no steps that stay in the $x = 0$ hyperplane then x essentially acts as a parameter in the problem and we can ignore it. Otherwise we'll have

$$\begin{aligned} LESQ = LSQ &= (1 + b[x^0])SQ && (9.8)\\ &= SQ + b(S_{-1}Q_1 + S_0Q_0 + S_1Q_{-1}) && (9.9)\\ &= SQ + bS_0Q_0 \neq SQ. && (9.10) \end{aligned}$$

3) If $S_{-1} \neq 0$ but $S_1 = 0$ then ESQ is non-trivial but since we cannot take any steps away from the $x = 0$ hyperplane we can eliminate the S_{-1} term in the analysis of the problem.

In all the cases the random walk model behaves as if $S_{-1} = 0$. In the non-interacting version of the problem we can fully ignore the x coordinate in the analysis since it acts like a parameter. The problem is essentially one of lower dimension. In the interacting case we have a bit extra because of the marking operator. In either case we'll define the dimension of the group to be the number of non-trivial generators.

Definition 47. *Let the* dimension *of the model or the group be the number of non-trivial generators of G_S in the defining representation.*

Without loss of generality we will speak of models set in d dimensions that have a full set of generators. Firstly, we can easily show that there are infinitely many different group sizes that are manifested among the different dimensions. Let S be a $(d-1)$-dimensional model with group G of largest finite order (number of elements) for that dimension. Set this model in d dimensions and define a new model $S' = S(x_d + \overline{x}_d)$. This model has a non-trivial generator g_d that S lacks. Therefore the group G' of this new model satisfies

$$G' \cong G \times \langle g_d \rangle \tag{9.11}$$

and $|G'| = 2|G|$ and continuing in this manner allows us to write an infinite sequence of distinct groups. That is, not only are there infinitely many distinct groups but also, there is a group in d dimensions not encountered before in $(d-1)$ dimensions (neither in order nor structure). For example the orders of the finite groups set in 2 dimensions are 1, 2, 4, 6 and 8. So in 3 dimensions we have groups of order 1, 2, 4, 6, 8, 12 and 16 at least. In fact Steve's paper shows that there are even more - of orders 24 and 48 too.

This discussion would be short without delving into the subject of Coxeter groups. The d-dimensional groups defined above are Coxeter groups - a generalization of the dihedral groups encountered in dimension 2 random walk models. Coxeter groups are defined as

$$G = \langle r_1, \ldots, r_d | (r_i r_j)^{m_{i,j}} = 1 \rangle \tag{9.12}$$

where $m_{i,i} = 1$ and $m_{i,j} \geqslant 2$ for all $i \neq j$, and $m_{i,j}$ could also be infinite. We can call these r's reflections. When $m_{i,j} = 2$ the reflections commute. Also $m_{i,j} = m_{j,i}$. The *rank* of G is the number of generators of in this specific representation of G. Often G will have more than one representation with different numbers of generators. So in our discussion of groups for random walks with small steps the rank of G is the dimension of the model. We can draw a graph for each Coxeter group

- label vertices with the generators of G
- connect the vertices g_i and g_j if $m_{i,j} \geqslant 3$
- label the edge with $m_{i,j}$ if $m_{i,j} \geqslant 4$.

If the graph has different components, the Coxeter group factors as a direct product of Coxeter groups represented by the individual components. We can see this in terms of the step polynomial

S. Suppose that S factors as

$$S = T(x_1, \ldots, x_k)R(x_{k+1}, \ldots, x_d) \tag{9.13}$$

For a fixed $1 \leqslant i \leqslant k$ we have

$$S_i^j = [x_i^j]S = R[x_i^j]T = RT_i^j \tag{9.14}$$

and therefore

$$g_i(x_i) = \overline{x}_i \frac{T_i^{-1}R}{T_i^1 R} = \overline{x}_i \frac{T_i^{-1}}{T_i^1} \tag{9.15}$$

Similarly if $k + 1 \leqslant j \leqslant d$ then

$$g_j(x_j) = \overline{x}_j \frac{R_i^{-1}}{R_i^1} \tag{9.16}$$

Clearly

$$g_i g_j = g_j g_i \tag{9.17}$$

since they don't interfere with each other. Since i and j were arbitrary, this means that the generators from different sets commute and the Coxeter graph of the group factors into two pieces. We can write

$$G_S = G_T \times G_R \tag{9.18}$$

where the subscripts highlight which parts of S they groups are generated by. This factoring also allows us to write the generating function Q as a Hadamard product of lower-dimensional generating functions. We can call each component of the graph a prime model. We can say that a simple small component is one whose group doesn't split non-trivially (its graph has one component).

9.2 Non-interacting Models

9.2.1 Walks with small steps

At the core of the Algebraic Kernel method we have the idea that we can apply group actions from G onto the master functional equation and obtain non-trivially different equations for each $g \in G$.

After we sum these equations in a nice way we manage to eliminate the functions Q_1^0, Q_2^0 and $Q_{1,2}^{0,0}$ and their orbits leaving only the orbit of the kernel term KQ and a term on the right-hand side. What makes this possible is pre-multiplying the whole equation by $x_1 \dots x_2$. This also works in higher dimensions as we'll show later. Although the generalization has been done before we will present it in a compact and easy-to-generalize way. Instead of thinking just in terms of equations and summing them we'll encapsulate the process into an operator D_G that will take the functional equation and eliminate the terms we don't want by applying the group actions in a nice way. We start by defining D_G in order to formalize the process in the Algebraic Kernel method.

Definition 48. *Let G be a group represented and generated by g_k's preserving the kernel K for a model for small steps. Then let D_G be the operator defined as*

$$D_G = \sum_{g \in G} (-1)^{|g|} g \tag{9.19}$$

in the sense of group actions. Here $|g|$ is the sign of the group element g.

In two dimensions we can obtain the whole group by iterating over the two generators 1, g_i, $g_j g_i$, $g_i g_j g_i$. It's cleaner to write g_{iji} instead of $g_i g_j g_i$ so we have the following notation

Definition 49. *For a sequence $L = (l_1, l_2 \dots)$ of positive integers define*

$$g_L = g_{l_1 l_2 \dots} = g_{l_1} g_{l_2} \cdots \tag{9.20}$$

whenever this makes sense - when the list is finite or when there is a convergence in the case of an infinite list.

In the case of a finite group in two dimensions we have $(g_1 g_2)^n = 1$ for some minimal n. This allows us to re-write D_G in a useful way.

Lemma 50. *For a two-dimensional random walk model with small steps possessing a finite group of order $2n$ and generators g_1, g_2 we can write the operator D_G as*

$$D_G = (1 + g_{ij} + \cdots + (g_{ij})^{n-1})(1 - g_i) \tag{9.21}$$

Proof. We can just check the identity by expanding it

$$\begin{aligned} D_G &= (1 + g_{ij} + \cdots + (g_{ij})^{n-1})(1 - g_i) && (9.22) \\ &= 1 - g_i + g_{ij} - g_{iji} + \cdots - (g_{ij})^{n-1} g_i && (9.23) \end{aligned}$$

giving us one of the possible orbits □

Here are some examples of small groups

Example 51. When $|G| = 4$ the elements commute so the orbit is $1, g_1, g_{12} = g_{21}, g_2$ and the operator is

$$D_G = 1 - g_1 - g_2 + g_{12} = (1 - g_1)(1 - g_2) \tag{9.24}$$

When $|G| = 6$ the orbit is $1, g_1, g_{21}, g_{121} = g_{212}, g_{12}, g_2$. We can write the operator as

$$\begin{aligned} D_G &= (1 + g_{12} + g_{1212})(1 - g_1) && (9.25)\\ &= 1 + g_{12} + g_{1212} - g_1 - g_{121} - g_{12121} && (9.26)\\ &= 1 - g_1 - g_2 + g_{12} + g_{21} - g_{121} && (9.27) \end{aligned}$$

We can generalize the lemma for a multi-dimensional group G.

Lemma 52. *Let $\mathcal{M}$ be a model with small steps in $\mathbb{Z}^d$ and let G be its asociated group. For a non-trivial generator g_k we can write the operator as $D_G = D_H(1 - g_k)$ for a subgroup H of G.*

Proof. Simpler proof: each element in the fixed representation of G either ends in g_k or doesn't. We have a bijection between the two sets by using the involution g_k applied on the right. The elements ending in g_k have opposite parity to those not ending on g_k so we can split G into two parts H and Hg_k. We can split the defining sum into two parts and keep track of the sign changes via $e_g = (-1)^{|g|}$. We have

$$\begin{aligned} D_G &= \sum_{g \in H} e_g g + \sum_{g \in H g_k} e_g g && (9.28)\\ &= \sum_{g \in H} e_g g + \sum_{g \in H} (-1) e_g g g_k && (9.29)\\ &= \left(\sum_{g \in H} e_g g \right) (1 - g_k) && (9.30)\\ &= D_H(1 - g_k) && (9.31) \end{aligned}$$

I'll have to provide a different proof maybe. Take $H = G/\langle g_k \rangle$. Since g_k is of order 2 the resulting group has $|G|/2$ elements. Consider the elements of G that commute with g_k, that is $h \in G$ where $g_k h = h g_k$. Multiplying by $g_k^{-1} = g_k$ we see that $h = g_k h g_k$. Let H be the subgroup of G generated by those elements. There are two cosets H and Hg_k. □

Lemma 53. *The operator D_G eliminates the term $tS_I^{-\mathbb{I}}Q_I^{\mathbb{O}}$ for $I \neq \emptyset$.*

Proof. By the previous lemma, for some $i \in I$, $D_G(tS_I^{-\mathbb{I}}Q_I^{\mathbb{O}}) = D_H((1-g_i)tS_I^{-\mathbb{I}}Q_I^{\mathbb{O}})) = D_H(0) = 0$ since the variable x_i is not present in the function. □

We obtain the d-dimensional generalization to the 2-dimensional Algebraic Kernel method.

Theorem 54. *For a model with small steps we have*

$$D_G(x_1 \cdots x_d KQ) = D_G(x_1 \cdots x_d) \tag{9.32}$$

where the RHS may be zero sometimes.

Proof. When we multiply the master functional equation by $x_1 \cdots x_d$ we obtain a lot of terms like the ones in the lemma. The only term that survives the application of D_G on the LHS is $x_1 \cdots x_d KQ$ and the equality follows. □

Corollary 55. *Suppose that there is a group element $h \in G$ with odd parity that permutes the variables $x_1, \ldots, x_d$ in some way and has order 2. Then the regular Algebraic Kernel method fails. That is*

$$D_G(x_1 \cdots x_d KQ) = 0. \tag{9.33}$$

Proof. Since h is an involution the same method applies as before and we can factor $D_G = D_H(1-h)$ for some $H \subseteq G$. The action $1-h$ eliminates the monomial $x_1 \cdots x_d$. □

Example 56. The 2D models where the Algebraic Kernel method fails in its straight-forward construction are precisely those where we have a permutation as in the corollary. That is $h(x_1, x_2) = h(x_2, x_1)$ and $|h| = 3$.

Example 57. In Steve's paper there is a model with group size 12 for which the Algebraic Kernel method fails. For that model one of the group element is a permutation of two of the variables - in that case x_2 and x_3. Strangely enough G contain the group seen in the previous example for the variables x_2 and x_3. This means that every higher-dimensional model that has group G containing the group above will fail in the regular Algebraic Kernel method.

A natural question is to consider taking D_H instead of $D_G = D_H(1-h)$ and checking if we can perfom a half-orbit summation in general. The answer is yes. The operatro D_H eliminates some of the Q variables and introduces others but the result is a function symmetric under permutation by h. One may hope to perform a method analogous to the half-orbit summation method to prove algebraicity or D-finiteness.

Example 58. There is another way we can prove that D_G does what it advertises. We start with the kernel form $EKQ = 1$ where we make the substitution $Q = W/(x_1 \cdots x_d K)$ and re-write the equation as

$$E'W = x_1 \cdots x_d, \qquad E'_k = 1 - [x_k^0] \tag{9.34}$$

Therefore

$$W = x_1 \cdots x_d - (1 - E')W = x_1 \cdots x_d - \sum_{\varnothing \neq I \subseteq [d]} W_I^{\mathbb{O}} \tag{9.35}$$

We immediately see that

$$D_G(x_1 \cdots x_d KQ) = D_G(W) = D_G\left(x_1 \cdots x_d - \sum_{\varnothing \neq I \subseteq [d]} W_I^{\mathbb{O}}\right) \tag{9.36}$$

$$= D_G(x_1 \cdots x_d) \tag{9.37}$$

since the $W_I^{\mathbb{O}}$ terms have missing variables. Let's think about this for a moment. The form of W is the same for all random walk models with small steps so if there are two models whose operators both annihilate expressions of the form $W_I^{\mathbb{O}}$ for all non-empty $I \subseteq [d]$ then both models have equations of the above type with each other's operators! In particular we may take the group G of the highly-symmetric models and apply it via the operator D_G for any other model and we obtain an equation of the above type! We'll look into this further.

Definition 59. *Let $g_k(x_k) = \overline{x}_k$. Define the operator $\Delta_k = 1 - g_k$ for this specific g_k. Let $\Delta = \Delta_1 \cdots \Delta_d$.*

Theorem 60. *Let G be the group generated by $g_k(x_k) = \overline{x}_k$ for $1 \leqslant k \leqslant d$. Let $\Delta_k = 1 - g_k$. Then we have $D_G = \Delta$ and*

$$\Delta(x_1 \cdots x_d KQ) = \Delta(x_1 \cdots x_d) = (x_1 - \overline{x}_1) \cdots (x_d - \overline{x}_d) \tag{9.38}$$

for any small step random walk model in dimension d with no interactions.

Proof. It's enough to see that Δ annihilates any term of the form $W_I^{\mathbb{O}}$ for $I \neq \varnothing$. Since the operatos Δ_k commute we only need to apply one Δ_i for some i first and find

$$\Delta_i(W_I^{\mathbb{O}}) = W_I^{\mathbb{O}} - g_i(W_I^{\mathbb{O}}) = 0 \tag{9.39}$$

since the term is missing variable x_i. The rest follows by linearity. □

This won't help show a specific model has a D-finite solution because in general we cannot factor out the K term from the LHS. However the idea is important and it allws us to write the following.

Corollary 61. *All random walk models of dimension d satisfy the functional equation $E'(x_1 \cdots x_d KQ) = x_1 \cdots x_d$. Therefore the only difference between the models is in the initial conditions that go with this functional equation.*

We'll extend these ideas in the next section but first we'll look into generalizing the ideas involved in the algebraic kernel method.

Definition 62. *Let G be a group acting on $x_1, \ldots, x_d$. Then we define a weighted orbit summation operator to be any operator of the form*

$$D = \sum_{g \in G} R_g g \tag{9.40}$$

where R_g are functions in the variables $x_1, \ldots, x_d$.

The orbit summation operator is an example of this with $R'_g = 1$ for the groups considered so far. We can capture various strategies in this notation

Example 63. Since the Algebraic Kernel method fails to give a nice result when G contains permutations we may try to apply the operator D_G on a slightly different equation - one where we multiply the whole equation by a function $T \neq x_1 \cdots x_d$. For a function f we have

$$D_G(Tf) = \sum_{g \in G} g(Tf) = \sum_{g \in G} (g(T))g(f) = \sum_{g \in G} R_g g(f). \tag{9.41}$$

The situation of applying D_G on T times the functional equation is the same as applying a weighted orbit summation operator on the original functional equation with weights $g(T)$ for $g \in G$.

We may also consider the strategy of applying a weighted operator several times in the hopes of obtaining an easier equation to work with. However as we shall see this is no different than looking at a weighted operator in the first place. This is because G is a group and the composition D_G^2 leads to another weighted operator. Also, since some of the operators D_G have non-trivial kernels they are not invertible, so the set of operators is not a group. Howeve we can show that the set is a monoid under composition.

Proposition 64. *For a group G, the set of weighted orbit summation operators form a monoid under composition.*

Proof. Since G is a group we have an identity element 1 which can also be considered as a weighted orbit summation operator with $R_g = 0$ for all $g \neq 1$ and $R_1 = 1$. Thus the set of operator has an identity element. Since composition of operators is associative just need to check that the set of operators is closed under composition. We have

$$D_1 D_2 = \sum_{g \in G} \sum_{h \in H} R_g g T_h h = \sum_{g \in G} \sum_{h \in H} R_g T_{gh} gh \sum_{g \in G} (\sum_{h \in H} R_{gh_{-1}} T_g) g \tag{9.42}$$

which is also an operator of this form once we collect terms corresponding to the same group elements. □

We hope to apply a weighted operator D to a problem in order to gain more insight into the structure of the generating function Q. Since all models with small steps and finite groups also have a regular orbit summation operator we have two main interesting cases to investigate. These are

(1) When is $D_G(x_1 \cdots x_d) = 0$ and can we find weights so that a new operator D that acts like D_G but spares $x_1 \cdots x_d$.

(2) Can we create a useful orbit summation operator when G is infinite.

We can answer the first question in the negative when it comes to un-marked random walk models. In general there is no other weighted orbital operator except the one with weights $R_g = 1$ that doesn't also eliminate $x_1 \cdots x_d$. The second case remain unclear.

Here is our strategy. If we apply the operator D on the master functional equation we need to eliminate all terms except KQ. That is D needs to annihilate all the Q terms that are missing variables. It is enough to work on the fundamental functional equation for $W = x_1 \cdots x_d KQ$. To go back to the master functional equation case we recognize that $x_1 \cdots x_d$ just make it a weighted operator and we can remove the weight by pre-dividing by $x_1 \cdots x_d$. The resulting operator is also weighted.

Lemma 65. *A weighted operator D exists such that $D(KQ) = D(1)$ if and only if a weighted operator D' exists such that $D'(f_I^{\mathbb{O}})$ for generic functions $f_I^{\mathbb{O}}$ that are missing the variables indexed by I, for all I.*

Proof. The discussion above provides the direction from left to right for $f = W$. To prove it for generic functions f it suffices to note that $f_I^{\mathbb{O}}$ and $f_J^{\mathbb{O}}$ are independent from each other if $I \neq J$.

That is, $D(f_I^{\mathbb{O}})$ and $D(f_J^{\mathbb{O}})$ do not share terms. The converse direction is proved by setting $f = W$ and setting $D = D'\overline{x}_1 \cdots \overline{x}_d$ where $(D'\overline{x}_1 \cdots \overline{x}_d)(f) = D'(\overline{x}_1 \cdots \overline{x}_d f)$. □

This essentially turns this into a linear algebra problem. We just need to apply D on generic functions $W_I^{\mathbb{O}}$ that are missing the variables indexed by I, collect the different results and make their coefficients zero. That is, for every $f = W_I^{\mathbb{O}}$ we look at the orbit

$$\mathrm{Orbit}_G(f) = \{g(f) | g \in G\} \tag{9.43}$$

and collect coefficients in

$$0 = D(f) = \sum_{g \in G} R_g g(f) = \sum_{f' \in Orbit(f)} R'_g f' \tag{9.44}$$

where we set

$$R'_g = \sum_{\substack{g(f)=f' \\ g \in G}} R_g = 0 \tag{9.45}$$

Thus we obtain a linear system of $|\mathrm{Orbit}_G(f)|$ equations in $|G|$ unknowns. Of course we can do this for all $2^d - 1$ terms $f = W_I^{\mathbb{O}}$ and hopefully obtain a system with a non-trivial solution. Here is a simple example

Example 66. In $d = 2$ dimensions we have the generic functions $W(0,0) = const$, $W(0, x_2)$, $W(x_1, 0)$ where at least one variable is missing. We can write them as $1, f(x_1), f(x_2)$. Now we consider the group generated by $g_k(x_k) = \overline{x}_k$

$$G = \{1, g_1, g_2, g_{12} = g_{21}\} \tag{9.46}$$

We'll write the operator D as

$$D = R - R_1 g_1 - R_2 g_2 + R_{12} g_{12} \tag{9.47}$$

We consider the different orbits. The one with the constant is trivial since the function doesn't change under the group action. Next we have

$$\mathrm{Orbit}_G(f(x_1)) = \{f(x_1), f(\overline{x}_1), f(\overline{x}_1), f(\overline{x}_1)\} = \{f(x_1), f(\overline{x}_1)\} \tag{9.48}$$

and similarly

$$\mathrm{Orbit}_G(f(x_2)) = \{f(x_2), f(\overline{x}_2)\} \tag{9.49}$$

This means we have the equations

$$R - R_1 + R_2 - R_{12} = 0 \tag{9.50}$$

$$R = R_2, \quad R_1 = R_{12} \tag{9.51}$$

$$R = R_1, \quad R_2 = R_{12} \tag{9.52}$$

Folowing the implications provided from these operators we find $R = R_1 = R_2 = R_{12}$. The simplest choice is to set $R = 1$ and we find that every weighted orbit summation operator for this group G must be trivially-weighted if it were to annihilate generic functions with missing variables.

Definition 67. *A weighted operator D is trivial if all the weights are the same. Then we call the operator unweighted or regular.*

We have the following lemma

Lemma 68. *Let G be a finite group with d generators for a random walk model in d dimensions with small steps and no interactions. Then any orbital operator is trivial.*

We need to check this proof.

Proof. It is enough to show that $D(W_k^0) = 0$ for all k implies that the operator is trivial. To this end suppose that $D(f) = 0$ where $f = W_k^0$. Then we can partition the group G as before into sets H and Hg_k. We see that $hg_k(f) = h(f)$ so the orbit of f contains at most $|G|/2$ elements. However these elements must all be distinct otherwise they would come from the same operations but the group elements in H are all distinct. So $|Orbit(f) = |G|/2$. This means that we have the equations $R_h = R_{hg_k}$ for every $h \in H$. Once we go through every k we have overlaps $R = R_1 = \cdots = R_d$ and $R_1 = R_{12} = R_{13}$ etc. This means that all the R's are the same and the operator is trivial. □

This shows that in general there is no clever weighted summation operator that also satisfies the condition $D(x_1 \cdots x_d) \neq 0$. However, the idea will be fruitful in that we may find non-trivial partial solution in the interacting case.

9.2.2 Walks with unconstrained steps

We may go through the process again but this time we take

$$Q = \frac{W}{x_{[d]}^N K} \tag{9.53}$$

where N_k is the maximum negative size of steps in the set $\mathcal{S}$ in the direction x_k or in other words the maximum N_k for which $S_k^{N_k}$ is non-zero. Then the operators that keep the walks in the quarter plane are

$$E_k = 1 - \sum_{i=1}^{N_k} \overline{x}_k^i [\overline{x}_k^i] \tag{9.54}$$

Their conjugates are

$$E_k' = x_k^{N_k} E_k \overline{x}_k^{N_k} = 1 - \sum_{i=0}^{N_k - 1} x_k^i [x_k^i] = 1 - \mathcal{E}_k \tag{9.55}$$

and so in the kernel equation we obtain

$$x_{[d]}^N = x_{[d]}^N E \overline{x}_{[d]}^N W = E'W = W + \sum_{\varnothing \neq I \subseteq [d]} (-1)^{|I|} \mathcal{E}_I(W) \tag{9.56}$$

Here we have various powers x_k^i along with terms with $x_k = 0$ for $k \in I$. So if we seek an operator D that does the same job as the orbit summation operator we need D to annihilate all the terms with x_k's missing in W on the RHS. So then we'll have

$$D(x_{[d]} K Q) = D(W) = D\left(x_{[d]}^N - \sum_{\varnothing \neq I \subseteq [d]} (-1)^{|I|} \mathcal{E}_k(W) \right) = D(x_{[d]}^N) \tag{9.57}$$

In other words we want D to behave (in some sense) similarly to the derivative

$$\partial_{[d]}^N = \partial_1^{N_1} \cdots \partial_d^{N_d} \tag{9.58}$$

This suggests some definitions.

Definition 69. *For a sequence $N = (N_1, \ldots, N_d)$ of positive integers we say that a model $\mathcal{M}$ is of type N that the longest step in $\mathcal{S}$ in the negative k-th direction has length N_k in that coordinate. In particular we may say that a model is of type n if $N_k = n$ for all k.*

For example, full random walks with small steps are of type 1.

Definition 70. *For sequences $N = (N_1, \ldots, N_d)$ and $M = (M_1, \ldots, M_d)$ of positive integers we say that a model $\mathcal{M}$ is of type (N, M) if it's of type N and the longest step in $\mathcal{S}$ in the positive k-th direction has length N_k in that coordinate. In particular we may say that a model is of type (n, m) is of type n and $M_k = m$ for all k.*

For example, full random walks with small steps are of type $1, 1$. The notation is still a little awful. Anyway, it allows us to write things in the following form

Theorem 71. *A random walk model of type N satisfies the functional equation*

$$E'W = x_{[d]}^N \tag{9.59}$$

where $E' = x_{[d]}^N E \overline{x}_{[d]}^N$ and $W = x_{[d]}^N KQ$.

Corollary 72. *For a random walk model of type N the generating function Q for the non-interacting case satisfies the PDE*

$$\partial_{[d]}^N \left(x_{[d]}^N KQ \right) = 1 \tag{9.60}$$

and is therefore D-finite in the x variables.

Here $\partial_{[d]}^N$ is an example of operator D that kills off all the unnecessary terms and doesn't kill off $x_1 \cdots x_d$. It's possible to discuss weighted operators in this setting too but the situation becomes more complicated. However we may investigate type (N, M) problems by finding a suitable model whose orbital operator is nice enough. This will show that all type (N, M) satisfy an orbital equation with that particular operator, not just a PDE.

Example 73. Consider the random walk model of type (N, N) with $S = \sum_{k=1}^d \overline{x}_k^{n_k} + x_k^{n_k}$. Its group G will is generated by the elements of groups G_k with $2n_k$ elements each - each of the elements $g_{k,j,l} = \omega_{n_k}^j x_k^l$ where $\omega_{n_k} = \exp(2\pi i / n_k)$ and $l \in \{-1, 1\}$. These elements commute and allow us to find an orbital operator that annihilates all terms on the LHS except $x_{[d]}^N KQ$ and also spares the $x_{[d]}^N$ term on the RHS.

9.3 Interacting Models

It is natural to consider extending the algebraic kernel method to one working for interacting models but we quickly see that the regular orbit summation operator is not sufficient. This is where the

power of the weighted orbital summation operators starts to show. The analysis for two-dimensional models is particularly nice because we end up having the same number of equations as unknowns and we are able to find the required weights. The linear systems can be written as matrix equations where the determinant of the matrix tells us when we have a useful orbit summation operator. In the end the method doesn't extend fully because we do not eliminate the $Q^{0,0}_{1,2}$ term. The D-finiteness of the full generating function Q depends on the D-finiteness of the function $Q^{0,0}_{1,2}$. Also we hope to say something about the initial goal of establishing relationship between the full and partial interactions for a given model. We'll only consider the interesting finite cases with 2 non-trivial generators. The group G has order 4, 6 or 8 and the functional equation in two dimensions is

$$KQ - K_1Q^0_1 - K_2Q^0_2 + K_{1,2}Q^{0,0}_{1,2} = \alpha \tag{9.61}$$

9.3.1 Groups of order 4

From a previous section we found that

$$D_G = 1 - g_1 - g_2 + g_1g_2 = (1 - g_1)(1 - g_2) = \Delta_1\Delta_2 \tag{9.62}$$

where we defined $\Delta_i = 1 - g_i$. In the non-interacting orbit summation method we had a direct application of D_G on the equations to perform the required cancellations. Here we will attempt to do something similar. Since the cancellations don't happen automatically if we use D_G directly, we'll instead obtain a linear system of equations by applying the group operations on the original functional equation. Then we'll use linear algebra to find a solution where our variables will be the ones we wish to eliminate Q^0_i and their images under the group actions. If the associated matrix has non-zero determinant we have a unique solution and unfortunately we won't be able to eliminate variables. However there are some cases where the determinant will be zero. Because of symmetry we expect to be able eliminate the desired variables. The group operations commute so we have

$$g_iQ^0_i = Q^0_i, \quad g_ig_jQ^0_i = g_jQ^0_i \tag{9.63}$$

and therefore we have the variables that we wish to eliminate

$$E = \{Q^0_1, Q^0_2, g_2Q^0_1, g_1Q^0_2\}. \tag{9.64}$$

This is looking promising since we have four equations and four unknowns that we wish to eliminate. We re-write the original equation in a format centered on these variables

$$K_1 Q_1^0 + K_2 Q_2^0 = KQ + K_{1,2} Q_{1,2}^{0,0} - \alpha = \beta \tag{9.65}$$

For brevity let $g_{12} = g_1 g_2$

$$K_1 Q_1^0 + K_2 Q_2^0 = \beta \tag{9.66}$$
$$g_1 K_1 Q_1^0 + g_1 K_2 g_1 Q_2^0 = g_1 \beta = \beta_1 \tag{9.67}$$
$$g_2 K_1 g_2 Q_2^0 + g_2 K_2 Q_2^0 = g_2 \beta = \beta_2 \tag{9.68}$$
$$g_{12} K_1 g_2 Q_1^0 + g_{12} K_2 g_1 Q_2^0 = g_{12} \beta = \beta_{12} \tag{9.69}$$
$$\tag{9.70}$$

Using the displayed order of variables in E as our order for the system we can write the associated matrix M as

$$M_4 = \begin{bmatrix} K_1 & K_2 & 0 & 0 \\ g_1 K_1 & 0 & 0 & g_1 K_2 \\ 0 & g_2 K_2 & g_2 K_1 & 0 \\ 0 & 0 & g_{12} K_1 & g_{12} K_2 \end{bmatrix} \tag{9.71}$$

Its determinant is

$$\det M_4 = K_1 g_{12} K_1 g_1 K_2 g_2 K_2 - g_1 K_1 g_2 K_1 K_2 g_{12} K_2 \tag{9.72}$$
$$= K_1 g_{12} K_1 g_1 K_2 g_2 K_2 - g_1 (g_1 K_1 g_{12} K_1 g_1 K_2 g_2 K_2) \tag{9.73}$$
$$= \Delta_1 (K_1 g_{12} K_1 g_1 K_2 g_2 K_2) \tag{9.74}$$
$$= \Delta_2 (g_1 K_1 g_2 K_1 K_2 g_{12} K_2) \tag{9.75}$$

where the last form comes from doing the process in the same way but with g_2 instead.

When is $\det M_4 = 0$**?**

Now our task is to find where this determinant is zero. Clearly the expression withing each Δ must be in the kernel of the operator. To that end let's find when that happens. For a Laurent polynomial

f in the variables x_i and t we have

$$\Delta_k f = \sum_n f_n(1-g_k)x_k^n = \sum_n (f_n - f_{-n}\mu_k^{-n})x_i^n \tag{9.76}$$

where $\mu_k = S_k^{-1}/S_k^1$. This implies that

$$f_n = f_{-n}\mu_k^{-n} \tag{9.77}$$

$$f_n(S_k^1)^n = f_{-n}(S_k^{-1})^n \tag{9.78}$$

for all $n \in \mathbb{Z}$. It's not clear if this will help us. However the models that exhibit groups of order 4 have at least one symmetry across a line - either vertical or horizontal symmetry. WLOG let the model have vertical symmetry. That is

$$g_1(x_1) = x_1, \quad g_1(S_2^i) = S_2^i, \quad g_1(S_1^i) = S_1^{-i} \tag{9.79}$$

This allows us to compute some things. For example

$$g_1 K_2 = c_2 - t\overline{x}_2 g_1 S_2^{-1} = c_2 - t\overline{x}_2 S_2^{-1} = K_2 g_1 g_2 K_2 = g_2 g_1 K_2 = g_2 \tag{9.80}$$

Therefore Δ_1 fixes both K_2 and $g_2 K_2$. We see that the determinant is

$$\det M_4 = \Delta_1(K_1 K_2 g_{12} K_1 g_2 K_2) = K_2 g_2 K_2 \Delta_1(K_1 g_{12} K_1) \tag{9.81}$$

This is zero if and only if $\Delta_1(K_1 g_{12} K_1) = 0$ so we have

$$\begin{aligned}
0 &= \Delta_1(K_1 g_{12} K_1) && (9.82)\\
&= K_1 g_{12} K_1 - g_1 K_1 g_2 K_2 && (9.83)\\
&= (c_1 - t\overline{x_1}S_1^{-1})(c_1 - t x_1 g_2 S_1^{-1}) - (c_1 - t x_1 S_1^{-1})(c_1 - t\overline{x_1} g_2 S_1^{-1}) && (9.84)\\
&= c_1 t\overline{x}_1(g_2 S_2^{-1} - S_2^{-1}) - c_1 t x_1(g_2 S_2^{-1} - S_2^{-1}) && (9.85)\\
&= c_1 t(\overline{x}_1 - x_1)(g_2 S_1^{-1} - S_1^{-1}) && (9.86)\\
&= -c_1 t(\overline{x}_1 - x_1)\Delta_2(S_1^{-1}) && (9.87)
\end{aligned}$$

This is true if and only if

$$g_2 S_1^{-1} = S_1^{-1} \tag{9.88}$$

We have

$$g_2(S_{1,2}^{-1,-1}\overline{x}_2 + S_{1,2}^{-1,0} + S_{1,2}^{-1,1}x_2) = S_{1,2}^{-1,-1}x_2\overline{\mu}_2 + S_{1,2}^{-1,0} + S_{1,2}^{-1,1}\overline{x}_2\mu_2 \tag{9.89}$$

and

$$S_{1,2}^{-1,-1}x_2\overline{\mu}_2 + S_{1,2}^{-1,1}\overline{x}_2\mu_2 = S_{1,2}^{-1,-1}\overline{x}_2 + S_{1,2}^{-1,1}x_2 \tag{9.90}$$

which can be re-written succinctly as

$$S_{1,2}^{-1,-1} = \mu_2 S_{-1,1} \tag{9.91}$$

If one of the S's is zero so is the other and therefore by symmetry the model doesn't have diagonal steps. Since we have two non-trivial generators this implies that both x_2 and $\overline{x}_2$ are present in S, so the model trivially exhibits a symmetry about the x_1 axis. If the S's are non-zero this implies μ_2 is a constant and therefore $S_2^{-1} = S_2^1$. Again the model exhibits symmetry about the x_1 axis.

Theorem 74. $\det M_4 = 0$ *if and only if the model is highly-symmetric.*

Corollary 75. *The algebraic kernel method extends to the non-interacting case for* $|G| = 4$ *if and only if the model is highly-symmetric.*

We are now ready to attempt using the orbit summation method in the highly symmetric cases. There are 8 such models but only 5 of them have dimension 2. Furthermore 2 of the models are reflections of each other so we are really left with 4 interesting cases - diagonal walks, full walks, full walks minus two horizontal steps. If we are able to eliminate all the variables then we'll be left with a linear relation

$$\gamma\beta - \gamma_1\beta_1 - \gamma_2\beta_2 + \gamma_{12}\beta_{12} = 0. \tag{9.92}$$

Here the choice of signs is inspired by the orbit summation operators. Let $\bullet$ and $*$ stand for applications of g_1 and g_2 respectively. First we have $K_2^\bullet = K_2$ and $K_1^* = K_1$. Then we let

$$x = Q_1^0, y = Q_2^0, u = Q_1^{0,*}, v = Q_2^{0,\bullet} \tag{9.93}$$

and

$$a = K_1, b = K_1^\bullet, c = K_2, d = K_2^* \tag{9.94}$$

The original equations become

$$\beta = ax + cy, \quad \beta_1 = bx + cv \tag{9.95}$$

$$\beta_2 = dy + au, \quad \beta_{12} = bu + dv \tag{9.96}$$

Therefore the above equation needs to satisfy

$$0 = (ax + cy)\gamma - (bx + cv)\gamma_1 - (dy + au)\gamma_2 + (bu + dv)\gamma_{12} \tag{9.97}$$

$$= (a\gamma - b\gamma_1)x + (c\gamma - d\gamma_2)y + (b\gamma_{12} - a\gamma_2)u + (d\gamma_{12} - c\gamma_1)v \tag{9.98}$$

and in order to eliminate the variables x, y, u, v we need to make all their coefficients zero. Therefore

$$a\gamma = b\gamma_1, \quad c\gamma = d\gamma_2, \quad b\gamma_{12} = a\gamma_2, \quad d\gamma_{12} = c\gamma_1 \tag{9.99}$$

If we set $\gamma = 1$ then $\gamma_1 = a/b$, $\gamma_2 = c/d$ and $\gamma_{12} = (ac)/(bd)$. So we can multiply out by bd to find the relation

$$bd\beta - ad\beta_1 - cb\beta_2 + ac\beta_{12} = 0 \tag{9.100}$$

where we have successfully eliminated $Q_1^0, Q_2^0, Q_1^{0,*}$, and $Q_2^{0,\bullet}$. Or written in terms of the K's we have

$$K_1^\bullet K_2^* \beta - K_1 K_2^* \beta^\bullet - K_1^\bullet K_2 \beta^* + K_1 K_2 \beta^{\bullet *} = 0 \tag{9.101}$$

We can re-write this equation in the following nicer way

Theorem 76. *For highly-symmetric models with small steps we have the functional equation*

$$\Delta(K_1^\bullet K_2^* \beta) = 0 \tag{9.102}$$

where $\beta = KQ + K_{1,2} Q_{1,2}^{0,0} - \alpha$.

We can expand this functional equation to obtain a different form. Towards this end let $K_0 =$

$K_1^\bullet K_2^*$. We obtain

$$\Delta(K_0\beta) = \Delta(K_0(KQ + K_{1,2}Q_{1,2}^{0,0} - \alpha)) \tag{9.103}$$
$$= \Delta(K_0KQ) + \Delta(K_0K_{1,2}Q_{1,2}^{0,0}) - \Delta(K_0\alpha) \tag{9.104}$$
$$= K\Delta(K_0Q) + \Delta(K_0K_{1,2})Q_{1,2}^{0,0} - \Delta(K_0)\alpha \tag{9.105}$$

where the last line follows since Δ commutes with K and functions with some variables x_i missing. All that remains is to expand the various quantities involved

$$\Delta g_i K_i = -\Delta K_i = -(c_i - tx_iS_i^{-1} - c_i + t\overline{x}_iS_i^{-1}) = -t(\overline{x}_i - x_i)S_i^{-1} \tag{9.106}$$

Therefore

$$\Delta K_0 = \Delta_1(K_1^\bullet)\Delta_2(K_2^*) = t^2(\overline{x}_1 - x_1)(\overline{x}_2 - x_2)S_1^{-1}S_2^{-1} \tag{9.107}$$

Next

$$\Delta(K_0K_{1,2}) = \Delta_1K_1^\bullet\Delta_2K_2^*K_{1,2} = \Delta_1K_1^\bullet(K_2^*K_{1,2} - K_2K_{1,2}^*) \tag{9.108}$$

The quantity in parentheses is

$$K_2^*K_{1,2} - K_2K_{1,2}^* = \tag{9.109}$$
$$=(c_2 - tx_2S_2^{-1})(c_1c_2 - t\overline{x_1x_2}S_{1,2}^{-1,-1}) - (c_2 - t\overline{x_2}S_2^{-1})(c_1c_2 - t\overline{x_1}x_2S_{1,2}^{-1,-1}) \tag{9.110}$$
$$=c_1c_2tS_2^{-1}(\overline{x_2} - x_2) - c_2t\overline{x_2}S_{1,2}^{-1,-1}(\overline{x_2} - x_2) \tag{9.111}$$
$$=c_2t(\overline{x_2} - x_2)(c_1S_2^{-1} - \overline{x_1}S_{1,2}^{-1,-1}) \tag{9.112}$$

and thus

$$\Delta(K_0K_{1,2}) = c_2t(\overline{x_2} - x_2)\Delta_1K_1^\bullet(c_1S_2^{-1} - \overline{x_1}S_{1,2}^{-1,-1}) \tag{9.113}$$
$$= c_2t(\overline{x_2} - x_2)\Delta_1K_1^\bullet(c_1S_2^{-1} - \overline{x_1}S_{1,2}^{-1,-1}) \tag{9.114}$$

To simplify this quantity note that

$$\Delta_1K_1^\bullet c_1S_2^{-1} = c_1S_2^{-1}\Delta_1K_1^\bullet = -c_1tS_2^{-1}(\overline{x}_1 - x_1) \tag{9.115}$$

and

$$\Delta_1 K_1^{\bullet}\overline{x_1} S_{1,2}^{-1,-1} = S_{1,2}^{-1,-1}\Delta_1 K_1^{\bullet}\overline{x_1} = c_1(\overline{x}_1 - x_1)S_{1,2}^{-1,-1}. \tag{9.116}$$

Finally

$$\Delta(K_0 K_{1,2}) = c_1 c_2 t(\overline{x}_1 - x_1)(\overline{x}_2 - x_2)(tS_1^{-1}S_2^{-2} - S_{1,2}^{-1,-1}). \tag{9.117}$$

Combining terms and manipulating things a bit leads us to the equation

$$K\Delta(K_0 Q) = \alpha t(\overline{x}_1 - x_1)(\overline{x}_2 - x_2)(tS_1^{-1}S_2^{-1} - b_1 b_2(tS_1^{-1}S_2^{-1} - S_{1,2}^{-1,-1})Q_{1,2}^{0,0}) \tag{9.118}$$

Corollary 77. *For a highly-symmetric model with $K_0 = K_1^{\bullet}K_2^{*}$ we have the functional equation*

$$K\Delta(K_0 Q) = \alpha t(\overline{x}_1 - x_1)(\overline{x}_2 - x_2)(tS_1^{-1}S_2^{-1}(1 - b_1 b_2 Q_{1,2}^{0,0}) + S_{1,2}^{-1,-1}Q_{1,2}^{0,0}) \tag{9.119}$$

9.3.2 Groups of order 6

Since we have 6 group actions it's worth taking the time to see what kinds of expressions we obtain by applying g on Q_i^0 or K_i for each i. For example $g_i Q_i^0 = Q_i^0$ so we will say the action of g_i is reduced to $g_{\varnothing} = 1$ on Q_i^0. Using the sequences L produced in one orbit of a representation of G and their alternatives we can tabulate their reductions on the different expresions. For example

$$g_{2121}Q_1^0 = g_{212}Q_1^0 = g_{121}Q_1^0 = g_{12}Q_1^0 \tag{9.120}$$

where on the second step we've used the alternative form $g_{212} = g_{121}$. The table has 12 in the row and column corresponding to 2121 and Q_1^0. We'll omit the row corresponding to the identity.

Sequence L	Alternative	Q_1^0	Q_2^0	K_1	K_2
1	21212	$\varnothing$	1	1	1
21	1212	2	21	21	21
121	212	12	21	121	212
2121	12	12	1	12	12
12121	2	2	$\varnothing$	2	2

Table 9.1: Reductions of group actions $g \in G$ on Q_i^0 and K_i

We only have 6 different functions results of applying G on Q_1^0 and Q_2^0 so we have only 6 variables that we may wish to eliminate. Let

$$u_1 = Q_1^0, \quad u_2 = g_2 Q_1^0, \quad u_3 = g_{12} Q_1^0 \tag{9.121}$$

$$v_1 = Q_2^0, \quad v_2 = g_1 Q_2^0, \quad v_3 = g_{21} Q_2^0 \tag{9.122}$$

$$\tag{9.123}$$

Also there are 12 different K_1's and K_2's which we label sequentially as $A_1, \ldots, A_6$ and $B_1, \ldots, B_6$ respectively. The associated matrix to the linear system of equations is

$$M_6 = \begin{bmatrix} A_1 & & & B_1 & & \\ A_2 & & & & B_2 & \\ & A_3 & & & & B_3 \\ & & A_4 & & & B_4 \\ & & A_5 & & B_5 & \\ & A_6 & & B_6 & & \end{bmatrix} \tag{9.124}$$

and it has determinant

$$\det M_6 = A_2 A_4 A_6 B_1 B_3 B_5 - A_1 A_3 A_5 B_2 B_4 B_6. \tag{9.125}$$

The even index $A's$ are obtained by applying g_1 on the odd indx $A's$ and vice versa for the $B's$. That is,

$$\det M_6 = \Delta_1(A_2 A_4 A_6 B_1 B_3 B_5) = \Delta_1(g_1 K_1 g_{121} K_1 g_{12} K_1 K_2 g_{21} K_2 g_{12} K_2) \tag{9.126}$$

We can easily check the relationships A's and B's for the two different groups of order 6. For the models with the first groupwe have

- $S = x_1\overline{x}_2 + \overline{x}_1 + x_2 : A_1 = A_6, A_2 = A_5, A_3 = A_4$ and $B_1 = B_4, B_2 = B_3, B_5 = B_6$.
- $S = \overline{x}_1 x_2 + x_1 + \overline{x}_2$: $A_1 = A_4, A_2 = A_3, A_5 = A_6$ and $B_1 = B_2, B_3 = B_6, B_4 = B_5$
- $S = x_1 + \overline{x}_1 + x_2 + \overline{x}_2 + x_1\overline{x}_2 + \overline{x}_1 x_2$: We have no non-trivial relations.

For the models with the second group we have

- $S = x_1 x_2 + \overline{x}_1 + \overline{x}_2 : A_1 = A_6, A_2 = A_5, A_3 = A_4$ and $B_1 = B_2, B_3 = B_6, B_4 = B_5$
- $S = \overline{x}_1\overline{x}_2 + x_1 + x_2$: $A_1 = A_4, A_2 = A_3, A_5 = A_6$ and $B_1 = B_4, B_2 = B_3, B_5 = B_6$.

- $S = x_1 + \overline{x}_1 + x_2 + \overline{x}_2 + x_1x_2 + \overline{x}_1\overline{x}_2$: We have no non-trivial relations.

It's clear that $\det M_6 = 0$ for the cases with relations. Using a computer we can easily check that the determinant is not zero in the cases with no nontrivial relations.

We seek the same type of relation on β and its orbit as before so we'll use D_G as inspiration for the sign changes. We have the following

$$\begin{aligned} 0 =& \gamma\beta - \gamma_1\beta_1 + \gamma_{21}\beta_{21} - \gamma_{121}\beta_{121} + \gamma_{12}\beta_{12} - \gamma_2\beta_2 && (9.127)\\ =& \gamma(A_1u_1 + B_1v_1) - \gamma_2(A_6u_2 + B_6v_1) - \gamma_1A_2u_1 + B_2v_2) && (9.128)\\ &+ \gamma_{12}(A_5u_3 + B_5v_2) + \gamma_{21}(A_3u_2 + B_3v_3) - \gamma_{121}(A_4u_3 + B_4v_3). && (9.129) \end{aligned}$$

Expanding out and collecting terms produces the system

$$A_1\gamma = A_2\gamma_1, \quad A_3\gamma_{21} = A_6\gamma_2, \quad A_5\gamma_{12} = A_4\gamma_{121} \tag{9.130}$$

$$B_1\gamma = B_6\gamma_2, \quad B_5\gamma_{12} = B_2\gamma_1, \quad B_3\gamma_{21} = B_4\gamma_{121}. \tag{9.131}$$

This system of equations doesn't have a nontrivial symbolic solution so we need to use the relations here. There is a rational solution in each case with relations where γ is a free variable. We set γ to be the least common multiple of the denominators. We find that $K_0 = \gamma$ in each of the four cases with relations.

- $S = x_1\overline{x}_2 + \overline{x}_1 + x_2 : \quad \gamma = K_0 = A_3A_5B_6 = g_{21}(K_1)g_{12}(K_1)g_2(K_2)$
- $S = \overline{x}_1x_2 + x_1 + \overline{x}_2: \quad \gamma = K_0 = A_2A_5B_6 = g_1(K_1)g_{12}(K_1)g_2(K_2)$
- $S = x_1x_2 + \overline{x}_1 + \overline{x}_2 : \quad \gamma = K_0 = A_2A_3B_5B_6 = g_1(K_1)g_{21}(K_1)g_{12}(K_2)g_2(K_2)$
- $S = \overline{x}_1\overline{x}_2 + x_1 + x_2: \quad \gamma = K_0 = A_2B_6 = g_1(K_1)g_2(K_2)$

Here we've picked the respresentations of A's and B' with the least number of group operations g_i to reach the resulting expression. We also tried finding a possible K_0 by brute force for the cases with no relations but we didn't find any among products of distinct A's and B's.

Theorem 78. *The algebraic kernel method extends for interacting random walk models with small steps and group G of size 6 if and only if* $\det M_6 = 0$. *In those cases we have the functional equation*

$$D_G(K_0\beta) = 0 \tag{9.132}$$

where $\beta = KQ + K_{1,2}Q_{1,2}^{0,0} - \alpha$ and K_0 depends on the specific model.

9.3.3 Groups of order 8

The analysis for groups of order 8 is quite similar so we will omit it. Here are the results. The two models are

$$S = x_1 + \overline{x}_1 + x_1\overline{x}_2 + \overline{x}_1 x_2 \qquad S = x_1 + \overline{x}_1 + x_1 x_2 + \overline{x}_1\overline{x}_2 \tag{9.133}$$

The relations are the same in both models

$$A_1 = A_6, \quad A_2 = A_5, \quad A_3 = A_4, \quad A_7 = A_8 \tag{9.134}$$

$$B_1 = B_4, \quad B_2 = B_3, \quad B_4 = B_8, \quad B_6 = B_7. \tag{9.135}$$

In both cases we obtain the same γ and we write it in the form with least group actions.

$$\gamma = K_0 = A_2A_3B_5B_6 = g_1(K_1)g_{21}(K_1)g_{212}(K_2)g_{1212}(K_2) \tag{9.136}$$

Theorem 79. *The algebraic kernel method extends for all interacting random walk models with small steps and group G of size 8. The functional equation is*

$$D_G(K_0\beta) = 0 \tag{9.137}$$

where $\beta = KQ + K_{1,2}Q^{0,0}_{1,2} - \alpha$ and $K_0 = g_1(K_1)g_{21}(K_1)g_{212}(K_2)g_{1212}(K_2)$.

9.3.4 Infinite Groups

A natural question is whether we can extend this analysis to handle infinite groups, at least in a purely formal sense. Here the orbits will be infinitely long and while there will be relations between the gQ^0_i variables from one orbit iteration to the next, there will be no point where we can stop that we'll have as many equations as variables - there will always be more variables. Thus we'll need to take an infinite summation of the equations. It's quite possible that any candidates for K_0 will be infinite products. Since the K_i's have t^0 coefficients, such products will not converge. However, there may be hope if there are enough relations between the gK_i's so that K_0 would be a finite product of gK_i's. The non-interacting orbit method doesn't need a K_0 to get rid of all terms Q^0_i and $Q^{0,0}_{1,2}$. One would hope that something like this will work for the infinite case.

Chapter 10

Obstinate Kernel Method

10.1 Non-Interacting Case

10.2 Interacting Case

10.2.1 Group of order 4

10.2.2 Group of order 6

10.2.3 Group of order 8

There are 2 non-isomorphic cases of random walk models on the plane that have group of order 8. They have step polynomials

$$S = \overline{x}_1 + x_1 + \overline{x}_1 x_2 + x_1 \overline{x}_2 \tag{10.1}$$

$$S = \overline{x}_1 + x_1 + \overline{x}_1 \overline{x}_2 + x_1 x_2 \tag{10.2}$$

We'll work on the first one since it has already been covered in the two-friendly-walks paper and this will provide us with an opportunity to compare. Here

$$S_1^{-1} = 1 + x_2, \quad S_1^0 = 0, \quad S_1^1 = 1 + \overline{x}_2 \tag{10.3}$$

$$S_2^{-1} = x_1, \quad S_2^0 = x_1 + \overline{x}_1, \quad S_2^1 = \overline{x}_1 \tag{10.4}$$

which give

$$A = K_1 = c_1 - t\overline{x}_1(1 + x_2) \qquad B = K_2 = c_2 - tx_1\overline{x}_2 \tag{10.5}$$

Note that since $S_{1,2}^{-1,-1} = 0$

$$K_{12} = c_1 c_2 \tag{10.6}$$

and we can combine the α term with $K_{1,2}Q_{1,2}^{0,0}$ in one

$$\gamma = \overline{a}_1\overline{a}_2(1 - b_1 b_2 Q_{1,2}^{0,0}) \tag{10.7}$$

and the master functional equation is

$$-KQ + K_1 Q_1^0 + K_2 Q_2^0 + \gamma = 0 \tag{10.8}$$

The roots of the kernel are

$$\sigma_1^{\pm} = \frac{1 \pm \sqrt{1 - 4t^2(1 + x_2)(1 + \overline{x}_2)}}{2t(1 + \overline{x}_2)} \tag{10.9}$$

$$\sigma_2^{\pm} = \frac{1 - t(x_1 + \overline{x}_1) \pm \sqrt{(1 - t(x_1 + \overline{x}_1))^2 - 4t^2}}{2t\overline{x}_1} \tag{10.10}$$

and as in the paper we pick $\sigma_k = \sigma_k^-$ because they are power series in t. Next we determine the birational transformations that leave the kernel unchanged. They are

$$g_1 : x_1 \mapsto \overline{x}_1 x_2 \qquad g_2 : x_2 \mapsto x_1^2 \overline{x}_2 \tag{10.11}$$

Simply iterating g_1 and g_2 shows that the group G has order 8. For the rest of the analysis we'll need the following notation.

Definition 80. *For a concatenated list $L = L_1 L_2 \dots L_n$of integer indices let $g_L = g_{L_1} \dots g_{L_n}$ and for some term X (for example A or B) we let $X_L = g_L(X)$.*

The elements in the orbit are id, g_1, g_{21}, g_{121}, etc. Since eventually we reach the identity again we can work backwards and have the g terms meet in the middle. The orbit is then given by

$$Orbit(G) = \{id, g_1, g_{21}, g_{121}, g_{2121} = g_{1212}, g_{212}, g_{12}, g_2\} \tag{10.12}$$

In the rest of this calculation we need to eliminate the kernel term via a substitution of σ_k in the

master equation (We'll overload the notation as before). Then we'll need some extra functional equations that are obtained by application of terms from the orbit. However not all such applications are valid. We require that the σ_k term appears in the numerator inside the functions Q. We can check that to use the sigma values we need to use the following transformations

$$\sigma_1 : id, g_2, g_{21}, g_{212} \tag{10.13}$$

$$\sigma_2 : id, g_1, g_{12}, g_{121} \tag{10.14}$$

For each σ_k we're going to obtain 3 additional functional equations to complement the master equation and then we'll substitute $x_k = \sigma_k$ into the results to eliminate the kernel term. Because of this we just won't include the kernel term from the start. The variable elimination method involving the different equations will work with or without the substitution $x_k = \sigma_k$. We'll also ignore the γ term for now and just work with the Q_1^0 and Q_2^0 terms. On this note we'll also drop the 0s from the exponents and write $Q_k^0 = Q_k$. The method is essentially linear algebra but because of the complexity and symmetry involved in the coefficients it's best to investigate the explicit simplification of the equations.

The σ_1 substitution

A simple but tedious check shows that there are no non-trivial relations for A, B and application of elements from the σ_1 orbit besides $A_{212} = A$.

$$g_{21}Q_1 = g_2Q_1, \quad g_2Q_2 = Q_2, \quad g_{212}Q_2 = g_{21}Q_2 \tag{10.15}$$

The functional equations have the terms

$$id: \quad AQ_1 + BQ_2 \tag{10.16}$$

$$g_2: \quad A_2g_2Q_1 + B_2Q_2 \tag{10.17}$$

$$g_{21}: \quad A_{21}g_2Q_1 + B_{21}g_{21}Q_2 \tag{10.18}$$

$$g_{212}: \quad A_{212}g_{212}Q_1 + B_{212}g_{21}Q_2 \tag{10.19}$$

To eliminate variables we'll take do the following transformations of the equations

$$(1) = B_2id - Bg_2 \Longrightarrow AB_2Q_1 - A_2Bg_2Q_1 \tag{10.20}$$

$$(2) = B_{212}g_{21} - B_{21}g_{212} \Longrightarrow A_{21}B_{212}g_2Q_1 - A_{212}B_{21}g_{212}Q_1 \tag{10.21}$$

and finally we'll take

$$(3) = A_{21}B_{212}(1) + A_2B(2) \tag{10.22}$$

$$\implies A_{21}B_{212}AB_2Q_1 - A_2BA_{212}B_{21}g_{212}Q_1 \tag{10.23}$$

The resulting expression can be written more compactly as

$$(1 - g_{212})(A_{21}B_{212}AB_2Q_1) = A(1 - g_{212})(A_{21}B_{212}B_2Q_1) \tag{10.24}$$

We obtain the expression for the coefficient of γ by following the transformations. We obtain the following term

$$[A_{21}B_{212}(B_2 - B) + A_2B(B_{212} - B_{21})]\gamma \tag{10.25}$$

The σ_2 substitution

A simple but tedious check shows that there are no non-trivial relations for A, B and application of elements from the σ_2 orbit besides $B_{121} = B$. The relations between the Q's are

$$g_1Q_1 = Q_1, \quad g_{121}Q_1 = g_{12}Q_1, g_{12}Q_2 = g_1Q_2 \tag{10.26}$$

The functional equations have the terms

$$id: \quad AQ_1 + BQ_2 \tag{10.27}$$

$$g_1: \quad A_1Q_1 + B_1g_1Q_2 \tag{10.28}$$

$$g_{12}: \quad A_{12}g_{12}Q_1 + B_{12}g_1Q_2 \tag{10.29}$$

$$g_{121}: \quad A_{121}g_{12}Q_1 + B_{121}g_{121}Q_2 \tag{10.30}$$

To eliminate variables we'll take do the following transformations of the equations

$$(1) = A_1id - Ag_1 \implies A_1BQ_2 - AB_1g_1Q_2 \tag{10.31}$$

$$(2) = A_{121}g_{12} - A_{12}g_{121} \implies A_{121}B_{12}g_1Q_2 - A_{12}B_{121}g_{121}Q_2 \tag{10.32}$$

and finally we'll take

$$(3) = A_{121}B_{12}(1) + AB_1(2) \tag{10.33}$$

$$\implies A_{121}B_{12}A_1BQ_2 - AB_1A_{12}B_{121}g_{121}Q_2 \tag{10.34}$$

The resulting expression may be written more compactly as

$$(1 - g_{121})(A_{121}B_{12}A_1BQ_2) = B(1 - g_{121})(A_1A_{121}B_{12}Q_2) \tag{10.35}$$

We obtain the expression for the coefficient of γ by following the transformations to

$$[A_{121}B_{12}(A_1 - A) + AB_1(A_{121} - A_{12})]\gamma \tag{10.36}$$

Partial Orbital Operators

We may think of the elimination of Q's with undesirable arguments as a transformation of the original functional equation. We take two functional equations and produce one with a function eliminated. Then we take both of these to produce the final equation. Each of these steps may be done with such an operator. For example the first transformation becomes

$$B_2id - Bg_2 = (1 - g_2)B_2 \tag{10.37}$$

that acts in the following way

$$((1 - g_2)B_2)(f) = g_2(B)f - Bg_2(f) \tag{10.38}$$

This inspires the following notation.

Definition 81. *Let L be a concatenated list and X be some function. We define the following operator via its action on any function F*

$$D_L(X)(F) = g_L(X)F - Xg_L(F) \tag{10.39}$$

The notation is reminiscent of a derivative, specifically in the numerator of the quotient rule. So the first transformation can be written as

$$B_2id - Bg_2 = D_2(B) \tag{10.40}$$

The second transformation is a little more involved

$$B_{212}g_{21} - B_{21}g_{212} \tag{10.41}$$
$$= g_{212}(Bg_2 - B_2) \tag{10.42}$$
$$= -g_{212}D_2(B) \tag{10.43}$$

The final transformation is

$$D^{(1)} = A_{21}B_{212}D_2(B) - A_2Bg_{212}D_2(B) \tag{10.44}$$
$$= (1 - g_{212})(A_2B)D_2(B) \tag{10.45}$$
$$= D_{212}(A_2B)D_2(B) \tag{10.46}$$

This operator is reminiscent of the Orbital operators that we've seen previously. It eliminates the Q_2 term with the undesirable argument and replaces it with a more desirable one. We can re-write the result in the following manner

$$A(1 - g_{212})(A_{21}B_{212}B_2Q_1) = AD_{212}(A_2BB_{21})(Q_1) \tag{10.47}$$

This leads to the following theorem

Theorem 82. *For the σ_1 substitution we have*

$$0 = D^{(1)}(AQ_1 + BQ_2 + \gamma) = AD_{212}(A_2BB_{21})(Q_1) + \gamma D_{212}(A_2B)(B_2 - B) \tag{10.48}$$

Similarly for the σ_2 substitution we have

$$0 = D^{(2)}(AQ_1 + BQ_2 + \gamma) = BD_{121}(AA_{12}B_1)(Q_2) + \gamma D_{121}(AB_1)(A_1 - A) \tag{10.49}$$

The resulting equations are actually better written out as

$$\sigma_1 : \quad D_{212}(A_2BB_{21})(Q_1) + \gamma\frac{D_{212}(A_2B)(B_2 - B)}{A} = 0 \tag{10.50}$$
$$\sigma_2 : \quad D_{121}(AA_{12}B_1)(Q_2) + \gamma\frac{D_{121}(AB_1)(A_1 - A)}{B} = 0 \tag{10.51}$$

The interesting fact (as calculated in the paper) is that the coefficients in front of the Q's become Laurent polynomials once we take into consideration the σ_k substitutions, even though σ_k contains a square root. The coefficients in front of γ can be reduced to a rational function plus a rational

function times a square root after properly conjugating. The calculations are exceedingly complicated and unlikely to be done easily by hand. The square root part does not depend on a_2 in the first case and a_1 in the second case.

10.3 Ore Operators

One can view $D_L(X)$ as an Ore operator via

$$g_L \cdot X = -Xg_L + g_L(X) \tag{10.52}$$

and

$$g_L(XY) = -Xg_L(Y) + g_L(X)Y = g_L(X)Y - Xg_L(Y) \tag{10.53}$$

Chapter 11

Highly-symmetric 2D models

When we consider the highly-symmetric cases the x-transformations are given by

$$\bullet : x_1 \mapsto \overline{x_1}, \quad * : x_2 \mapsto \overline{x_2} \tag{11.1}$$

In each case we have

$$K^{\bullet} = K^{*} = K, \quad K_2^{\bullet} = K_2, \quad K_1^{*} = K_1 \tag{11.2}$$

while the others change in some way. The solutions eliminating the kernel also have a nice form

$$\sigma_i = \frac{1 - S_i^0 - \sqrt{(1 - S_i^0)^2 - 4t^2 (S_i^1)^2}}{2tS_i^1}, \quad \overline{\sigma_i} = \frac{1 - S_i^0 + \sqrt{(1 - S_i^0)^2 - 4t^2 (S_i^1)^2}}{2tS_i^1} \tag{11.3}$$

because we have

$$\overline{\sigma}_i = \frac{S_i^1}{S_i^{-1}} conj(\sigma_i) = conj(\sigma_i) \tag{11.4}$$

We can take the master equation and apply $*$ to it

$$-KQ + K_1 Q_1^0 + K_2 Q_2^0 - K_{1,2} Q_{1,2}^{0,0} + \alpha = 0 \tag{11.5}$$

$$-KQ^{*} + K_1 Q_1^{0,*} + K_2^{*} Q_2^0 - K_{1,2}^{*} Q_{1,2}^{0,0} + \alpha = 0 \tag{11.6}$$

From here on we will input $x_1 = \sigma_1$ to rid ourselves of the kernel K. There will be an overloading of notation

$$K_1 Q_1^0 + K_2 Q_2^0 - K_{1,2} Q_{1,2}^{0,0} + \alpha = 0 \tag{11.7}$$
$$K_1 Q_1^{0,*} + K_2^* Q_2^0 - K_{1,2}^* Q_{1,2}^{0,0} + \alpha = 0 \tag{11.8}$$

We can multiply both equation by something to clear Q_2^0 because it involves σ_1 while the other term involves only x_2. We have

$$K_1 K_2^* Q_1^0 + K_2^* K_2 Q_2^0 - K_2^* K_{1,2} Q_{1,2}^{0,0} + \alpha K_2^* = 0 \tag{11.9}$$
$$K_1 K_2 Q_1^{0,*} + K_2 K_2^* Q_2^0 - K_2 K_{1,2}^* Q_{1,2}^{0,0} + \alpha K_2 = 0 \tag{11.10}$$

We subtract these two equations to obtain

$$K_1 (K_2^* Q_1 - K_2 Q_1^*) - (K_2^* K_{1,2} - K_2 K_{1,2}^*) Q_{1,2} + \alpha (K_2^* - K_2) = 0 \tag{11.11}$$
$$K_2^* Q_1 - K_2 Q_1^* - \frac{K_2^* K_{1,2} - K_2 K_{1,2}^*}{K_1} Q_{1,2} + \alpha \frac{K_2^* - K_2}{K_1} = 0 \tag{11.12}$$
$$\tag{11.13}$$

The differences in solutions will occur from here on out. We have

$$K_2^* - K_2 = c_2 - t x_2 S_2^{-1} - c_2 + t\overline{x_2} S_2^{-1} = t(\overline{x_2} - x_2) S_2^{-1}(\sigma_1) \tag{11.14}$$

Next

$$K_2^* K_{1,2} - K_2 K_{1,2}^* = (c_2 - t x_2 S_2^{-1})(c_1 c_2 - t\overline{x_1 x_2} S_{1,2}^{-1,-1}) \tag{11.15}$$
$$- (c_2 - t\overline{x_2} S_2^{-1})(c_1 c_2 - t\overline{x_1} x_2 S_{1,2}^{-1,-1}) \tag{11.16}$$
$$= c_1 c_2 t S_2^{-1}(\overline{x_2} - x_2) - c_2 t\overline{x_2} S_{1,2}^{-1,-1}(\overline{x_2} - x_2) \tag{11.17}$$
$$= c_2 t(\overline{x_2} - x_2)(c_1 S_2^{-1}(\sigma_1) - \overline{x_1} S_{1,2}^{-1,-1}) \tag{11.18}$$

11.1 Cartesian model: steps N, S, E, W

In this case $K_{1,2} = c_1 c_2$ so we have a simpler situation where we can re-write the above equation as

$$K_2^* Q_1 - K_2 Q_1^* + \alpha\beta \frac{K_2^* - K_2}{K_1} = 0 \tag{11.19}$$

where

$$\beta = 1 - b_1 b_2 Q_{1,2}^{0,0} \tag{11.20}$$

Furthermore

$$K_2^* - K_2 = t(\overline{x_2} - x_2) \tag{11.21}$$

So we can let

$$f = f(c_1, x_2) = \frac{t(\overline{x_2} - x_2}{K_1} \tag{11.22}$$

since K_1 doesn't depend on c_2 or x_1. The equation finally becomes

$$(c_2 - t x_2) Q_1^0(x_2) - (c_2 - t\overline{x_2}) Q_1^0(\overline{x_2}) + \alpha\beta f = 0 \tag{11.23}$$

We perform the extraction $[x_2^1]$ to get

$$c_2 Q_{1,2}^{0,1} - t Q_{1,2}^{0,0} + \alpha\beta f_2^1 = 0 \tag{11.24}$$

and the extraction $[b_2^0]$ to get

$$-t\breve{Q}_{1,2}^{0,0} + \overline{a_1} f_2^1 \tag{11.25}$$

Using the second equation within the first one and simplifying yields

$$b_2 Q_{1,2}^{0,1} = t(a_2 Q_{1,2}^{0,0} - \beta \breve{Q}_{1,2}^{0,0}) \tag{11.26}$$

$$b_1 Q_{1,2}^{1,0} = t(a_1 Q_{1,2}^{0,0} - \beta \widehat{Q}_{1,2}^{0,0}) \tag{11.27}$$

Here the second equation is obtained via the same method but with $\bullet$ instead of $*$. Using the same method we can also obtain

$$b_2 Q_{1,2}^{1,1} = t(a_2 Q_{1,2}^{1,0} - \beta \breve{Q}_{1,2}^{1,0}) \tag{11.28}$$

Multiplying by b_1 and using the above relations leads to

$$\begin{aligned} b_1 b_2 Q_{1,2}^{1,1} &= t(a_2 b_1 Q_{1,2}^{1,0} - \beta b_1 \breve{Q}_{1,2}^{1,0}) && (11.29)\\ &= t a_2 t(a_1 Q_{1,2}^{0,0} - \beta \widehat{Q}_{1,2}^{0,0}) - t\beta t(a_1 \breve{Q}_{1,2}^{0,0} - \beta \breve{\widehat{Q}}_{1,2}^{0,0}) && (11.30)\\ &= t^2(a_1 a_2 Q_{1,2}^{0,0} - a_2 \beta \widehat{Q}_{1,2}^{0,0} - a_1 \beta \breve{Q}_{1,2}^{0,0} + \beta^2 \breve{\widehat{Q}}_{1,2}^{0,0}) && (11.31) \end{aligned}$$

Let

$$G = Q_{1,2}^{0,0}, \quad \widehat{G} = G_1, \quad \breve{G} = G_2 \tag{11.32}$$

From before we have

$$Q_{1,2}^{0,0} = 1 + t a_1 a_2 (Q_{1,2}^{1,0} + Q_{1,2}^{0,1}) \tag{11.33}$$

Multiplying by $b_1 b_2$ and using the above relations leads to

$$\begin{aligned} b_1 b_2 G &= b_1 b_2 + t a_1 a_2 (b_2 b_1 Q_{1,2}^{1,0} + b_1 b_2 Q_{1,2}^{0,1}) && (11.34)\\ &= b_1 b_2 + t a_1 a_2 (b_2 t(a_1 G - \beta G_1) + b_1 t(a_2 G - \beta G_2)) && (11.35)\\ &= b_1 b_2 + t^2 a_1 a_2 (b_2 a_1 G + b_1 a_2 G - b_2 \beta G_1 - b_1 \beta G_2) && (11.36) \end{aligned}$$

Using the relation for β and simplifying leads

$$G = \frac{b_1 b - 2 - t^2 a_1 a_2 (b_2 G_1 + b_1 G_2)}{b_1 b_2 - t^2 a_1 a_2 (a_2 b_1 + a_1 b_2 + b_1 b_2 (b_2 G_1 + b_1 G_2))} \tag{11.37}$$

If we let $F = (b_2 G_1 + b_1 G_2)$ and perform the long division with respect to F we find that

$$Q_{1,2}^{0,0} = \frac{1}{b_1 b_2} \left(1 - \frac{b_1 b_2 (1 - b_1 b_2) - t^2 a_1 a_2 (a_1 b_2 + a_2 b_1)}{b_1 b_2 + t^2 a_1 a_2 (a_1 b_2 + a_2 b_1 + b_1 b_2 F)} \right) \tag{11.38}$$

Using arguments similar to those in the later section on diagonal steps we can find out more about the various Q's.

$$b_2 Q_{1,2}^{m,n+1} = t(a_2 Q_{1,2}^{m,n} - \beta \breve{Q}_{1,2}^{m,n}) \tag{11.39}$$

$$b_1 Q_{1,2}^{m+1,n} = t(a_1 Q_{1,2}^{m,n} - \beta \widehat{Q}_{1,2}^{m,n}) \tag{11.40}$$

Summing the equations multiplied by their respective powers of x_1 and x_2 we find the equations

$$b_2 Q_2^{n+1} = t(a_2 Q_2^n - \beta \breve{Q}_2^n) \tag{11.41}$$

$$b_1 Q_1^{m+1} = t(a_1 Q_1^m - \beta \widehat{Q}_1^m) \tag{11.42}$$

Now we can sum with respect to x_2 and x_1 to find

$$b_2(Q - Q_2^0) = x_2 t(a_2 Q - \beta \breve{Q}) \tag{11.43}$$

$$b_1(Q - Q_1^0) = x_1 t(a_1 Q - \beta \widehat{Q}) \tag{11.44}$$

We can solve these for Q_2^0 and Q_1^0 as follows

$$b_2 Q_2^0 = (b_2 - a_2 x_2 t)Q + x_2 t \beta \breve{Q} \tag{11.45}$$

$$b_1 Q_1^0 = (b_1 - a_1 x_1 t)Q + x_1 t \beta \widehat{Q} \tag{11.46}$$

$$\tag{11.47}$$

We can substitute these two expressions into the original functional equation and after some algebra we find that

$$Q = \frac{(1 - b_1 b_2 G)(\alpha + t x_1 \overline{b_1} K_1 \widehat{Q} + t x_2 \overline{b_2} K_2 \breve{Q})}{K - (1 - \overline{c_1} x_1 t) K_1 - (1 - \overline{c_2} x_2 t) K_2} \tag{11.48}$$

and after recognizing some of the quantities as modified K_1 and K_2's we finally stop at the solution for Q

$$Q = \frac{(1 - b_1 b_2 G)(\alpha - \overline{b}_1 K_1 \widehat{K}_1^{\bullet} \widehat{Q} - \overline{b}_2 K_2 \breve{K}_2^{*} \breve{Q})}{K - \overline{c}_1 K_1 K_1^{\bullet} - \overline{c}_2 K_2 K_2^{*}} \tag{11.49}$$

This expression is as perplexing as it is complicated. Why should the K's be combined in such a ridiculous way in order to give Q in terms of

11.2 Diagonal model: steps NE, NW, SE, SW

In this case the roots of the kernel become

$$\sigma_i = \frac{1 - \sqrt{1 - 4t^2(S_i^1)^2}}{2tS_i^1}, \quad \overline{\sigma_i} = \frac{1 + \sqrt{1 - 4t^2(S_i^1)^2}}{2tS_i^1} \tag{11.50}$$

with the extra property that

$$\sigma_1 + \overline{\sigma_1} = \frac{1}{tS_1^{-1}}. \tag{11.51}$$

First we have

$$K_2^* - K_2 = t(\overline{x_2} - x_2)S_2^{-1}(\sigma_1) \tag{11.52}$$

$$= t(\overline{x_2} - x_2)(\overline{\sigma_1} + \sigma_1) \tag{11.53}$$

$$= \frac{\overline{x_2} - x_2}{\overline{x_2} + x_2} \tag{11.54}$$

We have

$$K_2^* K_{1,2} - K_2 K_{1,2}^* = (c_2 - tx_2 S_2^{-1})\,(c_1 c_2 - t\overline{\sigma_1 x_2}) - (c_2 - t\overline{x_2} S_2^{-1})\,(c_1 c_2 - t\overline{\sigma_1} x_2) \tag{11.55}$$

$$= c_1 c_2 t(\overline{x_2} - x_2)S_2^{-1} - c_2 t\overline{\sigma_1}(\overline{x_2} - x_2) \tag{11.56}$$

$$= c_2 t(\overline{x_2} - x_2)(c_1(\sigma_1 + \overline{\sigma_1}) - \overline{\sigma_1}) \tag{11.57}$$

If we let $v = \overline{x_2} + x_2$ we find

$$\sigma_1 = \frac{1 - \sqrt{1 - 4t^2v^2}}{2tv}, \quad \overline{\sigma_1} = \frac{1 + \sqrt{1 - 4t^2v^2}}{2tv} \tag{11.58}$$

We have

$$(c_1(\sigma_1 + \overline{\sigma_1}) - \overline{\sigma_1}) = \frac{c1}{tv} - \frac{1 + \sqrt{1 - 4t^2v^2}}{2tv} \tag{11.59}$$

$$= \frac{1}{\overline{x_2} + x_2}\left(-\overline{a_1} + \frac{1 - \sqrt{1 - 4t^2v^2}}{2}\right) \tag{11.60}$$

Also

$$K_1 = c_1 - tv\overline{\sigma_1} \tag{11.61}$$

$$= c_1 - tv\frac{1+\sqrt{1-4t^2v^2}}{2tv} \tag{11.62}$$

$$= 1 - \overline{a_1} - \frac{1+\sqrt{1-4t^2v^2}}{2} \tag{11.63}$$

$$= -\overline{a_1} + \frac{1-\sqrt{1-4t^2v^2}}{2} \tag{11.64}$$

Thus we finally have

$$\frac{1}{K_1}(K_2^*K_{1,2} - K_2K_{1,2}^*) = -\frac{c_2t(\overline{x_2}-x_2)(c_1(\sigma_1+\overline{\sigma_1})-\overline{\sigma_1})}{c_1 - t\overline{\sigma_1}v} \tag{11.65}$$

$$= -\frac{c_2t(\overline{x_2}-x_2)(-1)}{tv} \tag{11.66}$$

$$= \frac{c_2(\overline{x_2}-x_2)}{\overline{x_2}+x_2} \tag{11.67}$$

Thus the equation will now read as

$$K_2^*Q_1 - K_2Q_1^* - \frac{c_2(\overline{x_2}-x_2)}{\overline{x_2}+x_2}Q_{1,2} + \alpha\frac{\overline{x_2}-x_2}{(\overline{x_2}+x_2)K_1} = 0 \tag{11.68}$$

We just need to take care of the remaining terms. We'll do this by multiplying by $\overline{x_2}+x_2$. We have

$$R^* = (\overline{x_2}+x_2)K_2^* = c_2(\overline{x_2}+x_2) - x_2 = c_2\overline{x_2} - \overline{a_2}x_2 \tag{11.69}$$

$$R = (\overline{x_2}+x_2)K_2 = c_2(\overline{x_2}+x_2) - \overline{x_2} = c_2x_2 - \overline{a_2x_2} \tag{11.70}$$

$$\tag{11.71}$$

We have a couple of relations here

$$R^* - R = v(K_2^* - K_2) = \overline{x_2} - x_2 \tag{11.72}$$

$$\overline{x_2}R^* - x_2R = \overline{x_2}c_2v - 1 - x_2c_2v + 1 \tag{11.73}$$

$$= c_2v(\overline{x_2} - x_2) \tag{11.74}$$

The equation becomes

$$(c_2\overline{x_2} - \overline{a_2}x_2)Q_1^0(x_2) + (\overline{a_2x_2} - c_2x_2)Q_1^0(\overline{x_2}) - c_2(\overline{x_2} - x_2)Q_{1,2}^{0,0} + (\overline{x_2} - x_2)\overline{K}_1\alpha = 0 \tag{11.75}$$

and it's ready for extraction. Let $\delta_0 = \overline{a_1}(\overline{x_2} - x_2)\overline{K}_1$. We re-write the equation as

$$R^*Q_1^0 - RQ_1^{0,*} - c_2(\overline{x_2} - x_2)Q_{1,2}^{0,0} + \overline{a_2}\delta_0 = 0 \tag{11.76}$$

We'll use the master equation

$$(1 - tS)Q - (c_1 - t\overline{x_1}S_1^{-1})Q_1^0 - (c_2 - t\overline{x_2}S_2^{-1})Q_2^0 + (c_1c_2 - t\overline{x_1x_2}S_{1,2}^{-1,-1})Q_{1,2}^{0,0} = \overline{a_1a_2} \tag{11.77}$$

to further our work here. For diagonal steps it is

$$(1 - tS)Q - (c_1 - tv\overline{x_1})Q_1^0 - (c_2 - t\overline{x_2}S_2^{-1})Q_2^0 + (c_1c_2 - t\overline{x_1x_2})Q_{1,2}^{0,0} = \overline{a_1a_2} \tag{11.78}$$

extracting the $[x_1^0]$ coefficient yields

$$Q_1^0 = \overline{a_2} + c_2Q_{1,2}^{0,0} + a_1tvQ_1^1 \tag{11.79}$$

We can extract higher coefficients too. Let $k \geqslant 1$. Then the $[x_2^k]$ coefficient is

$$\begin{aligned} 0 &= Q_1^k - tS_1^{-1}Q_1^{k+1} - tS_1^0Q_1^k - tS_1^1Q_1^{k-1} - c_2Q_{1,2}^{k,0} && (11.80)\\ &\quad + t\overline{x_2}(S_{1,2}^{-1,-1}Q_{1,2}^{k+1,0} + S_{1,2}^{0,-1}Q_{1,2}^{k,0} + S_{1,2}^{1,-1}Q_{1,2}^{k-1,0}) && (11.81)\\ &= Q_1^k - tvQ_1^{k+1} - tvQ_1^{k-1} - c_2Q_{1,2}^{k,0} + t\overline{x_2}(Q_{1,2}^{k+1,0} + Q_{1,2}^{k-1,0}) && (11.82) \end{aligned}$$

We can re-write the last one in a form more convenient for the next few steps

$$Q_1^k = tv(Q_1^{k+1} + Q_1^{k-1}) + c_2Q_{1,2}^{k,0} - t\overline{x_2}(Q_{1,2}^{k+1,0} + Q_{1,2}^{k-1,0}) \tag{11.83}$$

We'll use induction to prove a nice formula but for that we'll first need to establish another base case because of this splitting of cases for the coefficient extraction. We have

$$Q_1^0 = \overline{a_2} + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^1 \tag{11.84}$$

$$Q_1^{0,*} = \overline{a_2} + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^{1,*} \tag{11.85}$$

$$\tag{11.86}$$

Combining them with equation (11.76) gives

$$0 = R^* Q_1^0 - R Q_1^{0,*} - c_2(\overline{x_2} - x_2) Q_{1,2}^{0,0} + \overline{a_2}\delta_0 \tag{11.87}$$

$$= R^*(\overline{a_2} + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^1) - R(\overline{a_2} + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^{1,*}) \tag{11.88}$$

$$- c_2(\overline{x_2} - x_2) Q_{1,2}^{0,0} + \overline{a_2}\delta_0 \tag{11.89}$$

$$= a_1 tv(R^* Q_1^1 - R Q_1^{1,*}) + (R^* - R)(\overline{a_2} + c_2 Q_{1,2}^{0,0}) \tag{11.90}$$

$$- c_2(\overline{x_2} - x_2) Q_{1,2}^{0,0} + \overline{a_2}\delta_0 \tag{11.91}$$

$$= a_1 tv(R^* Q_1^1 - R Q_1^{1,*}) + \overline{a_2}(\overline{x_2} - x_2) + \overline{a_2}\delta_0 \tag{11.92}$$

$$= a_1 tv(R^* Q_1^1 - R Q_1^{1,*}) + \overline{a_2}((\overline{x_2} - x_2) + \delta_0) \tag{11.93}$$

So we can write the equation

$$R^* Q_1^1 - R Q_1^{1,*} + \overline{a_2}\delta_1 = 0 \tag{11.94}$$

where

$$\delta_1 = \frac{(\overline{x_2} - x_2) + \delta_0}{a_1 tv} \tag{11.95}$$

This equation together with equation (11.76) form the base cases for the following lemma

Lemma 83. *Let $m \geqslant 0$ be an integer. Then*

$$R^* Q_1^m - R Q_1^{m,*} - c_2(\overline{x_2} - x_2) Q_{1,2}^{m,0} + \overline{a_2}\delta_m = 0 \tag{11.96}$$

where δ_m is not a function of a_2.

Proof. We'll show this by induction. We have the base cases $m = 0$ and $m = 1$. The latter one fits the pattern because $Q_{1,2}^{1,0} = 0$ since we cannot have any walks end on coordinates $(1, 0)$. Now we

assume that the formula is true for all cases up to and including m. Then we use formula (11.83)

$$
\begin{aligned}
& R^* Q_1^m - R Q_1^{m,*} && (11.97)\\
=& tv(R^* Q_1^{m+1} - R Q_1^{m+1,*}) + tv(R^* Q_1^{m-1} - R Q_1^{m-1,*}) && (11.98)\\
& + c_2(R^* - R) Q_{1,2}^{m,0} - t(\overline{x_2} R^* - x_2 R)(Q_{1,2}^{m+1,0} + Q_{1,2}^{m-1,0}) && (11.99)\\
=& tv(R^* Q_1^{m+1} - R Q_1^{m+1,*}) + tv(c_2(\overline{x_2} - x_2) Q_{1,2}^{m-1,0} - \overline{a_2}\delta_{m-1}) && (11.100)\\
& + c_2(\overline{x_2} - x_2) Q_{1,2}^{m,0} - c_2 tv(\overline{x_2} - x_2)(Q_{1,2}^{m+1,0} + Q_{1,2}^{m-1,0}) && (11.101)\\
=& tv(R^* Q_1^{m+1} - R Q_1^{m+1,*}) + c_2(\overline{x_2} - x_2) Q_{1,2}^{m,0} && (11.102)\\
& - c_2 tv(\overline{x_2} - x_2) Q_{1,2}^{m+1,0}) - \overline{a_2} tv \delta_{m-1}) && (11.103)
\end{aligned}
$$

Combining this with equation (11.96) leads to

$$
\begin{aligned}
0 &= tv(R^* Q_1^{m+1} - R Q_1^{m+1,*}) - c_2 tv(\overline{x_2} - x_2) Q_{1,2}^{m+1,0} - \overline{a_2} tv \delta_{m-1} + \overline{a_2}\delta_m && (11.104)\\
0 &= R^* Q_1^{m+1} - R Q_1^{m+1,*} - c_2(\overline{x_2} - x_2) Q_{1,2}^{m+1,0} + \overline{a_2}\delta_{m+1} && (11.105)
\end{aligned}
$$

where

$$
\delta_{m+1} = \frac{\delta_m}{tv} - \delta_{m-1} \tag{11.106}
$$

This function δ_{m+1} is also not dependent on a_2 and this completes the proof! □

There is an equivalent statement for Q_2^n. We can expand R^* and R to a form we can use

$$
\begin{aligned}
0 &= (c_2\overline{x_2} - \overline{a_2}x_2) Q_1^m(x_2) - (c_2 x_2 - \overline{a_2 x_2}) Q_1^m(\overline{x_2}) && (11.107)\\
& - c_2(\overline{x_2} - x_2) Q_{1,2}^{m,0} + \overline{a_2}\delta_m && (11.108)
\end{aligned}
$$

We start by extracting the $[x_2^1]$ coefficient

$$
\begin{aligned}
& c_2 Q_{1,2}^{m,2} - \overline{a_2} Q_{1,2}^{m,0} - c_2 Q_{1,2}^{m,0} + c_2 Q_{1,2}^{m,0} + [x_2^1]\overline{a_2}\delta_m = 0 && (11.109)\\
& c_2 Q_{1,2}^{m,2} - \overline{a_2} Q_{1,2}^{m,0} + [x_2^1]\overline{a_2}\delta_m = 0 && (11.110)
\end{aligned}
$$

We can also extract the $[x_2^{n+1}]$ coefficient for $n \geqslant 2$

$$
c_2 Q_{1,2}^{m,n+2} - \overline{a_2} Q_{1,2}^{m,n} + [x_2^{n+1}]\overline{a_2}\delta_m = 0 \tag{11.111}
$$

This formula also contains the $n = 0$ case so we treat them together. We can set $a_2 = 1$ to get

$$-\breve{Q}_{1,2}^{m,n} + [x_2^{n+1}]\delta_m = 0 \tag{11.112}$$

Substituting this back into the above equation gives

$$b_2 Q_{1,2}^{m,n+2} = Q_{1,2}^{m,n} - \breve{Q}_{1,2}^{m,n} \tag{11.113}$$

An equivalent formula can be derived if the process is followed through the use of x_1 instead of x_2 in the analysis.

Corollary 84. *For integers $m, n \geqslant 0$ we have the following relations*

$$b_1 Q_{1,2}^{m+2,n} = Q_{1,2}^{m,n} - \widehat{Q}_{1,2}^{m,n} \tag{11.114}$$
$$b_2 Q_{1,2}^{m,n+2} = Q_{1,2}^{m,n} - \breve{Q}_{1,2}^{m,n} \tag{11.115}$$

We can sum up those equations to obtain functional equations

Corollary 85. *For integers $m \geqslant 0$ we have the following relations*

$$Q_1^m - \widehat{Q}_1^m = b_1 Q_1^{m+2} \tag{11.116}$$
$$\breve{Q}_1^m - \breve{\widehat{Q}}_1^m = b_1 \breve{Q}_1^{m+2} \tag{11.117}$$
$$x_1^2(Q_2^n - \widehat{Q}_2^n) = b_1(Q_2^n - Q_{1,2}^{1,n} x_1 - Q_{1,2}^{0,n}) \tag{11.118}$$
$$x_1^2(\breve{Q}_2^n - \breve{\widehat{Q}}_2^n) = b_1(\breve{Q}_2^n - \breve{Q}_{1,2}^{1,n} x_1 - \breve{Q}_{1,2}^{0,n}) \tag{11.119}$$
$$x_1^2(Q - \widehat{Q}) = b_1(Q - x_1 Q_1^1 - Q_1^0) \tag{11.120}$$
$$x_1^2(\breve{Q} - \breve{\widehat{Q}}) = b_1(\breve{Q} - x_1 \breve{Q}_1^1 - \breve{Q}_1^0) \tag{11.121}$$

Corollary 86. *For integers $n \geqslant 0$ we have the following relations*

$$Q_2^n - \breve{Q}_2^n = b_2 Q_2^{n+2} \tag{11.122}$$
$$\widehat{Q}_2^n - \breve{\widehat{Q}}_2^n = b_2 \widehat{Q}_2^{n+2} \tag{11.123}$$
$$x_2^2(Q_1^m - \breve{Q}_1^m) = b_2(Q_1^m - Q_{1,2}^{m,1} x_1 - Q_{1,2}^{m,0}) \tag{11.124}$$
$$x_2^2(\widehat{Q}_1^m - \breve{\widehat{Q}}_1^m) = b_2(\widehat{Q}_1^m - \widehat{Q}_{1,2}^{m,1} x_1 - \widehat{Q}_{1,2}^{m,0}) \tag{11.125}$$
$$x_2^2(Q - \breve{Q}) = b_2(Q - x_2 Q_2^1 - Q_2^0) \tag{11.126}$$
$$x_2^2(\widehat{Q} - \breve{\widehat{Q}}) = b_2(\widehat{Q} - x_2 \widehat{Q}_2^1 - \widehat{Q}_2^0) \tag{11.127}$$

Every other equation is obtained from the previous one by either setting $a_1 = 1$ or $a_2 = 1$. Here we can hope to set some telescoping sum type of thing in order to say more about $Q^{0,0}_{1,2}$

Now we can move onto the main result about the diagonal walks and it is that the function for labelled walks ending at $(0,0)$ is connected to the functions for semi-labelled and unlabelled walks endind at $(0,0)$. We can use the master equation to extract the $[x_1^0 x_2^0]$ and $[x_1^1 x_2^1]$ coefficients yielding the following two relations

$$Q^{0,0}_{1,2} = 1 + ta_1a_2Q^{1,1}_{1,2} \tag{11.128}$$
$$Q^{1,1}_{1,2} = t(Q^{0,0}_{1,2} + Q^{2,0}_{1,2} + Q^{0,2}_{1,2} + Q^{2,2}_{1,2}) \tag{11.129}$$

We also have

$$b_1b_2Q^{2,2}_{1,2} = b_1(Q^{2,0}_{1,2} - \breve{Q}^{2,0}_{1,2}) \tag{11.130}$$
$$= b_1Q^{2,0}_{1,2} - b_1\breve{Q}^{2,0}_{1,2} \tag{11.131}$$
$$= Q^{0,0}_{1,2} - \widehat{Q}^{0,0}_{1,2} - \breve{Q}^{0,0}_{1,2} + \breve{\widehat{Q}}^{0,0}_{1,2} \tag{11.132}$$

together with

$$b_1b_2Q^{2,0}_{1,2} = b_2Q^{0,0}_{1,2} - b_2\widehat{Q}^{0,0}_{1,2} \tag{11.133}$$
$$b_1b_2Q^{0,2}_{1,2} = b_1Q^{0,0}_{1,2} - b_1\breve{Q}^{0,0}_{1,2} \tag{11.134}$$

We we have

$$b_1b_2Q^{1,1}_{1,2} = t(b_1b_2Q^{0,0}_{1,2} + b_1b_2Q^{2,0}_{1,2} + b_1b_2Q^{0,2}_{1,2} + b_1b_2Q^{2,2}_{1,2}) \tag{11.135}$$
$$= t(b_1b_2Q^{0,0}_{1,2} + b_2Q^{0,0}_{1,2} - b_2\widehat{Q}^{0,0}_{1,2} + b_1Q^{0,0}_{1,2} \tag{11.136}$$
$$- b_1\breve{Q}^{0,0}_{1,2} + Q^{0,0}_{1,2} - \widehat{Q}^{0,0}_{1,2} - \breve{Q}^{0,0}_{1,2} + \breve{\widehat{Q}}^{0,0}_{1,2}) \tag{11.137}$$
$$= t((b_1b_2 + b_1 + b_2 + 1)Q^{0,0}_{1,2} - (b_2+1)\widehat{Q}^{0,0}_{1,2} \tag{11.138}$$
$$- (b_1+1)\breve{Q}^{0,0}_{1,2} + \breve{\widehat{Q}}^{0,0}_{1,2}) \tag{11.139}$$
$$= t(a_1a_2Q^{0,0}_{1,2} - a_2\widehat{Q}^{0,0}_{1,2} - a_1\breve{Q}^{0,0}_{1,2} + \breve{\widehat{Q}}^{0,0}_{1,2}) \tag{11.140}$$

We can finally write

$$b_1 b_2 Q^{0,0}_{1,2} = b_1 b_2 + t a_1 a_2 b_1 b_2 Q^{1,1}_{1,2} \tag{11.141}$$

$$= b_1 b_2 + t^2 a_1 a_2 (a_1 a_2 Q^{0,0}_{1,2} - a_2 \widehat{Q}^{0,0}_{1,2} - a_1 \breve{Q}^{0,0}_{1,2} + \breve{\widehat{Q}}^{0,0}_{1,2}) \tag{11.142}$$

Let $G(b_1, b_2)$ stand for $Q(0, 0, b_1, b_2)$ and G_i has $b_i = 0$. Then we can write it all as

$$b_1 b_2 G = b_1 b_2 + t^2 a_1 a_2 (a_1 a_2 G - a_2 G_1 - a_1 G_2 + G_{1,2}) \tag{11.143}$$

or written as a linear combination

$$(b_1 b_2 - t^2 a_1^2 a_2^2) G + t^2 a_1 a_2^2 G_1 + t^2 a_1^2 a_2 G_2 - t^2 a_1 a_2 G_{1,2} = b_1 b_2 \tag{11.144}$$

We can divide out by $a_1 a_2$ and obtain the equation

$$(c_1 c_2 - a_1 a_2 t^2) G + a_2 t^2 G_1 + a_1 t^2 G_2 - t^2 G_{1,2} = c_1 c_2 \tag{11.145}$$

from which we can read off

$$f(a_1, a_2, t) = c_1 c_2 - a_1 a_2 t^2 \tag{11.146}$$

and

$$G = \frac{(c_1 c_2 - t^2 (a_2 G_1 + a_1 G_2 - G_{1,2})}{f(a_1, a_2, t)} \tag{11.147}$$

together with $B(fG) = c_1 c_2$ Also, we can solve for Q^j_i and friends, in terms of various $[b^l_k]$ extractions, so overall it should lead to a relationship of the form $B(FQ) = 0$.

11.3 The almost-full model: missing steps E and W

11.4 The full model: all steps available

11.5 Conjectured relationships depending on $S^{-1,-1}_{1,2}$

As we have seen whenever $S^{-1,-1}_{1,2} = 0$, α and G tend to go together in the equations, so in the end we have a solution that depends on $\beta(G)$ multiplied by G_1, G_2 and/or $G_{1,2}$. In the end this leads to

a rational function solution for G in terms of G_1, G_2 and $G_{1,2}$. It's then easy to conjecture that this happens whenever we can have such an algebraic relationship. The other part of the conjecture is that we'd have a linear relationship between G, G_1, G_2 and $G_{1,2}$ when $S_{1,2}^{-1,-1} \neq 0$.

Chapter 12

Epilogue

12.1 Discussion

In this work we examined such and such things. This led to so and so stuff. Finally we see problems in this and that area. Some things remain a mystery.

12.2 Future Work

Obviously we need to solve the remaining problems. We also need to solve/explain the mysteries presented so far. Additionally we need to rigourize the subject some more and bring in some more abstraction to make the methods more powerful. Full treatment of flarvoured problems.

12.3 Conclusion

This work solved the challenges it presented itself and uncovered more challenges that await a good tackling.

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